Algebraic Topology, Part I

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1 Homotopy and a Handful of Constructions

1.1 Homotopies, relative homotopies, and equivalences

Definition 1.1. Homotopy

Let $f, g\colon X \to Y$ be maps. A **homotopy** from f to g is a map $H\colon X \times I \to Y$ such that $H(x,0) = f(x)$ and $H(x,1) = g(x)$ for all $x \in X$. We write $f \simeq g$ if such an H exists.

Definition 1.2. Relative homotopy

Let $A \subseteq X$ and let $f, g\colon X \to Y$ be maps with $f|_A = g|_A$. A **homotopy rel** A from f to g is a homotopy $H\colon X \times I \to Y$ such that $H(a,t) = f(a) = g(a)$ for all $a \in A$ and all $t \in I$. We write $f \simeq g \operatorname{rel} A$.

Definition 1.3. Homotopy equivalence

A map $f\colon X \to Y$ is a **homotopy equivalence** if there exists a map $g\colon Y \to X$ such that $g \circ f \simeq \mathrm{id}_X$ and $f \circ g \simeq \mathrm{id}_Y$. In this case we say X and Y are **homotopy equivalent** and write $X \simeq Y$.

Lemma 1.4 Homotopy equivalence is an equivalence relation on spaces.

Proof. Reflexivity is given by id_X. Symmetry is built into the definition. For transitivity, if $f\colon X \to Y$ and $f'\colon Y \to Z$ are homotopy equivalences with homotopy inverses g and g', then $g \circ g'$ is a homotopy inverse for $f' \circ f$. Indeed

$$(g \circ g') \circ (f' \circ f) = g \circ (g' \circ f') \circ f \simeq g \circ \mathrm{id}_Y \circ f = g \circ f \simeq \mathrm{id}_X,$$

and similarly $(f' \circ f) \circ (g \circ g') \simeq \mathrm{id}_Z$. □

Example. The projection $\pi\colon X \times I \to X$ is a homotopy equivalence. A homotopy inverse is $s\colon X \to X \times I$, $s(x) = (x, 0)$, and

$$H\colon (X \times I) \times I \to X \times I, \qquad H\big((x,t),u\big) = (x, (1-u)t)$$

is a homotopy from $\mathrm{id}_{X \times I}$ to $s \circ \pi$.

Definition 1.5. Deformation retract

A subspace $A \subseteq X$ is a **deformation retract** of X if there exists a map $r\colon X \to A$ with $r|_A = \mathrm{id}_A$ and a homotopy $H\colon X \times I \to X$ such that $H(-,0) = \mathrm{id}_X$, $H(-,1) = \iota \circ r$ (where $\iota\colon A \hookrightarrow X$ is inclusion), and $H(a,t) = a$ for all $a \in A$ and all $t \in I$.

Proposition 1.6

If A is a deformation retract of X, then the inclusion $\iota\colon A \hookrightarrow X$ is a homotopy equivalence with homotopy inverse r.

Proof. By definition, $r \circ \iota = \mathrm{id}_A$. The homotopy H shows that $\iota \circ r \simeq \mathrm{id}_X$. □

Example. The punctured space $\mathbb{R}^n \setminus \{0\}$ deformation retracts onto the sphere S^{n-1} by radial projection $x \mapsto x/\|x\|$. The homotopy $H(x,t) = ((1-t) + t/\|x\|)x$ slides each point along its ray to the unit sphere.

1.2 Quotients and pushouts

Quotients are the main way new spaces are assembled from old ones. The constructions below are all quotients of disjoint unions of simpler pieces.

Definition 1.7. Adjunction space

Let $A \subseteq X$ be a subspace and let $f\colon A \to Y$ be a map. The **adjunction space** $X \cup_f Y$ is the quotient of the disjoint union $X \sqcup Y$ by the identifications $a \sim f(a)$ for all $a \in A$.

It is often helpful to picture this via a pushout diagram. In a category-free phrasing, the defining property is that maps out of $X \cup_f Y$ are the same as pairs of maps out of X and Y that agree on A.

Proposition 1.8. Universal property of $X \cup_f Y$

For any space Z, giving a map $\Phi\colon X \cup_f Y \to Z$ is equivalent to giving maps $\alpha\colon X \to Z$ and $\beta\colon Y \to Z$ such that $\alpha|_A = \beta \circ f$.

Proof. A map Φ determines α and β by restriction to the images of X and Y. Conversely, if α and β agree on A in the stated sense, then the induced map $X \sqcup Y \to Z$ is constant on equivalence classes, hence descends uniquely to $X \cup_f Y$. □

Example. Let (X, x_0) and (Y, y_0) be based spaces. Their wedge $X \vee Y$ can be realized as an adjunction space by identifying the basepoints:

$$X \vee Y \;\cong\; X \cup_{x_0 \sim y_0} Y.$$

Concretely, it is the quotient of $X \sqcup Y$ obtained by forcing x_0 and y_0 to be the same point.

Example. Gluing two disks along their boundary gives a sphere:

$$S^n \;\cong\; D^n \cup_{\mathrm{id}} D^n,$$

where the map on the boundary is the identity $S^{n-1} = \partial D^n \to \partial D^n$.

1.3 Cones and suspension

Definition 1.9. Cone

The **cone** on a space X is the quotient

$$CX \;=\; (X \times I)/(X \times \{0\}),$$

where $X \times \{0\}$ is collapsed to a single point, called the **cone point**.

Proposition 1.10

For every space X, the cone CX is contractible.

Proof. Let $v \in CX$ be the cone point. The homotopy

$$H\colon CX \times I \to CX, \qquad H([x,t],u) = [x,(1-u)t]$$

is well defined and satisfies $H(-,0) = \mathrm{id}_{CX}$ and $H(-,1)$ is the constant map to v. □

Example. For $X = S^n$, the cone CS^n is homeomorphic to the closed ball D^{n+1}. Geometrically, one may think of CS^n as all line segments joining points of S^n to a new vertex.

Definition 1.11. Unreduced suspension

The **suspension** of a space X is the quotient

$$SX \;=\; (X \times I)/(X \times \{0\} \text{ collapsed to one point, and } X \times \{1\} \text{ collapsed to another}).$$

The two special points are the **suspension points**.

Lemma 1.12 There is a natural homeomorphism $SX \cong CX \cup_X CX$, where the two cones are glued along their copy of X.

Proof. Write the first cone as $(X \times [0, \frac{1}{2}])/(X \times \{0\})$ and the second cone as $(X \times [\frac{1}{2}, 1])/(X \times \{1\})$, then identify the two copies of $X \times \{\frac{1}{2}\}$. This produces the same quotient as in Definition 1.11. □

Example. For $X = S^0 = \{-1, +1\}$, the space SX is a circle. For $X = S^1$, the space SX is a 2-sphere with the suspension points as the north and south poles.

1.4 Mapping cylinders

Definition 1.13. Mapping cylinder

Given a map $f\colon X \to Y$, the **mapping cylinder** of f is the quotient

$$M_f \;=\; (X \times I) \sqcup Y \;/\; (x,0) \sim f(x).$$

We write $i\colon X \to M_f$ for the map $x \mapsto [(x,1)]$ and $j\colon Y \to M_f$ for the inclusion of Y.

This is an adjunction space: $M_f = (X \times I) \cup_f Y$, gluing $X \times \{0\}$ to Y via f.

Proposition 1.14

There is a deformation retraction of M_f onto the subspace $j(Y) \subseteq M_f$. In particular, $j\colon Y \hookrightarrow M_f$ is a homotopy equivalence.

Proof. Define $R\colon M_f \to Y$ by $R([y]) = y$ for $y \in Y$ and $R([(x,t)]) = f(x)$ for $(x,t) \in X \times I$. This is well defined since $(x,0) \sim f(x)$. Now define a homotopy $H\colon M_f \times I \to M_f$ by

$$H(y,u) = y \quad (y \in Y), \qquad H([(x,t)],u) = [(x,(1-u)t)].$$

At $u = 0$ this is the identity; at $u = 1$ it collapses the cylinder part onto the glued-in copy of Y. Moreover $H(y,u) = y$ for $y \in Y$, so this is a deformation retraction. □

Example. If $f\colon S^1 \to S^1$ is $z \mapsto z^n$, then M_f can be pictured as a cylinder whose bottom circle is wrapped n times around the target circle. The deformation retraction in Proposition 1.14 forgets the cylinder and keeps only the target end.

Remark. The mapping cylinder gives a convenient replacement of a map by an inclusion. The composite $R \circ i\colon X \to Y$ is exactly f. Thus the map f is realized as "include X as the top end of a cylinder, then retract the cylinder onto Y".

1.5 Mapping cones and cell attachments

Definition 1.15. Mapping cone

Given $f\colon X \to Y$, the **mapping cone** is the quotient

$$C_f \;=\; M_f/(X \times \{1\}).$$

Equivalently, C_f is obtained by gluing a cone on X to Y along f:

$$C_f \;\cong\; Y \cup_f CX.$$

Lemma 1.16 For the identity map $\mathrm{id}_X\colon X \to X$, the mapping cone C_{id_X} is homeomorphic to the cone CX.

Proof. The mapping cylinder M_{id_X} is $X \times I$ with $X \times \{0\}$ identified with X by the identity. Collapsing $X \times \{1\}$ to a point produces exactly $(X \times I)/(X \times \{1\})$, which is canonically homeomorphic to CX by the change of coordinate $t \mapsto 1-t$. □

Example. Let $f\colon S^{n-1} \to X$ be a map. Since $CS^{n-1} \cong D^n$, the adjunction space $X \cup_f D^n$ is the mapping cone C_f. So "attaching an n-cell along f" is a mapping cone construction.

Example. If $f\colon X \to Y$ is the constant map at a chosen point $y_0 \in Y$ and X is based at some x_0, then C_f is the quotient of $Y \sqcup (X \times I)$ obtained by collapsing $X \times \{0\}$ to y_0 and $X \times \{1\}$ to a new point. This makes C_f look like Y with a suspended copy of X attached at y_0. When y_0 is a basepoint of Y, this description becomes the familiar wedge picture $C_f \simeq Y \vee \Sigma X$.

1.6 Reduced suspension and wedges

Definition 1.17. Reduced suspension

Let (X, x_0) be a based space. The **reduced suspension** ΣX is the quotient

$$\Sigma X \;=\; (X \times I)/(X \times \{0\} \;\cup\; X \times \{1\} \;\cup\; \{x_0\} \times I).$$

There is a natural quotient map $q\colon SX \to \Sigma X$ collapsing the line segment $\{x_0\} \times I$.

Remark. For arbitrary based spaces, the quotient map $q : SX \to \Sigma X$ need not be a homotopy equivalence. Such pathologies come from badly based spaces. We mainly work with based CW complexes (and similar well-based spaces), and in that setting the segment $\{x_0\} \times I \subset SX$ is a contractible subcomplex, so collapsing it does not change the homotopy type. Hence q is a homotopy equivalence in our main examples.

Proposition 1.18

For based spaces (X, x_0) and (Y, y_0), there is a natural homeomorphism

$$\Sigma(X \vee Y) \;\cong\; (\Sigma X) \vee (\Sigma Y).$$

Proof. Write $X \vee Y$ as the quotient of $X \sqcup Y$ identifying $x_0 \sim y_0$. Then $(X \vee Y) \times I$ is the corresponding quotient of $(X \times I) \sqcup (Y \times I)$. In the reduced suspension we collapse $(X \vee Y) \times \{0\}$, $(X \vee Y) \times \{1\}$, and the line $(x_0 \sim y_0) \times I$. This collapses the X-part and the Y-part exactly as in ΣX and ΣY, and forces the two cone points to be shared. The resulting quotient is precisely $(\Sigma X) \vee (\Sigma Y)$. □

2 The Fundamental Group and Functoriality

2.1 Paths up to homotopy and concatenation

Fix a space X. A *path* is a map $f\colon I \to X$. If $f(0) = x$ and $f(1) = y$, we may think of f as a path from x to y.

Definition 2.1. Homotopy of paths

Let $f, g\colon I \to X$ be paths with $f(0) = g(0) = x$ and $f(1) = g(1) = y$. A **homotopy of paths** from f to g is a map $H\colon I \times I \to X$ such that

$$H(s, 0) = f(s), \qquad H(s, 1) = g(s), \qquad H(0, t) = x, \qquad H(1, t) = y.$$

We write $f \simeq g \operatorname{rel}\{0, 1\}$.

Lemma 2.2 Homotopy of paths rel $\{0, 1\}$ is an equivalence relation.

Proof. Reflexivity and symmetry are immediate. For transitivity, if $f \simeq g$ via H and $g \simeq k$ via K, then the pasted homotopy

$$L(s, t) = \begin{cases} H(s, 2t) & 0 \le t \le \frac{1}{2}, \\ K(s, 2t - 1) & \frac{1}{2} \le t \le 1, \end{cases}$$

is continuous and yields $f \simeq k \operatorname{rel}\{0, 1\}$. □

There are three basic operations on paths that we will use constantly: reversal, constants, and concatenation. For a path $f\colon I \to X$ we define the reverse path $\overline{f}$ by

$$\overline{f}(s) = f(1 - s).$$

For a point $x \in X$ we write $c_x\colon I \to X$ for the constant path $c_x(s) = x$.

If $f(1) = g(0)$, define the concatenation $fg\colon I \to X$ by traversing f then g:

$$(fg)(s) = \begin{cases} f(2s) & 0 \le s \le \frac{1}{2}, \\ g(2s - 1) & \frac{1}{2} \le s \le 1. \end{cases}$$

Lemma 2.3 If $f_0 \simeq f_1 \operatorname{rel}\{0,1\}$ and $g_0 \simeq g_1 \operatorname{rel}\{0,1\}$, and $f_i(1) = g_i(0)$, then $f_0 g_0 \simeq f_1 g_1 \operatorname{rel}\{0,1\}$.

Proof. Let H be a homotopy from f_0 to f_1 and K a homotopy from g_0 to g_1, both fixing endpoints. Define $L\colon I \times I \to X$ by

$$L(s,t) = \begin{cases} H(2s,t) & 0 \le s \le \frac{1}{2}, \\ K(2s-1,t) & \frac{1}{2} \le s \le 1. \end{cases}$$

Then $L(-,0) = f_0 g_0$ and $L(-,1) = f_1 g_1$, and endpoints are fixed since H and K fix endpoints. □

2.2 The definition of π_1

Definition 2.4. Fundamental group

Let $x_0 \in X$. A **loop** at x_0 is a path $f\colon I \to X$ with $f(0) = f(1) = x_0$. Define $\pi_1(X, x_0)$ to be the set of homotopy classes $[f]$ of loops at x_0, where homotopies fix endpoints.

Concatenation of loops descends to a binary operation on $\pi_1(X, x_0)$ by

$$[f]\,[g] = [fg].$$

Proposition 2.5

With the operation $[f][g] = [fg]$, the set $\pi_1(X, x_0)$ is a group. The identity element is $[c_{x_0}]$, and the inverse of $[f]$ is $[\overline{f}]$.

Proof. Well-definedness follows from Lemma 2.3. The constant loop c_{x_0} is a two-sided identity up to endpoint-fixing homotopy: $f\, c_{x_0} \simeq f$ and $c_{x_0}\, f \simeq f$, by slowing down f and inserting a constant segment.
Associativity holds up to endpoint-fixing homotopy: $(fg)h \simeq f(gh)$. Both paths traverse f, then g, then h; they differ only by a reparametrization of I. A homotopy between the two parametrizations gives the required homotopy of paths.
For inverses, the concatenations $f\,\overline{f}$ and $\overline{f}\,f$ are endpoint-fixing homotopic to c_{x_0}: shrink the traversed interval of f to length 0. □

Example. If X is contractible, then $\pi_1(X, x_0) = 0$. Indeed every loop is homotopic rel endpoints to the constant loop.

2.3 Maps induce homomorphisms

Proposition 2.6

Let $f\colon X \to Y$ be a map and let $x_0 \in X$. Then

$$f_*\colon \pi_1(X, x_0) \to \pi_1(Y, f(x_0)), \qquad f_*([g]) = [f \circ g]$$

is a well-defined group homomorphism. Moreover, $(\mathrm{id}_X)_* = \mathrm{id}$ and $(f \circ g)_* = f_* \circ g_*$.

Proof. If $g_0 \simeq g_1 \operatorname{rel}\{0,1\}$ via H, then $f \circ H$ is a homotopy rel endpoints from $f \circ g_0$ to $f \circ g_1$, so f_* is well defined. It is a homomorphism because $f \circ (gh) = (f \circ g)(f \circ h)$ as paths. The identities for id and composition are immediate from the definition. □

Proposition 2.7. Homotopy invariance

If $f, g\colon X \to Y$ satisfy $f \simeq g$, then $f_* = g_*\colon \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$, after identifying $f(x_0)$ and $g(x_0)$ by the path $t \mapsto H(x_0, t)$ coming from the homotopy.

Proof. Let $H\colon X \times I \to Y$ be a homotopy from f to g. For a loop α at x_0, consider the loop in Y based at $f(x_0)$ given by

$$\beta(t) = H(x_0, t), \qquad \beta\,(f \circ \alpha)\,\overline{\beta}.$$

There is an endpoint-fixing homotopy between this loop and $g \circ \alpha$ obtained by restricting H to the image of α and pasting along the edges of the square. Thus $[f \circ \alpha]$ and $[g \circ \alpha]$ differ by conjugation along β, which is exactly the stated identification. □

2.4 Change of basepoint

Let $h\colon I \to X$ be a path with $h(0) = x_0$ and $h(1) = x_1$. Given a loop f based at x_1, the concatenation $\overline{h}\, f\, h$ is a loop based at x_0.

Definition 2.8. Basepoint-change map

Define

$$\beta_h\colon \pi_1(X, x_1) \to \pi_1(X, x_0), \qquad \beta_h([f]) = [\overline{h}\, f\, h].$$

Proposition 2.9

The map β_h is a group isomorphism with inverse $\beta_{\overline{h}}$.

Proof. It is a homomorphism since

$$\beta_h([fg]) = [\overline{h}\,(fg)\,h] = [(\overline{h}\, f\, h)(\overline{h}\, g\, h)] = \beta_h([f])\,\beta_h([g]).$$

Also $\beta_{\overline{h}}\beta_h([f]) = [h(\overline{h}\, f\, h)\overline{h}] = [f]$, and similarly $\beta_h\beta_{\overline{h}} = \mathrm{id}$. □

Example. If X is path-connected, then $\pi_1(X, x_0)$ and $\pi_1(X, x_1)$ are isomorphic for any two basepoints. The isomorphism depends on the choice of path h, and changing h changes β_h by an inner automorphism of $\pi_1(X, x_0)$.

Basepoint change interacts cleanly with induced maps.

Proposition 2.10

Let $f\colon X \to Y$ and let $h\colon I \to X$ be a path from x_0 to x_1. Then the diagram

$$\begin{tikzcd}
\pi_1(X, x_1) \arrow[r, "f_*"] \arrow[d, "\beta_h"'] & \pi_1(Y, f(x_1)) \arrow[d, "\beta_{f\circ h}"] \\
\pi_1(X, x_0) \arrow[r, "f_*"'] & \pi_1(Y, f(x_0))
\end{tikzcd}$$

commutes.

Proof. For a loop α at x_1 we compute

$$f_*(\beta_h([\alpha])) = [\, f \circ (\overline{h}\,\alpha\, h)\,] = [\,\overline{(f \circ h)}\,(f \circ \alpha)\,(f \circ h)\,] = \beta_{f\circ h}(f_*([\alpha])).$$

□

2.5 Two computations

Theorem 2.11

$\pi_1(S^1, 1) \cong \mathbb{Z}$. An isomorphism sends the class of the loop $\omega(s) = e^{2\pi i s}$ to 1.

Proof. Let $p\colon \mathbb{R} \to S^1$ be $p(t) = e^{2\pi i t}$.
Step 1: path lifting for p. Let $\gamma\colon I \to S^1$ be a path and choose $t_0 \in \mathbb{R}$ with $p(t_0) = \gamma(0)$. Cover S^1 by two open arcs U_1, U_2 on which a continuous argument function exists, hence p admits a continuous local inverse. By compactness of I, there is a subdivision $0 = s_0 < s_1 < \cdots < s_m = 1$ such that $\gamma([s_{j-1}, s_j]) \subset U_{i(j)}$ for each j. Define $\widetilde{\gamma}$ inductively on each $[s_{j-1}, s_j]$ using the chosen local inverse, matching endpoints. Uniqueness follows because any two lifts differ by an integer-valued continuous function, hence are equal.
Step 2: winding number. For a loop γ based at 1, let $\widetilde{\gamma}$ be the unique lift with $\widetilde{\gamma}(0) = 0$. Then $\widetilde{\gamma}(1) \in \mathbb{Z}$ since $p(\widetilde{\gamma}(1)) = \gamma(1) = 1$. Define

$$\mathrm{wind}([\gamma]) = \widetilde{\gamma}(1) \in \mathbb{Z}.$$

This is a homomorphism since the lift of a concatenation is the concatenation of lifts, so endpoints add.
Step 3: surjectivity. The loop $\omega(s) = e^{2\pi i s}$ lifts to $\widetilde{\omega}(s) = s$, hence $\mathrm{wind}([\omega]) = 1$. Thus wind is surjective.
Step 4: injectivity. If $\mathrm{wind}([\gamma]) = 0$, then $\widetilde{\gamma}$ is a loop in $\mathbb{R}$ based at 0. Since $\mathbb{R}$ is convex, $\widetilde{\gamma}$ is homotopic rel endpoints to the constant loop at 0 via linear homotopy. Composing this homotopy with p gives a homotopy

rel endpoints from γ to the constant loop at 1. Hence $[\gamma] = 0$, so wind is injective. $\square$

Proposition 2.12

For $n \geq 2$, $\pi_1(S^n) = 0$.

Proof. Let $\alpha\colon I \to S^n$ be a loop. Its image $\alpha(I)$ is compact, hence a proper subset of S^n. Choose a point $a \in S^n \setminus \alpha(I)$. Then α is a loop in $S^n \setminus \{a\}$. Stereographic projection identifies $S^n \setminus \{a\}$ with $\mathbb{R}^n$, which is simply-connected for $n \geq 2$. Thus α is nullhomotopic in $S^n \setminus \{a\}$, hence also in S^n. $\square$

Proposition 2.13

For $n \geq 2$, $\pi_1(\mathbb{R}P^n) \cong \mathbb{Z}_2$.

Proof. Let $q\colon S^n \to \mathbb{R}P^n$ be the quotient map identifying antipodal points. Choose a basepoint $x_0 \in S^n$ and write $\bar{x}_0 = q(x_0)$.
Step 1: lifting loops to S^n (for this q). For each $x \in S^n$ there is an open neighborhood U such that $U \cap (-U) = \varnothing$. Then $q|_U\colon U \to q(U)$ is a homeomorphism. Given a loop $\gamma\colon I \to \mathbb{R}P^n$ based at $\bar{x}_0$, subdivide I so that each segment lands in some $q(U)$. Starting at x_0, lift segment-by-segment using the local inverses. Uniqueness holds for the same reason as for S^1: once a lift is fixed at one time, it is forced on a neighborhood.
Step 2: the endpoint invariant. Let $\tilde{\gamma}$ be the lift of γ starting at x_0. Then $\tilde{\gamma}(1)$ lies in the fiber $q^{-1}(\bar{x}_0) = \{x_0, -x_0\}$. Define

$$\varepsilon([\gamma]) = \begin{cases} 0 & \tilde{\gamma}(1) = x_0, \\ 1 & \tilde{\gamma}(1) = -x_0, \end{cases} \qquad \in \mathbb{Z}_2.$$

Concatenation of loops concatenates lifts, so ε is a homomorphism.
Step 3: surjectivity. Choose any path λ in S^n from x_0 to $-x_0$. Then $q \circ \lambda$ is a loop at $\bar{x}_0$ with $\varepsilon([q \circ \lambda]) = 1$.
Step 4: injectivity. If $\varepsilon([\gamma]) = 0$, then $\tilde{\gamma}$ is a loop in S^n based at x_0. By Proposition 2.12, this loop is nullhomotopic in S^n. Projecting the nullhomotopy along q gives a nullhomotopy of γ in $\mathbb{R}P^n$. Thus $\ker(\varepsilon) = 0$ and ε is an isomorphism. $\square$

Example. For $n = 1$, $\mathbb{R}P^1 \cong S^1$, hence $\pi_1(\mathbb{R}P^1) \cong \mathbb{Z}$ by Theorem 2.11.

3 van Kampen's Theorem and Standard Calculations

3.1 Free products and amalgamation

Definition 3.1. Free product

Let $\{G_\alpha\}$ be a family of groups. The **free product** $*_\alpha G_\alpha$ is the group whose elements are reduced words $g_1 g_2 \cdots g_m$ with $m \geq 0$, where each $g_i \in G_{\alpha_i} \setminus \{1\}$ and adjacent letters come from different groups, $\alpha_i \neq \alpha_{i+1}$. Multiplication is concatenation followed by reduction (cancelling identities and multiplying adjacent letters from the same factor).

Proposition 3.2. Universal property

Given homomorphisms $\varphi_\alpha\colon G_\alpha \to H$, there exists a unique homomorphism $\varphi\colon *_\alpha G_\alpha \to H$ whose restriction to each G_α is φ_α.

Proof. On a reduced word $g_1 \cdots g_m$ with $g_i \in G_{\alpha_i}$ set $\varphi(g_1 \cdots g_m) = \varphi_{\alpha_1}(g_1) \cdots \varphi_{\alpha_m}(g_m)$. This respects reduction, hence gives a well-defined homomorphism. Uniqueness follows because the factors G_α generate $*_\alpha G_\alpha$. $\square$

Definition 3.3. Normal closure

Let G be a group and $S \subseteq G$. The **normal closure** $\langle\langle S \rangle\rangle$ is the smallest normal subgroup of G containing S, equivalently the subgroup generated by all conjugates gsg^{-1} with $g \in G$ and $s \in S$.

Definition 3.4. Amalgamated free product

Given homomorphisms $i_1 \colon H \to G_1$ and $i_2 \colon H \to G_2$, define

$$G_1 *_H G_2 \;=\; (G_1 * G_2)/\langle\langle\, i_1(h)i_2(h)^{-1} \mid h \in H \,\rangle\rangle.$$

Proposition 3.5. Universal property

A homomorphism $G_1 *_H G_2 \to K$ is the same as a pair of homomorphisms $\phi_1 \colon G_1 \to K$ and $\phi_2 \colon G_2 \to K$ such that $\phi_1 \circ i_1 = \phi_2 \circ i_2$.

Proof. A pair (ϕ_1, ϕ_2) induces a unique $\phi \colon G_1 * G_2 \to K$ by Proposition 3.2. The condition $\phi_1 i_1 = \phi_2 i_2$ is exactly the condition that ϕ kills the normal subgroup in Definition 3.4, so ϕ descends to $G_1 *_H G_2$. Conversely, any map out of $G_1 *_H G_2$ restricts to maps out of G_1 and G_2 with the stated compatibility. □

3.2 The van Kampen theorem

Let X be path-connected, and choose a basepoint $x_0 \in X$. Suppose $X = U \cup V$ where U and V are path-connected open sets containing x_0, and $U \cap V$ is path-connected.

The inclusions induce homomorphisms

$$\pi_1(U \cap V, x_0) \longrightarrow \pi_1(U, x_0), \qquad \pi_1(U \cap V, x_0) \longrightarrow \pi_1(V, x_0),$$

and therefore, by Proposition 3.2, a canonical map

$$\Phi \colon \pi_1(U, x_0) * \pi_1(V, x_0) \longrightarrow \pi_1(X, x_0).$$

Theorem 3.6. van Kampen (two-set form)

With $X = U \cup V$ as above, the map Φ is surjective and induces an isomorphism

$$\pi_1(X, x_0) \;\cong\; \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0).$$

Equivalently, $\ker(\Phi)$ is the normal closure of all elements of the form $i_U(\omega)\, i_V(\omega)^{-1}$ with $\omega \in \pi_1(U \cap V, x_0)$, where i_U, i_V are the maps induced by inclusion.

Proof. *Surjectivity.* Let $\gamma \colon I \to X$ be a loop at x_0. Since I is compact and U, V cover $\gamma(I)$, there is a subdivision $0 = t_0 < t_1 < \cdots < t_m = 1$ such that each subpath $\gamma|_{[t_{j-1}, t_j]}$ lies entirely in U or entirely in V. Each subpath can be turned into a loop based at x_0 by adjoining paths in $U \cap V$ from x_0 to its endpoints and back. Thus $[\gamma]$ can be written as a product of elements coming from $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$, hence lies in the image of Φ.

The kernel. Consider a word in $\pi_1(U, x_0) * \pi_1(V, x_0)$ represented by a concatenation of loops that alternately lie in U and V. If its image in $\pi_1(X, x_0)$ is trivial, there is a homotopy $H \colon I \times I \to X$ to the constant loop. Choose a subdivision of the square into finitely many smaller squares, each mapped by H into either U or V. Reading along the boundaries of these small squares produces a finite list of relations, each occurring entirely inside U or V or coming from switching between the two along an edge in $U \cap V$. The relations inside U or V are already present in the factors $\pi_1(U, x_0)$ and $\pi_1(V, x_0)$. The remaining relations have the form $i_U(\omega) = i_V(\omega)$ with $\omega \in \pi_1(U \cap V, x_0)$, and conjugating these by partial words accounts for the change of basepoint along boundary edges. Hence the word lies in the normal closure of the elements $i_U(\omega) i_V(\omega)^{-1}$. Conversely, every such element maps trivially under Φ, so the normal closure is contained in $\ker(\Phi)$. □

Remark. A convenient way to remember the output is the commutative square

$$\begin{tikzcd}
\pi_1(U \cap V, x_0) \arrow[r] \arrow[d] & \pi_1(U, x_0) \arrow[d] \\
\pi_1(V, x_0) \arrow[r] & \pi_1(X, x_0)
\end{tikzcd}$$

with $\pi_1(X, x_0)$ characterized by the universal property of the pushout.

3.3 Wedge sums and free groups

A first clean case is when the overlap contributes no relations.

Corollary 3.7 (Wedge sums). Let (X_α, x_α) be a family of path-connected based spaces, and form the wedge $\bigvee_\alpha X_\alpha$ by identifying all basepoints. Assume each x_α has an open neighborhood in X_α that deformation retracts to x_α. Then

$$\pi_1\Big(\bigvee_\alpha X_\alpha\Big) \;\cong\; *_\alpha \pi_1(X_\alpha).$$

Proof. Let A_α be the union of X_α with the chosen neighborhoods from the other summands. Then $\{A_\alpha\}$ is an open cover of the wedge, each A_α deformation retracts to X_α, and any intersection of two or more distinct A_α deformation retracts to the wedge point. Applying Theorem 3.6 inductively, the overlap groups are trivial and the free product map is an isomorphism. □

Example. For a wedge of n circles $\bigvee_{i=1}^n S^1$, the group π_1 is the free group on n generators. A generator can be taken to be the loop that runs once around the ith circle. Words in these generators correspond to concatenations of these loops, reduced by cancelling consecutive inverse traversals.

3.4 Graphs

A connected graph is a 1-dimensional CW complex. One can compute its fundamental group by collapsing a maximal tree.

Proposition 3.8

Let G be a connected graph with vertex set V and edge set E. Then $\pi_1(G)$ is free of rank $|E| - |V| + 1$.

Proof. Choose a maximal tree $T \subseteq G$. Since T is contractible, it deformation retracts to a point by sliding along the unique path to a chosen root vertex. Collapsing T to a point turns each edge in $E \setminus E(T)$ into a circle attached at that point, and the quotient map $G \to G/T$ is a homotopy equivalence (it is a deformation retraction on a neighborhood built from small edge segments). Thus $G \simeq \bigvee_{|E|-|V|+1} S^1$, and Corollary 3.7 gives that $\pi_1(G)$ is free of the stated rank. □

3.5 Punctured planes

Proposition 3.9

Let $P = \{p_1, \ldots, p_n\} \subset \mathbb{R}^2$ be a finite set. Then $\mathbb{R}^2 \setminus P$ deformation retracts onto a wedge of n circles. Consequently, $\pi_1(\mathbb{R}^2 \setminus P)$ is a free group on n generators.

Proof. Choose disjoint closed disks D_i around each p_i, and choose disjoint embedded arcs from a common point x_0 to the boundary circles ∂D_i, meeting only at x_0. Let K be the union of the circles ∂D_i together with these arcs. A standard planar deformation retraction pushes $\mathbb{R}^2 \setminus P$ onto K by sliding points along straight segments until they hit K (outside the disks, points are pushed toward the arcs; near each puncture, points are pushed radially to ∂D_i). The space K is a wedge of n circles at x_0. The statement about π_1 follows from Corollary 3.7. □

Example. For $n = 1$ this recovers $\pi_1(\mathbb{R}^2 \setminus \{0\}) \cong \mathbb{Z}$. A generator is represented by a positively oriented circle around the puncture.

3.6 Surface groups

The computation for compact surfaces is a direct application of the idea that attaching a 2-cell adds one relation.

Proposition 3.10. Attaching a 2-cell adds a relation

Let X be obtained from a path-connected space A by attaching a 2-cell along a map $\varphi\colon S^1 \to A$. Let $\langle\langle[\varphi]\rangle\rangle$ be the normal closure of the element $[\varphi] \in \pi_1(A)$. Then the inclusion $A \hookrightarrow X$ induces an isomorphism

$$\pi_1(X) \;\cong\; \pi_1(A)/\langle\langle[\varphi]\rangle\rangle.$$

Proof. Write $X = A \cup D^2$, where D^2 is the attached disk. Choose a small open collar neighborhood N of ∂D^2 inside D^2, so that $N \simeq S^1$ and D^2 is simply connected. Then $A \cup N$ is an open neighborhood of A in X, and $(A \cup N) \cap D^2 = N$ is path-connected. Apply Theorem 3.6 to the cover $X = (A \cup N) \cup D^2$. Since $\pi_1(D^2) = 0$, the result is that $\pi_1(X)$ is the quotient of $\pi_1(A \cup N) \cong \pi_1(A) * \pi_1(N)$ by the normal subgroup identifying the image of $\pi_1(N)$ with its image in $\pi_1(A)$ under the attaching map. This forces the generator of $\pi_1(N) \cong \mathbb{Z}$ to become $[\varphi] \in \pi_1(A)$ and then to vanish, which is precisely quotienting by $\langle\langle[\varphi]\rangle\rangle$. □

Theorem 3.11. Orientable surface group

Let M_g be the closed orientable surface of genus $g \geq 1$. Then $\pi_1(M_g)$ admits the presentation

$$\pi_1(M_g) \;\cong\; \langle a_1, b_1, \ldots, a_g, b_g \mid [a_1, b_1] \cdots [a_g, b_g] = 1\rangle, \qquad [a, b] = aba^{-1}b^{-1}.$$

Proof. Using the standard $4g$-gon model with edge identifications, M_g has a CW structure with one 0-cell, $2g$ 1-cells, and one 2-cell. The 1-skeleton is a wedge of $2g$ circles, so its fundamental group is free on generators $a_1, b_1, \ldots, a_g, b_g$. The boundary of the 2-cell maps to the 1-skeleton by the word

$$[a_1, b_1] \cdots [a_g, b_g].$$

Applying Proposition 3.10 gives the stated presentation. □

Theorem 3.12. Nonorientable surface group

Let N_k be the closed nonorientable surface obtained as a connected sum of $k \geq 1$ copies of $\mathbb{R}P^2$. Then

$$\pi_1(N_k) \;\cong\; \langle x_1, \ldots, x_k \mid x_1^2 x_2^2 \cdots x_k^2 = 1\rangle.$$

Proof. The usual $2k$-gon model for N_k has one 0-cell, k 1-cells, and one 2-cell. The 1-skeleton is a wedge of k circles, giving a free group on generators $x_1, \ldots, x_k$. Tracing the boundary of the polygon yields the attaching loop $x_1^2 \cdots x_k^2$. Now apply Proposition 3.10. □

4 Covering Spaces and Classification

4.1 Covering maps

Definition 4.1. Covering map

A map $p\colon \widetilde{X} \to X$ is a **covering map** if every point $x \in X$ has an open neighborhood U such that

$$p^{-1}(U) = \bigsqcup_{\alpha \in A} U_\alpha$$

as a disjoint union of open sets $U_\alpha \subset \widetilde{X}$, and each restriction $p|_{U_\alpha}\colon U_\alpha \to U$ is a homeomorphism. Such a U is called **evenly covered**.

Evenly covered neighborhoods force a strong uniqueness principle for lifts.

Lemma 4.2. Uniqueness of lifts on connected domains Let $p\colon \widetilde{X} \to X$ be a covering map and let Y be connected. If $\widetilde{f}_1, \widetilde{f}_2\colon Y \to \widetilde{X}$ satisfy $p \circ \widetilde{f}_1 = p \circ \widetilde{f}_2$ and $\widetilde{f}_1(y_0) = \widetilde{f}_2(y_0)$ for some $y_0 \in Y$, then $\widetilde{f}_1 = \widetilde{f}_2$.

Proof. Let $E = \{y \in Y \mid \widetilde{f}_1(y) = \widetilde{f}_2(y)\}$. Fix $y \in Y$ and choose an evenly covered neighborhood U of $p(\widetilde{f}_1(y)) = p(\widetilde{f}_2(y))$. Let $\widetilde{U}_1, \widetilde{U}_2$ be the sheets containing $\widetilde{f}_1(y)$ and $\widetilde{f}_2(y)$. By continuity, after shrinking to a neighborhood of y we may assume $\widetilde{f}_i$ lands in $\widetilde{U}_i$. If $\widetilde{U}_1 \neq \widetilde{U}_2$, then $\widetilde{f}_1 \neq \widetilde{f}_2$ near y. If $\widetilde{U}_1 = \widetilde{U}_2$, then p is injective on this sheet, so $p\widetilde{f}_1 = p\widetilde{f}_2$ forces $\widetilde{f}_1 = \widetilde{f}_2$ near y. Thus E is open and closed. Since Y is connected

and $y_0 \in E$, we have $E = Y$. $\square$

4.2 Path lifting and homotopy lifting

Proposition 4.3. Path lifting

Let $p\colon \widetilde{X} \to X$ be a covering map. Given a path $\gamma\colon I \to X$ with $\gamma(0) = x_0$ and a point $\widetilde{x}_0 \in p^{-1}(x_0)$, there exists a unique lift $\widetilde{\gamma}\colon I \to \widetilde{X}$ such that $\widetilde{\gamma}(0) = \widetilde{x}_0$ and $p \circ \widetilde{\gamma} = \gamma$.

Proof. For each $t \in I$ choose an evenly covered neighborhood U_t of $\gamma(t)$. Compactness gives a subdivision $0 = t_0 < \cdots < t_m = 1$ such that $\gamma([t_{j-1}, t_j]) \subset U_{t_j}$. Assume $\widetilde{\gamma}$ is defined on $[0, t_{j-1}]$. Let $\widetilde{U}$ be the unique sheet over U_{t_j} containing the point $\widetilde{\gamma}(t_{j-1})$. Define $\widetilde{\gamma}$ on $[t_{j-1}, t_j]$ by composing $\gamma|_{[t_{j-1},t_j]}$ with the inverse of $p|_{\widetilde{U}}$. These pieces fit together by construction, giving existence.
For uniqueness, if $\widetilde{\gamma}_1, \widetilde{\gamma}_2$ are lifts with the same initial value, apply Lemma 4.2 to $Y = I$. $\square$

Proposition 4.4. Homotopy lifting

Let $p\colon \widetilde{X} \to X$ be a covering map and let $H\colon Y \times I \to X$ be a homotopy. If $\widetilde{H}_0\colon Y \to \widetilde{X}$ is a lift of $H(-, 0)$, then there exists a unique lift $\widetilde{H}\colon Y \times I \to \widetilde{X}$ with $\widetilde{H}(-, 0) = \widetilde{H}_0$ and $p \circ \widetilde{H} = H$.

Proof. Fix $y \in Y$. By Proposition 4.3, the path $t \mapsto H(y, t)$ has a unique lift starting at $\widetilde{H}_0(y)$. This defines a set-theoretic map $\widetilde{H}\colon Y \times I \to \widetilde{X}$.
Continuity is local. Fix (y, t) and choose an evenly covered neighborhood U of $H(y, t)$. After shrinking to a product neighborhood $W \times J$ of (y, t) in $Y \times I$, we may assume $H(W \times J) \subset U$. Let $\widetilde{U}$ be the sheet containing $\widetilde{H}(y, t)$. For $(y', t') \in W \times J$, the path $s \mapsto H(y', (1-s)t + st')$ stays inside U. Its lift starting at $\widetilde{H}(y', t)$ must stay inside $\widetilde{U}$, because $p|_{\widetilde{U}}$ is a homeomorphism onto U. This shows $\widetilde{H}(W \times J) \subset \widetilde{U}$, and there $\widetilde{H}$ is given by $(p|_{\widetilde{U}})^{-1} \circ H$, hence continuous. Uniqueness follows from Lemma 4.2 applied to each connected component of $Y \times I$. $\square$

Proposition 4.5. Injectivity of p_*

Let $p\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a covering map. Then $p_*\colon \pi_1(\widetilde{X}, \widetilde{x}_0) \to \pi_1(X, x_0)$ is injective. Moreover, a loop γ at x_0 represents an element in the image of p_* if and only if its lift starting at $\widetilde{x}_0$ is a loop.

Proof. Suppose $[\widetilde{\gamma}] \in \pi_1(\widetilde{X}, \widetilde{x}_0)$ and $p_*([\widetilde{\gamma}]) = \{e\}$. Then $p \circ \widetilde{\gamma}$ is homotopic rel endpoints to the constant loop at x_0. Lift this homotopy using Proposition 4.4 with initial lift $\widetilde{\gamma}$. The lifted homotopy ends at a lift of the constant loop that starts at $\widetilde{x}_0$, hence is the constant loop at $\widetilde{x}_0$ by uniqueness. Thus $\widetilde{\gamma}$ is nullhomotopic, proving injectivity.
For the image statement, if $\gamma = p \circ \widetilde{\gamma}$ then the lift of γ starting at $\widetilde{x}_0$ is $\widetilde{\gamma}$ and closes. Conversely, if the lift $\widetilde{\gamma}$ closes, then $\gamma = p \circ \widetilde{\gamma}$, so $[\gamma] = p_*([\widetilde{\gamma}])$. $\square$

Example. For $p\colon \mathbb{R} \to S^1$, $p(t) = e^{2\pi i t}$, every loop in S^1 lifts to a path in $\mathbb{R}$. It closes if and only if the endpoint differs from the start by an integer. This is the geometric reason the image subgroup is $0 \leq \mathbb{Z} \leq \mathbb{Z}$, and also the reason the deck group is generated by $t \mapsto t + 1$.

4.3 Deck transformations and the action on a fiber

Definition 4.6. Deck transformations

Let $p\colon \widetilde{X} \to X$ be a covering map. A **deck transformation** is a homeomorphism $\varphi\colon \widetilde{X} \to \widetilde{X}$ with $p \circ \varphi = p$. The set of deck transformations is a group, denoted $\mathrm{Deck}(\widetilde{X}/X)$.

Lemma 4.7. A deck transformation is determined by one value If $\widetilde{X}$ is path-connected and $\varphi, \psi \in \mathrm{Deck}(\widetilde{X}/X)$ satisfy $\varphi(\widetilde{x}) = \psi(\widetilde{x})$ for some $\widetilde{x} \in \widetilde{X}$, then $\varphi = \psi$.

Proof. Both φ and ψ are lifts of $p\colon \widetilde{X} \to X$. Apply Lemma 4.2 with $Y = \widetilde{X}$. $\square$

Fix $x_0 \in X$ and write $F = p^{-1}(x_0)$. Every loop at x_0 permutes F by lifting.

Proposition 4.8. Monodromy action

Let $p\colon (\widetilde{X}, \widetilde{x}) \to (X, x_0)$ be a covering map. For $[\gamma] \in \pi_1(X, x_0)$ define

$$\widetilde{x} \cdot [\gamma] = \widetilde{\gamma}(1),$$

where $\widetilde{\gamma}$ is the lift of γ starting at $\widetilde{x}$. Then $\cdot$ defines a right action of $\pi_1(X, x_0)$ on F. If $\widetilde{X}$ is path-connected, the action is transitive, and the stabilizer of $\widetilde{x}$ is the subgroup $p_*\pi_1(\widetilde{X}, \widetilde{x}) \subseteq \pi_1(X, x_0)$.

Proof. If $\gamma \simeq \gamma'$ rel endpoints, then lifts with the same initial value have the same endpoint by Proposition 4.4, so the endpoint depends only on $[\gamma]$. For the action law, lift $\gamma_1\gamma_2$ starting at $\widetilde{x}$. On $[0, \frac{1}{2}]$ it agrees with the lift of γ_1 starting at $\widetilde{x}$. On $[\frac{1}{2}, 1]$ it is the lift of γ_2 starting at $\widetilde{x} \cdot [\gamma_1]$. Thus $(\widetilde{x} \cdot [\gamma_1]) \cdot [\gamma_2] = \widetilde{x} \cdot ([\gamma_1][\gamma_2])$.
If $\widetilde{X}$ is path-connected and $\widetilde{y} \in F$, choose a path $\widetilde{\alpha}$ in $\widetilde{X}$ from $\widetilde{x}$ to $\widetilde{y}$. Then $\alpha = p\widetilde{\alpha}$ is a loop at x_0, and lifting α from $\widetilde{x}$ returns $\widetilde{\alpha}$, so $\widetilde{x} \cdot [\alpha] = \widetilde{y}$. Finally, $\widetilde{x} \cdot [\gamma] = \widetilde{x}$ iff the lift of γ from $\widetilde{x}$ closes, which is equivalent to $[\gamma] \in p_*\pi_1(\widetilde{X}, \widetilde{x})$ by Proposition 4.5. □

Corollary 4.9 (Sheets as cosets)**.** Assume X and $\widetilde{X}$ are path-connected and let $H = p_*\pi_1(\widetilde{X}, \widetilde{x}_0) \subseteq \pi_1(X, x_0)$. Then the fiber $F = p^{-1}(x_0)$ is in bijection with the set of left cosets $H\backslash\pi_1(X, x_0)$, and in particular the number of sheets is the index $[\pi_1(X, x_0) : H]$.

Proof. Send the coset $H[\gamma]$ to the endpoint of the lift of γ starting at $\widetilde{x}_0$. The map is well defined because elements of H lift to loops at $\widetilde{x}_0$. Surjectivity follows as in Proposition 4.8. Injectivity is equivalent to: two lifted endpoints agree iff the loop $\gamma_1\gamma_2^{-1}$ lifts to a loop at $\widetilde{x}_0$, which is equivalent to $[\gamma_1][\gamma_2]^{-1} \in H$. □

Example. For $p_n\colon S^1 \to S^1$, $p_n(z) = z^n$, the subgroup is $n\mathbb{Z} \leq \mathbb{Z}$. The index is n, matching the fact that $p_n^{-1}(1)$ consists of the nth roots of unity. The monodromy action of $\mathbb{Z}$ on the fiber is rotation by one step, generated by lifting the standard loop once around S^1.

4.4 Lifting criterion

The lifting problem is controlled by a subgroup condition.

Theorem 4.10. Lifting criterion

Let $p\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a covering map, and let $f\colon (Y, y_0) \to (X, x_0)$ be a based map with Y path-connected and locally path-connected. A lift $\widetilde{f}\colon (Y, y_0) \to (\widetilde{X}, \widetilde{x}_0)$ exists if and only if

$$f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \widetilde{x}_0) \subseteq \pi_1(X, x_0).$$

If a lift exists, it is unique.

Proof. If $\widetilde{f}$ exists, then $f = p \circ \widetilde{f}$ implies $f_* = p_* \circ \widetilde{f}_*$, hence the inclusion of subgroups.
Conversely, assume $f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(\widetilde{X}, \widetilde{x}_0)$. For each $y \in Y$ choose a path α_y from y_0 to y and set $\widetilde{f}(y)$ to be the endpoint of the lift of $f \circ \alpha_y$ starting at $\widetilde{x}_0$. If α'_y is another such path, then $\alpha'_y\overline{\alpha_y}$ is a loop at y_0. By assumption, its image under f_* lies in $p_*\pi_1(\widetilde{X}, \widetilde{x}_0)$, so the lift of $f \circ (\alpha'_y\overline{\alpha_y})$ starting at $\widetilde{x}_0$ is a loop. It follows that the lifts of $f \circ \alpha_y$ and $f \circ \alpha'_y$ have the same endpoint, so $\widetilde{f}$ is well defined.
To check continuity, fix $y \in Y$ and choose an evenly covered neighborhood U of $f(y)$. Since Y is locally path-connected, there is a path-connected neighborhood W of y with $f(W) \subset U$. Let $\widetilde{U}$ be the sheet containing $\widetilde{f}(y)$. For $y' \in W$, the path $\alpha_{y'}$ can be chosen as α_y followed by a path in W from y to y'. Its image under f stays in U on the last segment, so the lifted endpoint varies continuously and stays in $\widetilde{U}$. On W the map is given by $(p|_{\widetilde{U}})^{-1} \circ f$, hence continuous.
Uniqueness follows from Lemma 4.2 with Y connected. □

Example. Take $p\colon \mathbb{R} \to S^1$ and a based map $f\colon (Y, y_0) \to (S^1, 1)$. Then a lift $Y \to \mathbb{R}$ exists exactly when $f_*\pi_1(Y, y_0) = \{e\}$. In particular, if Y is simply connected then every based map to S^1 lifts uniquely once the basepoint lift is fixed.

4.5 Universal covers

A universal cover is a simply connected cover. Its existence requires mild local control on loops.

Definition 4.11. Semilocally simply connected

A space X is **semilocally simply connected** if for every $x \in X$ there exists an open neighborhood U of x such that the inclusion-induced map $\pi_1(U, x) \to \pi_1(X, x)$ is trivial.

Remark. Local path-connectedness and the condition in Definition 4.11 matter. There are spaces that are path-connected and locally path-connected but fail to admit a universal cover, because arbitrarily small neighborhoods carry nontrivial loops that survive in $\pi_1(X)$.

Theorem 4.12. Existence of a universal cover

If X is path-connected, locally path-connected, and semilocally simply connected, then there exists a covering map $q\colon \widetilde{X} \to X$ with $\pi_1(\widetilde{X}) = 0$.

Proof. Fix $x_0 \in X$. Let $\widetilde{X}$ be the set of endpoint-preserving homotopy classes rel endpoints of paths $\gamma\colon I \to X$ with $\gamma(0) = x_0$. Define $q([\gamma]) = \gamma(1)$.

For each $x \in X$, choose a path-connected open neighborhood U of x whose image in $\pi_1(X, x)$ is trivial. For each class $[\gamma]$ with endpoint $\gamma(1) \in U$, define

$$\widetilde{U}_{[\gamma]} = \{[\gamma\eta] \mid \eta \text{ is a path in } U \text{ with } \eta(0) = \gamma(1)\}.$$

Because loops in U become nullhomotopic in X, the class $[\gamma\eta]$ depends only on the endpoint $\eta(1)$. Hence q restricts to a bijection $\widetilde{U}_{[\gamma]} \to U$.

Use the sets $\widetilde{U}_{[\gamma]}$ as a basis for a topology on $\widetilde{X}$. If $[\gamma'] \in \widetilde{U}_{[\gamma]} \cap \widetilde{V}_{[\delta]}$, then $q([\gamma'])$ lies in $U \cap V$, which is open and locally path-connected; a small path-connected neighborhood $W \subset U \cap V$ of $q([\gamma'])$ yields a basic open set $\widetilde{W}_{[\gamma']}$ contained in both, showing the basis is closed under intersections. With this topology, each restriction $q|_{\widetilde{U}_{[\gamma]}}$ is a homeomorphism onto U, so q is a covering map.

To see $\pi_1(\widetilde{X}) = 0$, let $\widetilde{\alpha}$ be a loop in $\widetilde{X}$ based at the class of the constant path at x_0. Its projection $\alpha = q\widetilde{\alpha}$ is a loop in X based at x_0. Lift α to a path in $\widetilde{X}$ starting at the constant-path class. By construction of $\widetilde{X}$, that lift is exactly $\widetilde{\alpha}$. The endpoint of the lifted path records the endpoint-homotopy class of α. Since $\widetilde{\alpha}$ closes, α is homotopic rel endpoints to the constant loop, hence $[\alpha] = 0$ in $\pi_1(X, x_0)$. Now apply Proposition 4.5 to q to conclude $[\widetilde{\alpha}] = 0$. □

Example. For $X = \mathbb{R}P^n$ with $n \geq 2$, the antipodal quotient map $S^n \to \mathbb{R}P^n$ is a 2-sheeted covering. Since S^n is simply connected for $n \geq 2$, this is the universal cover. The deck group is generated by the antipodal map.

4.6 Classification of covering spaces

Fix X path-connected, locally path-connected, and semilocally simply connected, and fix a basepoint $x_0 \in X$. Let $q\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ be a universal cover.

Theorem 4.13. Classification of connected coverings

1. Pointed, connected covering spaces $p\colon (E, e_0) \to (X, x_0)$ up to isomorphism over X are in bijection with subgroups $H \leq \pi_1(X, x_0)$, via $H = p_*\pi_1(E, e_0)$.
2. Connected covering spaces $p\colon E \to X$ up to isomorphism over X are in bijection with conjugacy classes of subgroups of $\pi_1(X, x_0)$.
3. If $H \leq \pi_1(X, x_0)$ corresponds to $p\colon (E, e_0) \to (X, x_0)$, then E is isomorphic over X to the quotient $\widetilde{X}/H$ for the induced action of H on $\widetilde{X}$.

Proof. The group $\pi_1(X, x_0)$ acts on $\widetilde{X}$ by lifting loops based at x_0. More explicitly, for $[\gamma] \in \pi_1(X, x_0)$ and $\widetilde{x} \in q^{-1}(x_0)$, let $\widetilde{x} \cdot [\gamma]$ be the endpoint of the lift of γ starting at $\widetilde{x}$. This extends to an action on all of $\widetilde{X}$ by lifting paths.

Given a subgroup $H \leq \pi_1(X, x_0)$, restrict the action to H and form the orbit space $\widetilde{X}/H$. Since the action is properly discontinuous and free, the quotient map $\widetilde{X} \to \widetilde{X}/H$ is a covering map, and the composite

$$\widetilde{X} \longrightarrow \widetilde{X}/H \longrightarrow X$$

exhibits $\widetilde{X}/H \to X$ as a connected covering. Its associated subgroup is H by Proposition 4.8 and Corollary 4.9. Conversely, let $p\colon (E, e_0) \to (X, x_0)$ be a connected covering with subgroup $H = p_*\pi_1(E, e_0)$. Apply Theorem 4.10 to the universal cover $q\colon (\widetilde{X}, \widetilde{x}_0) \to (X, x_0)$ and the map $p\colon (E, e_0) \to (X, x_0)$: since $q_*\pi_1(\widetilde{X}, \widetilde{x}_0) = \{e\} \subseteq H$, there exists a unique lift $\widetilde{p}\colon (\widetilde{X}, \widetilde{x}_0) \to (E, e_0)$ with $p \circ \widetilde{p} = q$. This $\widetilde{p}$ is a covering map. Two points $\widetilde{x}, \widetilde{y} \in \widetilde{X}$ map to the same point of E exactly when they differ by the H-action, because a path from $\widetilde{x}$ to $\widetilde{y}$ projects to a loop whose class lies in H precisely when the endpoints coincide in E. Therefore $E \cong \widetilde{X}/H$ over X.

Changing the basepoint e_0 in E changes H by conjugation in $\pi_1(X, x_0)$, giving the unpointed statement. □

Proposition 4.14. Normal coverings and deck groups

Let $H \leq \pi_1(X, x_0)$ correspond to the covering $\widetilde{X}/H \to X$. Let $N(H) = \{g \in \pi_1(X, x_0) \mid gHg^{-1} = H\}$ be the normalizer of H. Then

$$\mathrm{Deck}((\widetilde{X}/H)/X) \cong N(H)/H.$$

In particular, the covering is normal if and only if H is normal in $\pi_1(X, x_0)$, and then $\mathrm{Deck}(\widetilde{X}/H\,/\,X) \cong \pi_1(X, x_0)/H$.

Proof. A deck transformation of $\widetilde{X}/H$ lifts to a deck transformation of $\widetilde{X}$ that permutes H-orbits. Deck transformations of $\widetilde{X}$ correspond to elements of $\pi_1(X, x_0)$, and preserving H-orbits is exactly the condition of lying in $N(H)$. Two elements of $N(H)$ induce the same map on $\widetilde{X}/H$ precisely when they differ by an element of H. □

Example. Let $X = S^1 \vee S^1$. Then $\pi_1(X)$ is the free group on two generators. Connected coverings of X can be drawn as connected graphs mapping to X so that each vertex has one incoming and one outgoing edge above each oriented loop of X. The universal cover is an infinite tree. Finite-sheeted coverings correspond to finite-index subgroups, and the index is the number of vertices in the fiber over the wedge point.

5 CW Complexes and (Δ)-Complexes

5.1 Cells, skeleta, and attaching maps

A CW complex is built by attaching disks one dimension at a time. What matters in practice is that the space comes with a filtration

$$X^{(0)} \subset X^{(1)} \subset \cdots \subset X^{(n)} \subset \cdots \subset X,$$

where each stage is obtained from the previous one by gluing in n-disks along their boundary spheres.

Definition 5.1. CW complex

A **CW complex** is a space X together with a sequence of subspaces $X^{(n)}$ $(n \geq 0)$ such that:

1. $X^{(0)}$ is a discrete set.
2. For each $n \geq 1$, the space $X^{(n)}$ is obtained from $X^{(n-1)}$ by attaching a (possibly empty) family of n-cells. Concretely, there is an indexing set J_n and maps (attaching maps)
$$\varphi_j\colon S^{n-1} \to X^{(n-1)} \qquad (j \in J_n)$$
such that
$$X^{(n)} \;\cong\; X^{(n-1)} \cup_{\coprod_{j\in J_n} \varphi_j} \coprod_{j \in J_n} D^n_j,$$
that is, $X^{(n)}$ is the quotient of $X^{(n-1)} \sqcup \coprod_{j\in J_n} D^n_j$ obtained by identifying $x \in \partial D^n_j$ with $\varphi_j(x)$.
3. $X = \bigcup_{n\geq 0} X^{(n)}$, endowed with the weak topology with respect to the inclusions $X^{(n)} \hookrightarrow X$.

The subspace $X^{(n)}$ is the **n-skeleton** of X.

Definition 5.2. Characteristic map

For an n-cell indexed by $j \in J_n$, the **characteristic map** is the composite

$$\Phi_j \colon D^n_j \longrightarrow X^{(n-1)} \sqcup \bigsqcup_{j \in J_n} D^n_j \longrightarrow X^{(n)} \hookrightarrow X.$$

Its restriction $\Phi_j|_{\mathring{D}^n_j}$ is a homeomorphism onto the open cell $e^n_j \subset X$.

Example. A connected graph is a CW complex with $X^{(0)}$ the vertices and 1-cells the edges. A maximal tree in the graph is a contractible subcomplex, and collapsing it to a point turns the graph into a wedge of circles. This gives a fast route to the computation of π_1 of a graph.

Example. The n-sphere has a CW structure with one 0-cell and one n-cell. The attaching map $S^{n-1} \to X^{(0)}$ is the constant map, so

$$S^n \cong X^{(0)} \cup D^n \cong D^n/\partial D^n.$$

Attaching cells is a pushout construction and therefore has a clean universal property. In the language of adjunction spaces from Lecture 1, $X^{(n)}$ is obtained from $X^{(n-1)}$ by gluing in disks.

Proposition 5.3. A relative cell attachment is a pushout

Let A be a subspace and let $\{\varphi_j : S^{n-1}_j \to A\}_{j \in J}$ be attaching maps. Set

$$X \;=\; A \cup_{\bigsqcup_j \varphi_j} \bigsqcup_{j \in J} D^n_j.$$

Then for every space Z, giving a map $X \to Z$ is equivalent to giving a map $A \to Z$ and maps $D^n_j \to Z$ such that the restrictions $S^{n-1}_j \to Z$ agree with the composites $S^{n-1}_j \xrightarrow{\varphi_j} A \to Z$.

Proof. This is the universal property of the quotient defining X as an adjunction space. A map from the disjoint union $A \sqcup \bigsqcup_j D^n_j$ descends to X exactly when it is constant on the equivalence relation generated by $x \sim \varphi_j(x)$ for $x \in S^{n-1}_j$. □

A useful flexibility is that, up to homotopy, the attaching maps only matter through their homotopy classes.

Proposition 5.4. Homotopic attaching maps give homotopy equivalent spaces

Let A be a space and let $\varphi, \psi \colon S^{n-1} \to A$ be homotopic maps. Form the adjunction spaces

$$X_\varphi = A \cup_\varphi D^n, \qquad X_\psi = A \cup_\psi D^n.$$

Then there is a homotopy equivalence $X_\varphi \simeq X_\psi$ whose restriction to the copy of A is the identity.

Proof. Choose a homotopy $H \colon S^{n-1} \times I \to A$ from φ to ψ. Consider the mapping cylinder M_H of H and its two ends $S^{n-1} \times \{0\}$ and $S^{n-1} \times \{1\}$. Glue a disk D^n to $S^{n-1} \times \{0\}$ by the identity, producing a space W obtained from A by attaching the cylinder $S^{n-1} \times I$ along H and then capping off the 0-end by D^n. Collapsing the cylinder direction retracts W onto X_φ, while collapsing it the other way retracts W onto X_ψ. Both collapses are deformation retractions that fix A, hence $X_\varphi \simeq W \simeq X_\psi$ under A. □

5.2 Cellular maps and the cellular approximation theorem

The filtration by skeleta suggests a class of maps that respect dimension.

Definition 5.5. Cellular map

Let X and Y be CW complexes. A map $f \colon X \to Y$ is **cellular** if $f(X^{(n)}) \subset Y^{(n)}$ for all n.

Example. For a CW structure on the torus with one 0-cell v, two 1-cells e_1, e_2, and one 2-cell, a cellular map $T^2 \to S^1$ is determined up to homotopy by what it does on the 1-skeleton, since S^1 has no 2-cells. On π_1, this is the same as choosing images of the two generators of $\pi_1(T^2) \cong \mathbb{Z}^2$ in $\pi_1(S^1) \cong \mathbb{Z}$.

To make the next theorem work smoothly, one uses a basic connectivity property of the skeleton filtration.

Lemma 5.6. Skeleton connectivity Let Y be a CW complex. Then for each $n \geq 0$ the pair $(Y, Y^{(n)})$ is n-connected. Equivalently, for every $k \leq n$ and every map of pairs

$$f\colon (D^k, S^{k-1}) \to (Y, Y^{(n)}),$$

there exists a homotopy rel S^{k-1} from f to a map with image in $Y^{(n)}$.

Proof. Write Y as obtained from $Y^{(n)}$ by attaching cells of dimensions $\geq n+1$. It suffices to show that attaching an m-cell with $m \geq n+1$ does not create new relative homotopy in degrees $\leq n$. So consider a single attachment $Z = B \cup_\varphi D^m$ with B a CW complex and $m \geq n+1$. Given $f\colon (D^k, S^{k-1}) \to (Z, B)$ with $k \leq n$, the image of f meets the open m-cell in a compact set. Choose a point p in the open m-cell outside this compact set. Since the open cell is homeomorphic to $\mathbb{R}^m$, removing p deformation retracts the open cell onto its boundary sphere. Using this deformation inside the attached cell and fixing B, one gets a homotopy rel S^{k-1} pushing the image of f off the point p. Iterating with a finite set of such points if necessary, one can push the entire image of f into B. Applying this one cell at a time along the attachment sequence from $Y^{(n)}$ to Y gives the claim. □

Theorem 5.7. Cellular approximation

Let X and Y be CW complexes and let $f\colon X \to Y$ be a map. Then f is homotopic to a cellular map $g\colon X \to Y$. More generally, if $A \subset X$ is a subcomplex and $f|_A$ is cellular, then g may be chosen so that $g|_A = f|_A$ and the homotopy from f to g is stationary on A.

Proof. The construction is inductive over skeleta of X.
Start with $g_0 = f$ on $X^{(0)} \cup A$. Assume by induction that we have defined a map $g_{n-1}\colon X \to Y$ and a homotopy from f to g_{n-1} that is stationary on A, and such that $g_{n-1}(X^{(n-1)}) \subset Y^{(n-1)}$.
Let e^n_j be an n-cell of X not contained in A, with characteristic map $\Phi_j\colon D^n \to X$. The boundary $\Phi_j(S^{n-1})$ lies in $X^{(n-1)}$, hence $g_{n-1} \circ \Phi_j$ maps S^{n-1} into $Y^{(n-1)} \subset Y^{(n)}$. Consider the map of pairs

$$g_{n-1} \circ \Phi_j\colon (D^n, S^{n-1}) \to (Y, Y^{(n)}).$$

By Lemma 5.6 with $k = n$, there is a homotopy rel S^{n-1} from $g_{n-1} \circ \Phi_j$ to a map $h_j\colon D^n \to Y^{(n)}$. Use h_j to redefine g_{n-1} on the closed cell $\Phi_j(D^n)$, keeping g_{n-1} unchanged on $X^{(n-1)} \cup A$. Doing this simultaneously for all n-cells not in A produces a new map $g_n\colon X \to Y$ with

$$g_n(X^{(n)}) \subset Y^{(n)}, \qquad g_n|_{X^{(n-1)} \cup A} = g_{n-1}|_{X^{(n-1)} \cup A}.$$

The corresponding homotopies on each closed n-cell paste together to a homotopy on $X^{(n)}$ that is stationary on $X^{(n-1)} \cup A$. Extend this homotopy to all of X by keeping points outside $X^{(n)}$ fixed.
Now define $g\colon X \to Y$ by setting $g|_{X^{(n)}} = g_n|_{X^{(n)}}$. This is well defined because the g_n agree on overlaps. The resulting map is cellular since each point of X lies in some skeleton, and on that skeleton the map lands in the corresponding skeleton of Y. The homotopies at each stage fit together to a homotopy from f to g that is stationary on A. □

A useful refinement controls the size of the image during a homotopy between cellular maps.

Corollary 5.8 (Cellular homotopies)**.** Let $f_0, f_1\colon X \to Y$ be cellular maps between CW complexes that are homotopic. Then there exists a homotopy $H\colon X \times I \to Y$ from f_0 to f_1 such that

$$H(X^{(n)} \times I) \subset Y^{(n+1)} \qquad \text{for all } n \geq 0.$$

If the original homotopy is stationary on a subcomplex $A \subset X$, then H may be chosen stationary on A as well.

Proof. Let $\widehat{H}\colon X \times I \to Y$ be any homotopy from f_0 to f_1, stationary on A if required. Give $X \times I$ the CW structure whose cells are products $e^n \times I$ together with the cells of $X \times \partial I$. Then $X \times \partial I \cup A \times I$ is a subcomplex on which $\widehat{H}$ is cellular (its image lies in $Y^{(n)}$ there). Apply Theorem 5.7 to the pair

$$(X \times I,\ X \times \partial I \cup A \times I)$$

to homotope $\widehat{H}$ rel $X \times \partial I \cup A \times I$ to a cellular map. Cellularity means that $(n+1)$-cells of $X \times I$ map

into $Y^{(n+1)}$, and since $X^{(n)} \times I$ is contained in the $(n+1)$-skeleton of $X \times I$, we obtain the desired inclusion $H(X^{(n)} \times I) \subset Y^{(n+1)}$. $\square$

5.3 (Δ)-complexes as a convenient source of cell structures

A (Δ)-complex structure is a way of gluing simplices along faces, allowing different faces and even different vertices to be identified. For low-dimensional spaces, it is often faster to write down a quotient of a polygon and then add a diagonal to see simplices.

Definition 5.9. (Δ)**-complex**

A (Δ)**-complex** structure on a space X consists of a collection of maps

$$\sigma_\alpha \colon \Delta^n \to X$$

(one for each n-simplex of X) such that:

1. σ_α restricts to a homeomorphism from the interior of Δ^n onto its image.
2. The restriction of σ_α to each face of Δ^n is one of the characteristic maps σ_β for an $(n-1)$-simplex of X.
3. The space X is obtained from the disjoint union of simplices by identifying faces according to these restrictions.

Vertices are allowed to map to the same point of X.

Every (Δ)-complex gives a CW structure by declaring the open simplices to be open cells. In this direction the skeleton $X^{(n)}$ is the union of simplices of dimension $\leq n$.

Example. Start with a square $I \times I$ and identify opposite edges by translations to obtain the torus T^2. This produces a CW structure with one 0-cell v, two 1-cells e_1, e_2, and one 2-cell. Adding a diagonal to the square yields a (Δ)-complex structure with one vertex, three edges (the two sides and the diagonal), and two 2-simplices (the two triangles in the subdivided square). Both structures describe the same space; the (Δ)-picture is convenient when one wants explicit simplices, while the CW picture is convenient when one wants a minimal number of cells.

Example. The projective plane $\mathbb{R}P^2$ is obtained by identifying antipodal points on the boundary of a disk. Equivalently, it is a quotient of a square with the usual edge identifications. This exhibits $\mathbb{R}P^2$ as a CW complex with one 0-cell, one 1-cell e^1, and one 2-cell e^2. If one instead starts from a triangle and identifies all three edges by the same orientation-reversing identification, one gets a (Δ)-complex structure with one vertex, one edge, and one 2-simplex. The induced CW structure is the same minimal one: the 1-skeleton is a circle and the 2-cell is attached by the map of degree 2 on that circle.

Example. For $n \geq 1$, the space $\mathbb{R}P^n$ admits a CW structure with one cell in each dimension $0, 1, \ldots, n$. The m-skeleton is $\mathbb{R}P^m$, and the attaching map of the m-cell is the standard double covering

$$S^{m-1} \longrightarrow \mathbb{R}P^{m-1}.$$

In low dimensions this recovers the description of $\mathbb{R}P^2$ above, and it also matches the universal covering map $S^n \to \mathbb{R}P^n$ for $n \geq 2$.

6 Singular Homology and Homotopy Invariance

6.1 Singular chains and the boundary operator

Recall the standard n-simplex

$$\Delta^n = \Big\{(t_0, \ldots, t_n) \in \mathbb{R}^{n+1} \;\Big|\; t_i \geq 0,\ \sum_{i=0}^{n} t_i = 1\Big\}.$$

A *singular n-simplex* in X is a map $\sigma \colon \Delta^n \to X$.

Definition 6.1. Singular chain groups

For $n \geq 0$, let $C_n(X)$ be the free abelian group on the set of singular n-simplices in X. For $n < 0$ set $C_n(X) = 0$.

For each $0 \leq i \leq n$, the inclusion of the ith face is the affine map

$$\phi_i \colon \Delta^{n-1} \longrightarrow \Delta^n, \qquad \phi_i(t_0, \ldots, t_{n-1}) = (t_0, \ldots, t_{i-1}, 0, t_i, \ldots, t_{n-1}).$$

Given a singular n-simplex σ, the ith face of σ is $\sigma \circ \phi_i$.

Definition 6.2. Boundary operator

Define $\partial \colon C_n(X) \to C_{n-1}(X)$ on generators by

$$\partial[\sigma] = \sum_{i=0}^{n} (-1)^i [\sigma \circ \phi_i],$$

and extend $\mathbb{Z}$-linearly.

Lemma 6.3 $\partial \circ \partial = 0$.

Proof. It suffices to check this on a generator $[\sigma]$. The composite $\partial\partial[\sigma]$ is a signed sum of codimension-2 faces of σ. Each such face occurs exactly twice with opposite signs because the two ways of deleting distinct vertices $i < j$ produce the same map $\Delta^{n-2} \to \Delta^n$, and the corresponding signs differ by a minus sign. Hence all terms cancel. □

Thus $(C_*(X), \partial)$ is a chain complex.

Definition 6.4. Singular homology

Let $Z_n(X) = \ker(\partial \colon C_n(X) \to C_{n-1}(X))$ and $B_n(X) = \operatorname{im}(\partial \colon C_{n+1}(X) \to C_n(X))$. The **$n$th singular homology group** is

$$H_n(X) = Z_n(X)/B_n(X).$$

A class in $Z_n(X)$ is an **n-cycle**, and a class in $B_n(X)$ is an **n-boundary**.

Example. Let $X = *$ be a point, and let $\sigma_n \colon \Delta^n \to *$ be the unique singular n-simplex. Then $C_n(*) \cong \mathbb{Z}\langle[\sigma_n]\rangle$ and

$$\partial[\sigma_n] = \Big(\sum_{i=0}^{n} (-1)^i\Big)[\sigma_{n-1}] = \begin{cases} [\sigma_{n-1}] & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$

So the chain complex alternates between $\mathbb{Z} \xrightarrow{\text{id}} \mathbb{Z}$ and $\mathbb{Z} \xrightarrow{0} \mathbb{Z}$, and

$$H_0(*) \cong \mathbb{Z}, \qquad H_n(*) = 0 \text{ for } n \geq 1.$$

6.2 Functoriality

A map $f \colon X \to Y$ sends singular simplices in X to singular simplices in Y by composition.

Definition 6.5. Induced chain map

Define $f_\# \colon C_n(X) \to C_n(Y)$ on generators by $f_\#[\sigma] = [f \circ \sigma]$ and extend $\mathbb{Z}$-linearly.

Lemma 6.6 $f_\# \partial = \partial f_\#$.

Proof. On a generator $[\sigma]$ we compute

$$\partial f_\#[\sigma] = \partial[f \circ \sigma] = \sum_{i=0}^{n} (-1)^i [f \circ \sigma \circ \phi_i] = f_\#\Big(\sum_{i=0}^{n} (-1)^i [\sigma \circ \phi_i]\Big) = f_\# \partial[\sigma].$$

□

Hence $f_\#$ sends cycles to cycles and boundaries to boundaries, so it induces homomorphisms

$$f_* \colon H_n(X) \to H_n(Y).$$

The identities $(\mathrm{id}_X)_* = \mathrm{id}$ and $(g \circ f)_* = g_* \circ f_*$ follow from the definitions.

Example. Let $i\colon A \hookrightarrow X$ be an inclusion. If a cycle α in A is regarded as a cycle in X, the induced class $i_*[\alpha] \in H_n(X)$ is exactly its class after viewing α inside $C_n(X)$. This is the basic mechanism behind comparing the homology of subspaces and ambient spaces.

6.3 Chain homotopies and homotopy invariance

Two chain maps can induce the same map on homology even when they are not equal on the nose.

Definition 6.7. Chain homotopy

Let $f_\#, g_\#\colon C_*(X) \to C_*(Y)$ be chain maps. A **chain homotopy** from $f_\#$ to $g_\#$ is a family of homomorphisms

$$P_n\colon C_n(X) \to C_{n+1}(Y)$$

such that for all $n \geq 0$,

$$\partial P_n + P_{n-1}\partial = g_\# - f_\#,$$

where $P_{-1} = 0$.

Lemma 6.8 If $f_\#$ and $g_\#$ are chain homotopic, then $f_* = g_*\colon H_n(X) \to H_n(Y)$ for all n.

Proof. Let $z \in Z_n(X)$ be a cycle. Then

$$(g_\# - f_\#)(z) = \partial P_n(z) + P_{n-1}\partial(z) = \partial P_n(z),$$

so $g_\#(z) - f_\#(z)$ is a boundary in $C_n(Y)$. Therefore $[g_\#(z)] = [f_\#(z)]$ in $H_n(Y)$, which is exactly $g_*[z] = f_*[z]$. □

The point is that a homotopy of maps produces a chain homotopy of the induced chain maps.

Theorem 6.9. Homotopy invariance

If $f, g\colon X \to Y$ are homotopic, then $f_* = g_*\colon H_n(X) \to H_n(Y)$ for all n.

Proof. Let $H\colon X \times I \to Y$ be a homotopy from f to g. For a singular n-simplex $\sigma\colon \Delta^n \to X$, consider the prism

$$\Delta^n \times I \xrightarrow{\sigma \times \mathrm{id}} X \times I \xrightarrow{H} Y.$$

Decompose $\Delta^n \times I$ into $(n+1)$ simplices in the standard way (each simplex has vertices $0, 1, \ldots, i, i', \ldots, n'$ in the usual prism notation). This yields a singular $(n+1)$-chain $P_n([\sigma]) \in C_{n+1}(Y)$. Extending $\mathbb{Z}$-linearly defines $P_n\colon C_n(X) \to C_{n+1}(Y)$.
A direct comparison of the combinatorial boundary of the prism with the formula for a chain homotopy shows that

$$\partial P_n + P_{n-1}\partial = g_\# - f_\#.$$

Thus $f_\#$ and $g_\#$ are chain homotopic, and Lemma 6.8 gives $f_* = g_*$. □

Example. If X is contractible, then every map $X \to Y$ is homotopic to a constant map. Hence all maps $H_n(X) \to H_n(Y)$ factor through the homology of a point. In particular, $H_n(X) = 0$ for $n \geq 1$.

6.4 Reduced homology and the suspension viewpoint

Reduced homology modifies degree 0 so that a connected space carries no extra $\mathbb{Z}$ in degree 0.

Definition 6.10. Augmentation and reduced chains

Assume $X \neq \varnothing$. Define the **augmentation** $\epsilon\colon C_0(X) \to \mathbb{Z}$ on generators by $\epsilon([x]) = 1$. Set $\widetilde{C}_n(X) = C_n(X)$ for $n \geq 1$, and $\widetilde{C}_0(X) = \ker(\epsilon)$. The boundary ∂ restricts to $\widetilde{C}_*(X)$, giving a subcomplex. Define $\widetilde{H}_n(X) = H_n(\widetilde{C}_*(X))$.

Lemma 6.11 For $n \geq 1$, the natural map $\widetilde{H}_n(X) \to H_n(X)$ is an isomorphism. There is a natural splitting $H_0(X) \cong \widetilde{H}_0(X) \oplus \mathbb{Z}$.

Proof. For $n \geq 1$ we have $\widetilde{C}_n(X) = C_n(X)$, so cycles and boundaries agree in those degrees. In degree 0, the short exact sequence of chain complexes

$$0 \to \widetilde{C}_*(X) \to C_*(X) \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

has trivial differential on $\mathbb{Z}$, and the induced long exact sequence shows that $H_0(X) \twoheadrightarrow \mathbb{Z}$ with kernel $\widetilde{H}_0(X)$. A section is induced by the projection $X \to *$, which sends a chosen point to $1 \in \mathbb{Z}$ on H_0. □

Example. If X is path-connected, then $\widetilde{H}_0(X) = 0$ and $H_0(X) \cong \mathbb{Z}$. For a space with k path components, $\widetilde{H}_0(X) \cong \mathbb{Z}^{k-1}$.

7 Relative Homology, Excision, and Long Exact Sequences

7.1 Relative chains

Let $A \subset X$ be a subspace. Singular chains in A sit inside singular chains in X, and the quotient keeps track of what remains after we declare chains in A to be invisible.

Definition 7.1. Relative chain complex

For $n \geq 0$, define

$$C_n(X, A) = C_n(X)/C_n(A),$$

and for $n < 0$ set $C_n(X, A) = 0$. The boundary operator on $C_n(X)$ preserves the subgroup $C_n(A)$, hence descends to a boundary operator

$$\partial \colon C_n(X, A) \longrightarrow C_{n-1}(X, A).$$

Lemma 7.2 The induced boundary on $C_*(X, A)$ satisfies $\partial^2 = 0$.

Proof. This follows from $\partial^2 = 0$ on $C_*(X)$ and the fact that $C_*(A)$ is a subcomplex. □

Definition 7.3. Relative homology

Define

$$H_n(X, A) = H_n(C_*(X, A)).$$

A class $[\alpha] \in H_n(X, A)$ can be represented by an n-chain $\alpha \in C_n(X)$ with $\partial\alpha \in C_{n-1}(A)$, and two such representatives define the same class if their difference is of the form $\partial\beta + \gamma$ with $\beta \in C_{n+1}(X)$ and $\gamma \in C_n(A)$.

This description is often the right geometric picture: a relative cycle is a chain in X whose boundary lies in A, and we are allowed to modify it by adding boundaries in X and by adding chains already living in A.

Example. If A is a deformation retract of an open neighborhood $U \subset X$, then relative classes in $H_n(X, A)$ can be represented by chains supported away from A. One way to see this is to use excision later, with $B = X \setminus U$.

7.2 Functoriality for pairs

A map of pairs is a map $f \colon X \to Y$ with $f(A) \subset B$.

Definition 7.4. Induced maps on relative chains

If $f \colon (X, A) \to (Y, B)$ is a map of pairs, define $f_\# \colon C_n(X) \to C_n(Y)$ by $f_\#[\sigma] = [f \circ \sigma]$. Since $f_\#(C_n(A)) \subset C_n(B)$, this induces a quotient map

$$f_\# \colon C_n(X, A) \to C_n(Y, B),$$

and hence a homomorphism $f_*\colon H_n(X,A) \to H_n(Y,B)$.

Lemma 7.5 For a map of pairs $f\colon (X,A) \to (Y,B)$ we have $f_\# \partial = \partial f_\#$ on $C_*(X,A)$.

Proof. The identity holds on $C_*(X)$, and passing to quotients preserves it. □

Example. If $i\colon (A,\varnothing) \hookrightarrow (X,\varnothing)$ is the inclusion, then $i_*\colon H_n(A) \to H_n(X)$ is the usual map on absolute homology. If $j\colon (X,\varnothing) \to (X,A)$ is the identity on X, the induced map $j_*\colon H_n(X) \to H_n(X,A)$ is the canonical projection at the level of cycles.

7.3 The long exact sequence of a pair

The short exact sequence of chain complexes

$$0 \longrightarrow C_*(A) \xrightarrow{i_\#} C_*(X) \xrightarrow{q_\#} C_*(X,A) \longrightarrow 0$$

produces a long exact sequence on homology, with a connecting map that is easy to write down.

Theorem 7.6. Long exact sequence in homology for a pair

For every pair (X,A) there is a natural long exact sequence

$$\cdots \to H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X,A) \xrightarrow{\partial} H_{n-1}(A) \to \cdots \to H_0(X,A) \to 0.$$

The connecting homomorphism $\partial\colon H_n(X,A) \to H_{n-1}(A)$ is given as follows: if $[\alpha] \in H_n(X,A)$ is represented by a chain $\alpha \in C_n(X)$ with $\partial\alpha \in C_{n-1}(A)$, then $\partial[\alpha] = [\partial\alpha] \in H_{n-1}(A)$.

Proof. We define the connecting homomorphism and then verify exactness.

Definition of ∂. Let $[\alpha] \in H_n(X,A)$ be represented by a cycle $\overline{\alpha} \in C_n(X,A)$. Choose a lift $\alpha \in C_n(X)$ of $\overline{\alpha}$ under $q_\#$. Since $\overline{\alpha}$ is a cycle, $q_\#(\partial\alpha) = \partial(q_\#\alpha) = \partial\overline{\alpha} = 0$, hence $\partial\alpha \in \ker(q_\#) = \operatorname{im}(i_\#)$. Thus there exists $a \in C_{n-1}(A)$ with $i_\#(a) = \partial\alpha$. Since $i_\#$ is injective, a is unique. We claim a is a cycle: $\partial a = 0$ because

$$i_\#(\partial a) = \partial(i_\# a) = \partial(\partial\alpha) = 0,$$

and $i_\#$ is injective. Define $\partial[\overline{\alpha}] = [a] \in H_{n-1}(A)$.

Well-definedness. If $\overline{\alpha}$ is changed by a boundary $\partial\overline{\beta}$ in $C_*(X,A)$, lift $\overline{\beta}$ to $\beta \in C_{n+1}(X)$. Then a lift of $\overline{\alpha} + \partial\overline{\beta}$ is $\alpha + \partial\beta$, and

$$\partial(\alpha + \partial\beta) = \partial\alpha,$$

so $[a]$ does not change. If the lift α is changed by adding $i_\#(c)$ with $c \in C_n(A)$, then the new boundary is

$$\partial(\alpha + i_\# c) = \partial\alpha + i_\#(\partial c),$$

so the new a differs from the old one by ∂c, hence defines the same class in $H_{n-1}(A)$. Thus ∂ is well defined.

Exactness. Exactness at $H_n(X)$ and $H_n(X,A)$ is checked as usual.

1. $\operatorname{im}(i_*) \subset \ker(j_*)$ since $j_\# i_\# = 0$ on chains. Conversely, if $[x] \in H_n(X)$ maps to 0 in $H_n(X,A)$, choose a cycle $x \in C_n(X)$ representing it. The condition $j_*[x] = 0$ means $q_\#(x) = \partial\overline{\beta}$ for some $\overline{\beta} \in C_{n+1}(X,A)$. Lifting $\overline{\beta}$ to $\beta \in C_{n+1}(X)$ gives $q_\#(x - \partial\beta) = 0$, hence $x - \partial\beta = i_\#(a)$ for some $a \in C_n(A)$. Thus $[x] = i_*[a]$, so $\ker(j_*) \subset \operatorname{im}(i_*)$.
2. $\operatorname{im}(j_*) \subset \ker(\partial)$ because if $\overline{x} = q_\#(x)$ with x a cycle, then $\partial[\overline{x}] = [\partial x] = 0$. Conversely, suppose $[\overline{\alpha}] \in H_n(X,A)$ satisfies $\partial[\overline{\alpha}] = 0$. Choose a lift $\alpha \in C_n(X)$ as above, so $\partial\alpha = i_\#(a)$ for some cycle $a \in C_{n-1}(A)$. The condition $\partial[\overline{\alpha}] = [a] = 0$ means $a = \partial c$ for some $c \in C_n(A)$. Then

$$\partial(\alpha - i_\# c) = \partial\alpha - i_\#(\partial c) = i_\#(a - \partial c) = 0,$$

so $\alpha - i_\# c$ is a cycle in $C_n(X)$. Its image under $q_\#$ equals $\overline{\alpha}$ since $q_\# i_\# = 0$. Hence $[\overline{\alpha}]$ lies in the image of j_*.

Exactness at $H_{n-1}(A)$ is similar and follows from $i_*\partial = 0$ and the definition of ∂. □

Proposition 7.7

If $x_0 \in X$, then $H_n(X,x_0) \cong \widetilde{H}_n(X)$ for all n.

Proof. Apply Theorem 7.6 to the pair (X, x_0). Since $\widetilde{H}_n(x_0) = 0$ for all n, the reduced and relative groups match degree by degree. $\square$

Example. Let $A \subset X$ be a deformation retract of X. Then $H_n(X, A) = 0$ for all n, and the long exact sequence forces $i_*\colon H_n(A) \to H_n(X)$ to be an isomorphism for all n. This matches the homotopy invariance from the previous lecture.

7.4 A geometric reading of the connecting map

The formula $\partial[\alpha] = [\partial\alpha]$ is already geometric, but one can sharpen the picture in common situations.

Example. Consider the pair (D^n, S^{n-1}). A relative n-cycle is an n-chain in D^n whose boundary lies in S^{n-1}. The connecting homomorphism

$$\partial\colon H_n(D^n, S^{n-1}) \to H_{n-1}(S^{n-1})$$

sends the class of such a chain to the class of its boundary cycle on the sphere. When $n \geq 2$, this is exactly how the boundary of an oriented n-dimensional region produces an oriented $(n-1)$-cycle.

For later use it helps to record the same mechanism for triples.

Definition 7.8. Triple and the connecting map

Given $B \subset A \subset X$, the inclusion of chain complexes $C_*(A, B) \hookrightarrow C_*(X, B)$ yields a long exact sequence

$$\cdots \to H_n(A, B) \to H_n(X, B) \to H_n(X, A) \xrightarrow{\partial} H_{n-1}(A, B) \to \cdots .$$

Example. Let X be a manifold with boundary and let $A = \partial X$. A relative class in $H_n(X, \partial X)$ can often be represented by a chain filling the interior. The connecting map sends it to its boundary cycle in $H_{n-1}(\partial X)$. This is one of the reasons $H_n(X, \partial X)$ behaves like "top-dimensional classes with boundary allowed".

7.5 Excision

Excision says that relative homology only depends on what happens near the part of the space where the boundary of a relative chain is allowed to live.

Theorem 7.9. Excision

1. Let $B \subset A \subset X$ and assume $\overline{B} \subset \operatorname{int}(A)$. Then the inclusion of pairs
$$(X \setminus B,\ A \setminus B) \hookrightarrow (X, A)$$
induces isomorphisms
$$H_n(X \setminus B,\ A \setminus B) \xrightarrow{\cong} H_n(X, A) \qquad \text{for all } n.$$
2. Equivalently, if $X = Y_1 \cup Y_2$ and Y_1, Y_2 cover X by interiors, then the inclusion
$$(Y_2,\ Y_1 \cap Y_2) \hookrightarrow (X,\ Y_1)$$
induces isomorphisms $H_n(Y_2, Y_1 \cap Y_2) \cong H_n(X, Y_1)$ for all n.

Proof. We prove the second formulation. Let $X = Y_1 \cup Y_2$ with Y_1, Y_2 covering X by interiors.

Step 1: chains supported in Y_1 and Y_2. Let $C_n(Y_1 + Y_2) \subset C_n(X)$ be the subgroup consisting of chains that are sums of chains in Y_1 and chains in Y_2. This is a subcomplex since ∂ preserves it. A standard subdivision argument produces a chain map

$$\rho\colon C_*(X) \to C_*(Y_1 + Y_2)$$

and a chain homotopy D with

$$\partial D + D\partial = \mathrm{id} - \iota\rho,$$

where ι is the inclusion $C_*(Y_1 + Y_2) \hookrightarrow C_*(X)$. Hence ι induces an isomorphism on homology.

Step 2: pass to quotients. All maps in Step 1 take chains in Y_1 to chains in Y_1, so they descend to the quotient complexes by $C_*(Y_1)$. Thus the inclusion

$$C_*(Y_1 + Y_2)/C_*(Y_1) \hookrightarrow C_*(X)/C_*(Y_1)$$

induces an isomorphism on homology, i.e.

$$H_n(C_*(Y_1+Y_2)/C_*(Y_1)) \cong H_n(X, Y_1).$$

Step 3: identify the quotient with $(Y_2, Y_1 \cap Y_2)$. The natural map $C_*(Y_2) \to C_*(Y_1+Y_2)/C_*(Y_1)$ has kernel $C_*(Y_1 \cap Y_2)$ and is surjective, so it induces an isomorphism

$$C_*(Y_2)/C_*(Y_1 \cap Y_2) \;\cong\; C_*(Y_1+Y_2)/C_*(Y_1).$$

Therefore $H_n(Y_2, Y_1 \cap Y_2) \cong H_n(X, Y_1)$ for all n. □

Example. Let $X = S^1$ and let A be a small arc, homeomorphic to an interval. Since A deformation retracts to a point, the long exact sequence for (S^1, A) and excision can be used to identify

$$H_1(S^1, A) \cong \mathbb{Z},$$

with a generator represented by a 1-chain traversing the circle once, whose boundary lies in A.

7.6 The pair (D^n, S^{n-1}) and the homology of spheres

We now compute $H_*(D^n, S^{n-1})$ and recover $H_*(S^n)$ in a way that makes the connecting map concrete.

Proposition 7.10

For $n \geq 1$,

$$H_i(D^n, S^{n-1}) \cong \begin{cases} \mathbb{Z} & i = n, \\ 0 & i \neq n. \end{cases}$$

Proof. Since D^n is contractible, $H_i(D^n) = 0$ for $i \geq 1$ and $H_0(D^n) \cong \mathbb{Z}$. Apply the long exact sequence of the pair (D^n, S^{n-1}). For $i \geq 2$ the relevant segment is

$$H_i(D^n) \to H_i(D^n, S^{n-1}) \xrightarrow{\partial} H_{i-1}(S^{n-1}) \to H_{i-1}(D^n),$$

so for $2 \leq i \leq n$ we get

$$H_i(D^n, S^{n-1}) \xrightarrow{\cong} H_{i-1}(S^{n-1})$$

since the outer terms vanish. In particular, if we already know $H_{i-1}(S^{n-1}) = 0$ for $i-1 \notin \{0, n-1\}$, this forces $H_i(D^n, S^{n-1}) = 0$ for $i \neq n$. To pin down the top group, use the quotient description from excision: the quotient D^n/S^{n-1} is S^n, and for a CW pair (or more generally a good pair) one has an isomorphism

$$H_i(D^n, S^{n-1}) \cong \widetilde{H}_i(D^n/S^{n-1}) \cong \widetilde{H}_i(S^n).$$

Thus $H_i(D^n, S^{n-1})$ vanishes for $i \neq n$ and is $\mathbb{Z}$ for $i = n$.
Concretely, a generator in degree n can be represented by any singular n-simplex filling the disk whose boundary lies in S^{n-1}. □

Proposition 7.11

For $n \geq 1$,

$$H_i(S^n) \cong \begin{cases} \mathbb{Z} & i = 0 \text{ or } i = n, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The space S^n is path-connected, so $H_0(S^n) \cong \mathbb{Z}$. For reduced groups, $\widetilde{H}_0(S^n) = 0$.
Using $D^{n+1}/S^n \cong S^{n+1}$ and the identification

$$H_i(D^{n+1}, S^n) \cong \widetilde{H}_i(S^{n+1}),$$

the connecting homomorphism in the long exact sequence of (D^{n+1}, S^n) gives isomorphisms

$$\partial\colon H_i(D^{n+1}, S^n) \xrightarrow{\cong} \widetilde{H}_{i-1}(S^n) \qquad (i > 0),$$

because the remaining reduced groups of D^{n+1} vanish. With the base case $\widetilde{H}_0(S^0) \cong \mathbb{Z}$, induction yields

$$\widetilde{H}_k(S^n) \cong \begin{cases} \mathbb{Z} & k = n, \\ 0 & k \neq n. \end{cases}$$

Passing from reduced to unreduced finishes the computation of $H_*(S^n)$. □

Example. In the pair (D^n, S^{n-1}), the connecting map

$$\partial\colon H_n(D^n, S^{n-1}) \to H_{n-1}(S^{n-1})$$

is an isomorphism onto the top homology of the boundary sphere. If $[D^n] \in H_n(D^n, S^{n-1})$ denotes a chosen generator, then $\partial[D^n]$ is the corresponding generator of $H_{n-1}(S^{n-1})$. This is the homological shadow of "boundary of a fundamental class is the fundamental class of the boundary".

8 Mayer–Vietoris and First Computations

8.1 Excisive decompositions

Let X be a space and write $X = A \cup B$. The Mayer–Vietoris sequence is available when A and B cover X "from the inside", so that singular chains in X can be replaced, up to homology, by chains that split into an A-part and a B-part.

For such a decomposition, define $C_n(A+B) \subset C_n(X)$ to be the subgroup consisting of chains of the form $x+y$ with $x \in C_n(A)$ and $y \in C_n(B)$. The boundary operator preserves this subgroup, so $C_*(A+B)$ is a subcomplex.

Lemma 8.1 For each n there is a short exact sequence

$$0 \to C_n(A \cap B) \xrightarrow{\varphi} C_n(A) \oplus C_n(B) \xrightarrow{\psi} C_n(A+B) \to 0,$$

where $\varphi(z) = (z, -z)$ and $\psi(x, y) = x + y$.

Proof. Injectivity of φ is clear. We have $\psi \circ \varphi = 0$, so $\operatorname{im}\varphi \subseteq \ker\psi$. If $(x, y) \in \ker\psi$ then $x + y = 0$, hence $x = -y$ is a chain in both A and B, so $x \in C_n(A \cap B)$ and $(x, y) = (x, -x) \in \operatorname{im}\varphi$. Surjectivity of ψ is immediate from the definition of $C_n(A+B)$. □

Under the usual hypotheses (for instance, A and B are open and X is the union of their interiors), the inclusion $C_*(A+B) \hookrightarrow C_*(X)$ induces an isomorphism on homology. Taking the long exact sequence in homology of Lemma 8.1 and then identifying $H_*(C_*(A+B))$ with $H_*(X)$ gives Mayer–Vietoris.

8.2 The Mayer–Vietoris long exact sequence

Theorem 8.2. Mayer–Vietoris

Assume $X = A \cup B$ and the inclusion $C_*(A+B) \hookrightarrow C_*(X)$ induces an isomorphism on homology. Then there is a natural long exact sequence

$$\cdots \to H_n(A \cap B) \xrightarrow{\Phi} H_n(A) \oplus H_n(B) \xrightarrow{\Psi} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \to \cdots \to H_0(X) \to 0,$$

where $\Phi(u) = (i_{A*}u, -i_{B*}u)$ and $\Psi(v, w) = j_{A*}v + j_{B*}w$.

Proof. Apply homology to the short exact sequence of chain complexes from Lemma 8.1 to obtain a long exact sequence with third term $H_n(C_*(A+B))$. Compose with the isomorphism $H_n(C_*(A+B)) \cong H_n(X)$ induced by $C_*(A+B) \hookrightarrow C_*(X)$. The displayed formulas for Φ and Ψ are induced by φ and ψ. □

The connecting homomorphism ∂ has a concrete description that is often the quickest way to keep signs straight. It is exactly the connecting map associated to the short exact sequence in Lemma 8.1.

Proposition 8.3. A chain-level description of the boundary map

Let $\alpha \in H_n(X)$ and choose a cycle $z \in C_n(A+B)$ representing α. Write $z = x + y$ with $x \in C_n(A)$ and $y \in C_n(B)$. Then $\partial x = -\partial y$ is a cycle in $C_{n-1}(A \cap B)$, and the connecting class $\partial(\alpha) \in H_{n-1}(A \cap B)$ is represented by this cycle.

Proof. Since $\partial z = 0$ and $z = x + y$, we have $\partial x = -\partial y$. Because ∂x lies in both A and B, it is a chain in $A \cap B$. The definition of the connecting map in the long exact sequence of a short exact sequence of chain complexes gives exactly this representative. □

There is also a reduced Mayer–Vietoris sequence obtained by working with augmented chain complexes. In practice, this is the version that makes inductive computations on spheres particularly clean.

8.3 Spheres from hemispheres

Let S^n be covered by the northern and southern hemispheres A and B. Each hemisphere deformation retracts to a point, while $A \cap B$ deformation retracts onto the equatorial S^{n-1}. The reduced Mayer–Vietoris sequence therefore collapses to isomorphisms shifting degree by one.

Proposition 8.4

For $n \geq 1$ there are natural isomorphisms

$$\widetilde{H}_i(S^n) \cong \widetilde{H}_{i-1}(S^{n-1}) \qquad \text{for all } i.$$

Proof. In reduced homology, $\widetilde{H}_i(A) = \widetilde{H}_i(B) = 0$ for all i since A and B are contractible. Hence the reduced Mayer–Vietoris sequence gives isomorphisms

$$\widetilde{H}_i(S^n) \xrightarrow{\cong} \widetilde{H}_{i-1}(A \cap B) \cong \widetilde{H}_{i-1}(S^{n-1}),$$

as claimed. □

Starting from $\widetilde{H}_0(S^0) \cong \mathbb{Z}$, this recovers $\widetilde{H}_n(S^n) \cong \mathbb{Z}$ and vanishing in other degrees. This matches the computation from the pair (D^{n+1}, S^n) and is a useful alternative viewpoint when a space decomposes into pieces that look like disks.

8.4 The torus via Mayer–Vietoris

Let $T^2 = S^1 \times S^1$. Choose two open arcs $U, V \subset S^1$ such that $U \cup V = S^1$ and $U \cap V$ is the disjoint union of two open arcs. Set

$$A = S^1 \times U, \qquad B = S^1 \times V.$$

Then A and B deformation retract onto S^1 (for instance onto $S^1 \times \{u_0\}$ and $S^1 \times \{v_0\}$), while

$$A \cap B = S^1 \times (U \cap V)$$

deformation retracts onto a disjoint union of two circles. Thus

$$H_0(A) \cong H_0(B) \cong \mathbb{Z}, \quad H_1(A) \cong H_1(B) \cong \mathbb{Z}, \quad H_i(A) = H_i(B) = 0 \text{ for } i \geq 2,$$

and

$$H_0(A \cap B) \cong \mathbb{Z} \oplus \mathbb{Z}, \quad H_1(A \cap B) \cong \mathbb{Z} \oplus \mathbb{Z}, \quad H_i(A \cap B) = 0 \text{ for } i \geq 2.$$

Proposition 8.5

$H_1(T^2) \cong \mathbb{Z}^2$.

Proof. From Theorem 8.2 we extract the segment

$$H_1(A \cap B) \xrightarrow{\Phi} H_1(A) \oplus H_1(B) \xrightarrow{\Psi} H_1(T^2) \xrightarrow{\partial} H_0(A \cap B) \xrightarrow{\Phi} H_0(A) \oplus H_0(B).$$

The last map Φ is $(i_{A*}, -i_{B*})$ on H_0. Since A and B are path-connected, $H_0(A) \oplus H_0(B) \cong \mathbb{Z} \oplus \mathbb{Z}$. The space $A \cap B$ has two components, so $H_0(A \cap B) \cong \mathbb{Z} \oplus \mathbb{Z}$. Under inclusions, each component maps to the unique component of A and of B, so Φ on H_0 is

$$\Phi(a, b) = (a + b, -(a + b)).$$

Hence $\ker(\Phi) \cong \mathbb{Z}$, generated by $(1, -1)$, and therefore $\operatorname{im}(\partial) = \ker(\Phi) \cong \mathbb{Z}$.
Next, describe Φ on H_1. Each circle component of $A \cap B$ includes into A and B as a generator of H_1 (up to sign). With respect to the decompositions $H_1(A \cap B) \cong \mathbb{Z} \oplus \mathbb{Z}$ and $H_1(A) \oplus H_1(B) \cong \mathbb{Z} \oplus \mathbb{Z}$, the map Φ sends the two generators to $(1, -1)$ and $(1, -1)$. Thus $\operatorname{im}(\Phi)$ is the diagonal copy of $\mathbb{Z}$ inside $\mathbb{Z} \oplus \mathbb{Z}$ generated by $(1, -1)$. Exactness gives

$$0 \to (\mathbb{Z} \oplus \mathbb{Z})/\mathbb{Z} \xrightarrow{\overline{\Psi}} H_1(T^2) \xrightarrow{\partial} \mathbb{Z} \to 0,$$

so $H_1(T^2)$ fits into a short exact sequence

$$0 \to \mathbb{Z} \to H_1(T^2) \to \mathbb{Z} \to 0.$$

Since all groups are free abelian, this sequence splits, hence $H_1(T^2) \cong \mathbb{Z} \oplus \mathbb{Z}$. □

Proposition 8.6

$H_2(T^2) \cong \mathbb{Z}$ and $H_i(T^2) = 0$ for $i \geq 3$.

Proof. For $i \geq 2$, $H_i(A) = H_i(B) = H_i(A \cap B) = 0$. The Mayer–Vietoris segment

$$0 = H_2(A) \oplus H_2(B) \to H_2(T^2) \xrightarrow{\partial} H_1(A \cap B) \xrightarrow{\Phi} H_1(A) \oplus H_1(B)$$

shows $H_2(T^2) \cong \ker(\Phi)$. From the computation in the proof of Proposition 8.5, the map Φ on H_1 sends (u, v) to $(u+v, -(u+v))$, so $\ker(\Phi) \cong \mathbb{Z}$ generated by $(1, -1)$. Thus $H_2(T^2) \cong \mathbb{Z}$. For $i \geq 3$ the long exact sequence gives $H_i(T^2) = 0$ since all neighboring terms vanish. □

Together with $H_0(T^2) \cong \mathbb{Z}$, this recovers the familiar homology of the torus using only Mayer–Vietoris and a carefully chosen open cover.

8.5 Connected sums

Let M and N be connected n-manifolds with $n \geq 2$. Choose embedded closed n-disks $D_M \subset M$ and $D_N \subset N$. Remove the interiors and glue the resulting boundary spheres by a homeomorphism to form the connected sum

$$M \# N = (M \setminus \mathring{D}_M) \cup_{S^{n-1}} (N \setminus \mathring{D}_N).$$

Set $A = M \setminus \mathring{D}_M$ and $B = N \setminus \mathring{D}_N$ inside $M \# N$. Then $A \cap B$ is a neighborhood of the glued sphere and deformation retracts onto S^{n-1}.

Proposition 8.7

Let $n \geq 2$. For $0 < k < n-1$ there are natural isomorphisms

$$H_k(M \# N) \cong H_k(M) \oplus H_k(N).$$

In degree $n-1$ there is a natural exact sequence

$$H_{n-1}(S^{n-1}) \xrightarrow{\Phi} H_{n-1}(A) \oplus H_{n-1}(B) \to H_{n-1}(M \# N) \to 0,$$

and for $k \geq n$ the groups are determined by those of A and B through Mayer–Vietoris.

Proof. Apply Mayer–Vietoris to $M \# N = A \cup B$. For $0 < k < n-1$, we have $H_k(S^{n-1}) = 0$, hence the exact segment

$$0 \to H_k(A) \oplus H_k(B) \xrightarrow{\Psi} H_k(M \# N) \to H_{k-1}(S^{n-1}) = 0$$

shows Ψ is an isomorphism.

It remains to identify $H_k(A)$ with $H_k(M)$ in this range, and similarly for B and N. Since $M = A \cup D_M$ with $A \cap D_M \simeq S^{n-1}$ and D_M contractible, another Mayer–Vietoris argument shows $H_k(A) \cong H_k(M)$ for $k < n-1$, because $H_k(S^{n-1}) = 0$ in that range. The same holds for B and N. Combining these identifications gives the direct sum statement.

For $k = n-1$, the relevant segment of the Mayer–Vietoris sequence is

$$H_{n-1}(S^{n-1}) \xrightarrow{\Phi} H_{n-1}(A) \oplus H_{n-1}(B) \to H_{n-1}(M \# N) \to H_{n-2}(S^{n-1}) = 0,$$

which yields the displayed exact sequence. □

For surfaces, this takes a familiar form. If M_g is the closed orientable surface of genus g, then $M_g = M_{g-1} \# T^2$. Since $H_1(T^2) \cong \mathbb{Z}^2$ and $H_1(S^1) \cong \mathbb{Z}$, the connected sum computation shows that $H_1(M_g) \cong \mathbb{Z}^{2g}$, consistent with the presentation of $\pi_1(M_g)$ from van Kampen and abelianization.

9 Cellular homology

9.1 The cellular chain complex

Let X be a CW complex with skeleta

$$X^{(0)} \subset X^{(1)} \subset \cdots \subset X^{(n)} \subset \cdots \subset X.$$

Fix an abelian coefficient group G.

Definition 9.1. Cellular chain groups

For $n \geq 0$, define the **cellular n-chains** (with coefficients in G) by

$$C_n^{\text{cell}}(X;G) := H_n(X^{(n)}, X^{(n-1)}; G).$$

The point is that the pair $(X^{(n)}, X^{(n-1)})$ is built from $X^{(n-1)}$ by attaching n-cells, so the relative homology in degree n is forced to be free on the n-cells, and it vanishes in other degrees.

Lemma 9.2. Relative homology of successive skeleta Let X be a CW complex. Then for each $n \geq 0$,

$$H_k(X^{(n)}, X^{(n-1)}; G) \cong \begin{cases} \bigoplus_{\alpha \in I_n} G & k = n, \\ 0 & k \neq n, \end{cases}$$

where I_n indexes the n-cells of X.

Proof. Write $X^{(n)}$ as $X^{(n-1)}$ with n-disks attached along $\partial D^n = S^{n-1}$. By excision, the pair $(X^{(n)}, X^{(n-1)})$ splits into a disjoint union of pairs (D^n, S^{n-1}) over the n-cells. Thus

$$H_k(X^{(n)}, X^{(n-1)}; G) \cong \bigoplus_{\alpha \in I_n} H_k(D^n, S^{n-1}; G).$$

But $H_k(D^n, S^{n-1}; G) = 0$ unless $k = n$, and $H_n(D^n, S^{n-1}; G) \cong G$ with generator given by the fundamental class of the disk relative to its boundary. This gives the claim. □

In particular, for $G = \mathbb{Z}$ we may think of $C_n^{\text{cell}}(X) := C_n^{\text{cell}}(X;\mathbb{Z})$ as a free abelian group with one preferred basis element for each oriented n-cell.

Definition 9.3. Cellular boundary maps

For $n \geq 1$, define

$$d_n \colon C_n^{\text{cell}}(X;G) \longrightarrow C_{n-1}^{\text{cell}}(X;G)$$

as the composite

$$H_n(X^{(n)}, X^{(n-1)}; G) \xrightarrow{\partial} H_{n-1}(X^{(n-1)}; G) \xrightarrow{\text{quot}} H_{n-1}(X^{(n-1)}, X^{(n-2)}; G),$$

where ∂ is the connecting map for the pair $(X^{(n)}, X^{(n-1)})$ and quot is induced by the quotient map $X^{(n-1)} \to (X^{(n-1)}, X^{(n-2)})$. For $n = 0$ we set $d_0 = 0$.

Proposition 9.4

The maps d_n satisfy $d_{n-1} \circ d_n = 0$ for all $n \geq 1$. Hence $(C_*^{\text{cell}}(X;G), d_*)$ is a chain complex.

Proof. Consider the triple $(X^{(n)}, X^{(n-1)}, X^{(n-2)})$. The long exact sequence of the triple contains

$$H_n(X^{(n)}, X^{(n-1)}; G) \xrightarrow{d_n} H_{n-1}(X^{(n-1)}, X^{(n-2)}; G) \xrightarrow{d_{n-1}} H_{n-2}(X^{(n-2)}, X^{(n-3)}; G),$$

and exactness forces the composite to be zero. □

Definition 9.5. Cellular homology

The **cellular homology** groups are

$$H_n^{\text{cell}}(X;G) := H_n(C_*^{\text{cell}}(X;G)).$$

9.2 Incidence numbers and degrees

To compute d_n concretely, one expresses it in terms of the attaching maps. Let $\{e_\alpha^n\}_{\alpha \in I_n}$ be the n-cells, with characteristic maps

$$\phi_\alpha^n \colon (D^n, S^{n-1}) \longrightarrow (X^{(n)}, X^{(n-1)}).$$

Write $\psi_\alpha^n := \phi_\alpha^n|_{S^{n-1}} \colon S^{n-1} \to X^{(n-1)}$ for the attaching map.

For each $(n-1)$-cell e_β^{n-1}, choose a characteristic map ϕ_β^{n-1} and form the quotient

$$q_\beta \colon X^{(n-1)} \longrightarrow X^{(n-1)}/(X^{(n-1)} \setminus \mathring{e}_\beta^{n-1}) \;\cong\; S^{n-1}.$$

After choosing orientations of the cells, each $q_\beta \circ \psi_\alpha^n$ has a well-defined degree.

Definition 9.6. Incidence numbers

Assume $G = \mathbb{Z}$ and each cell is oriented. Define the **incidence number**

$$d(\beta, \alpha) \in \mathbb{Z}$$

to be the degree of the composite

$$S^{n-1} \xrightarrow{\psi_\alpha^n} X^{(n-1)} \xrightarrow{q_\beta} S^{n-1}.$$

Proposition 9.7

Under the identification $C_n^{\mathrm{cell}}(X) \cong \bigoplus_{\alpha \in I_n} \mathbb{Z}\langle e_\alpha^n \rangle$, the cellular boundary map is

$$d_n(e_\alpha^n) = \sum_{\beta \in I_{n-1}} d(\beta, \alpha)\, e_\beta^{n-1}.$$

Proof. By construction, d_n is built from connecting homomorphisms for skeleta and the functoriality of relative homology. When one pins down generators coming from oriented disks, the resulting map records exactly the degrees of the maps between the $(n-1)$-sphere quotients described above. This produces the stated matrix formula. □

Two quick remarks help in practice.

Remark. If X has no $(n-1)$-cells, then $C_{n-1}^{\mathrm{cell}}(X) = 0$, so $d_n = 0$ automatically. More generally, if $C_{n-1}^{\mathrm{cell}}(X; G) = 0$ for degree reasons, then every n-cell is a cycle.

Remark. If $f \colon X \to Y$ is cellular, meaning $f(X^{(n)}) \subset Y^{(n)}$ for all n, then functoriality of relative homology yields chain maps

$$f_\# \colon C_*^{\mathrm{cell}}(X; G) \to C_*^{\mathrm{cell}}(Y; G)$$

compatible with cellular boundaries.

9.3 Comparison with singular homology

The cellular complex is not a new invariant. It is a computational model for the singular homology of CW complexes.

Theorem 9.8. Cellular homology agrees with singular homology

Let X be a CW complex and G an abelian group. There are natural isomorphisms

$$H_n^{\mathrm{cell}}(X; G) \;\cong\; H_n(X; G) \qquad \text{for all } n \geq 0.$$

Proof. Fix n and look at the long exact sequences of pairs $(X^{(k)}, X^{(k-1)})$ as k varies. By Lemma 9.2, the groups $H_i(X^{(k)}, X^{(k-1)}; G)$ vanish unless $i = k$, so the long exact sequences splice together into a single chain complex in which the only potentially nonzero maps are the d_k of Definition 9.3. One then compares the homology of this chain complex with $H_*(X; G)$ by induction on skeleta, using exactness and the Five Lemma. □

In particular, all computations done with $C_*^{\mathrm{cell}}(X; G)$ are computations of singular homology.

9.4 Projective spaces

Real projective space

Real projective space $\mathbb{R}P^n$ has a CW structure with exactly one q-cell for each $0 \leq q \leq n$, and $\mathbb{R}P^q$ is the q-skeleton. Let e^q denote the unique q-cell. Then

$$C_q^{\mathrm{cell}}(\mathbb{R}P^n) \cong \mathbb{Z}\langle e^q \rangle \qquad (0 \leq q \leq n),$$

and the cellular boundary maps alternate between 0 and multiplication by 2:

$$d_q = \begin{cases} 0 & q \text{ odd}, \\ \times 2 & q \text{ even}. \end{cases}$$

Concretely, the cellular chain complex is

$$0 \to \mathbb{Z} \xrightarrow{d_n} \mathbb{Z} \xrightarrow{d_{n-1}} \cdots \xrightarrow{d_2} \mathbb{Z} \xrightarrow{d_1} \mathbb{Z} \to 0,$$

with $d_1 = 0$, $d_2 = \times 2$, $d_3 = 0$, $d_4 = \times 2$, and so on.

From this, the homology groups follow immediately.

Theorem 9.9. Homology of $\mathbb{R}P^n$

For $n \geq 1$,

$$H_k(\mathbb{R}P^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z}/2 & 1 \leq k < n \text{ and } k \text{ odd}, \\ \mathbb{Z} & k = n \text{ and } n \text{ odd}, \\ 0 & \text{otherwise}. \end{cases}$$

Proof. If k is odd with $1 \leq k < n$, then $d_k = 0$ and $d_{k+1} = \times 2$, hence

$$H_k \cong \ker(d_k)/\operatorname{im}(d_{k+1}) \cong \mathbb{Z}/2.$$

If k is even with $0 < k < n$, then $d_k = \times 2$ is injective and $d_{k+1} = 0$, hence $H_k = 0$. In top degree, if n is odd then $d_n = 0$ so $H_n \cong \mathbb{Z}$, while if n is even then $d_n = \times 2$ so $H_n = 0$. □

It is worth checking a couple of low-dimensional cases against geometric intuition.

Example. $\mathbb{R}P^2$ has $H_0 \cong \mathbb{Z}$, $H_1 \cong \mathbb{Z}/2$, and $H_2 = 0$.

Example. $\mathbb{R}P^3$ has $H_0 \cong \mathbb{Z}$, $H_1 \cong \mathbb{Z}/2$, $H_2 = 0$, and $H_3 \cong \mathbb{Z}$.

Complex projective space

Complex projective space has a particularly clean cell structure. One convenient way to remember it is that $\mathbb{C}P^\infty$ has one cell in each even dimension, and $\mathbb{C}P^n$ is its $2n$-skeleton. Write e^{2k} for the unique $2k$-cell of $\mathbb{C}P^n$.

Thus

$$C_{2k}^{\text{cell}}(\mathbb{C}P^n) \cong \mathbb{Z}\langle e^{2k} \rangle \quad (0 \leq k \leq n), \qquad C_{2k+1}^{\text{cell}}(\mathbb{C}P^n) = 0,$$

and every cellular boundary map is zero for degree reasons, since

$$d_{2k} \colon C_{2k}^{\text{cell}}(\mathbb{C}P^n) \to C_{2k-1}^{\text{cell}}(\mathbb{C}P^n) = 0.$$

Therefore the cellular chains are already cycles, and there are no nontrivial boundaries.

Theorem 9.10. Homology of $\mathbb{C}P^n$

For $n \geq 1$,

$$H_k(\mathbb{C}P^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k = 0, 2, 4, \ldots, 2n, \\ 0 & \text{otherwise}. \end{cases}$$

Proof. Since all differentials vanish, H_k is isomorphic to C_k^{cell}. □

Remark. Over $\mathbb{Z}/2$, the computation of $H_*(\mathbb{R}P^n; \mathbb{Z}/2)$ becomes even simpler. The same CW structure has $C_q^{\text{cell}}(\mathbb{R}P^n; \mathbb{Z}/2) \cong \mathbb{Z}/2$ for all $0 \leq q \leq n$, and all differentials vanish because multiplication by 2 becomes 0 in $\mathbb{Z}/2$.

10 Coefficients, Universal Coefficient Theorems, and the Künneth Formula

10.1 Homology with coefficients

Let $C_*(X)$ be the singular chain complex of X with integer coefficients. For an abelian group G, define

$$C_n(X;G) := C_n(X) \otimes_{\mathbb{Z}} G, \qquad \partial_n^G := \partial_n \otimes \mathrm{id}_G.$$

Then $\partial_{n-1}^G \partial_n^G = 0$, and we set

$$H_n(X;G) := H_n\big(C_*(X;G)\big).$$

For a pair (X, A) we similarly define

$$C_n(X,A;G) := C_n(X,A) \otimes_{\mathbb{Z}} G, \qquad H_n(X,A;G) := H_n\big(C_*(X,A;G)\big).$$

Proposition 10.1. Naturality

A map $f\colon X \to Y$ induces chain maps

$$f_\# \otimes \mathrm{id}_G\colon C_*(X;G) \to C_*(Y;G),$$

and hence homomorphisms $f_*\colon H_n(X;G) \to H_n(Y;G)$. A homomorphism $\varphi\colon G \to G'$ induces chain maps

$$\mathrm{id} \otimes \varphi\colon C_*(X;G) \to C_*(X;G'),$$

and hence homomorphisms $\varphi_*\colon H_n(X;G) \to H_n(X;G')$.

Proof. Everything is induced by functoriality of tensor product together with the functoriality of singular chains. □

10.2 Universal coefficient theorem for homology

The functor $- \otimes_{\mathbb{Z}} G$ is right exact, and its failure to be exact is measured by Tor. For later computations it is convenient to keep in mind

$$(\mathbb{Z}/m) \otimes G \cong G/mG, \qquad \mathrm{Tor}(\mathbb{Z}/m, G) \cong \{g \in G \mid mg = 0\}.$$

Theorem 10.2. Universal coefficient theorem for homology

For each abelian group G there is a natural short exact sequence

$$0 \longrightarrow H_n(X) \otimes_{\mathbb{Z}} G \longrightarrow H_n(X;G) \longrightarrow \mathrm{Tor}\big(H_{n-1}(X), G\big) \longrightarrow 0.$$

This sequence splits, not canonically in general.

Proof. Write $Z_n = \ker(\partial\colon C_n \to C_{n-1})$ and $B_{n-1} = \mathrm{im}(\partial\colon C_n \to C_{n-1})$ for $C_*(X)$. Since each C_n is free abelian, the short exact sequences

$$0 \to Z_n \to C_n \to B_{n-1} \to 0, \qquad 0 \to B_n \to Z_n \to H_n(X) \to 0$$

are short exact sequences of free resolutions in the sense needed for Tor.
Tensor the first sequence with G. Since C_n is free, $\mathrm{Tor}(C_n, G) = 0$, so we obtain an exact sequence

$$0 \to Z_n \otimes G \to C_n \otimes G \to B_{n-1} \otimes G \to 0.$$

Under the identification $C_n(X;G) = C_n \otimes G$, the subgroup of cycles is

$$Z_n(X;G) = \ker(\partial_n \otimes \mathrm{id}) = \ker\big(C_n \otimes G \to C_{n-1} \otimes G\big) = Z_n \otimes G,$$

because the map $C_n \otimes G \to C_{n-1} \otimes G$ factors as

$$C_n \otimes G \twoheadrightarrow B_{n-1} \otimes G \hookrightarrow C_{n-1} \otimes G.$$

Similarly, boundaries satisfy $B_n(X;G) = \mathrm{im}(\partial_{n+1} \otimes \mathrm{id}) = B_n \otimes G$.
Now tensor the second short exact sequence $0 \to B_n \to Z_n \to H_n(X) \to 0$ with G. Right exactness gives

$$B_n \otimes G \to Z_n \otimes G \to H_n(X) \otimes G \to 0.$$

The kernel of $B_n \otimes G \to Z_n \otimes G$ is, by definition, $\mathrm{Tor}(H_n(X), G)$. In degree n, the homology of $C_*(X;G)$ is

$$H_n(X;G) = \frac{Z_n \otimes G}{B_n \otimes G}.$$

Putting these identifications together yields the short exact sequence

$$0 \to H_n(X) \otimes G \to H_n(X;G) \to \mathrm{Tor}(H_{n-1}(X), G) \to 0,$$

with naturality coming from functoriality of all constructions on the chain level.
For the splitting, one may choose splittings $C_n \cong Z_n \oplus S_n$ in the category of abelian groups, which exist because C_n is free. Such choices produce a splitting of the short exact sequence after tensoring, but different choices give different splittings. □

Corollary 10.3. If F is a field, then $\mathrm{Tor}(H_{n-1}(X), F) = 0$ and hence

$$H_n(X;F) \cong H_n(X) \otimes F.$$

Corollary 10.4. If $H_{n-1}(X)$ is torsionfree, then $\mathrm{Tor}(H_{n-1}(X), G) = 0$ for every G, and

$$H_n(X;G) \cong H_n(X) \otimes G.$$

Proposition 10.5

For each $n \geq 1$,

$$H_k(\mathbb{R}P^n; \mathbb{Z}/2) \cong \mathbb{Z}/2 \qquad \text{for all } 0 \leq k \leq n.$$

Proof. From the cellular computation of integral homology one has

$$H_0(\mathbb{R}P^n) \cong \mathbb{Z}, \qquad H_k(\mathbb{R}P^n) \cong \begin{cases} \mathbb{Z}/2 & k \text{ odd and } 0 < k < n, \\ 0 & k \text{ even and } 0 < k < n, \end{cases}$$

and in top degree $H_n(\mathbb{R}P^n) \cong \mathbb{Z}$ if n is odd and 0 if n is even. Apply Theorem 10.2 with $G = \mathbb{Z}/2$.
If $0 < k < n$ is odd, then

$$H_k(\mathbb{R}P^n) \otimes \mathbb{Z}/2 \cong (\mathbb{Z}/2) \otimes \mathbb{Z}/2 \cong \mathbb{Z}/2, \qquad \mathrm{Tor}(H_{k-1}(\mathbb{R}P^n), \mathbb{Z}/2) = 0,$$

so $H_k(\mathbb{R}P^n; \mathbb{Z}/2) \cong \mathbb{Z}/2$.
If $0 < k \leq n$ is even, then $H_k(\mathbb{R}P^n) = 0$, while $H_{k-1}(\mathbb{R}P^n) \cong \mathbb{Z}/2$ (since $k-1$ is odd and $< n$ when $k \leq n$). Hence

$$H_k(\mathbb{R}P^n; \mathbb{Z}/2) \cong \mathrm{Tor}(\mathbb{Z}/2, \mathbb{Z}/2) \cong \mathbb{Z}/2.$$

Finally $H_0(\mathbb{R}P^n; \mathbb{Z}/2) \cong \mathbb{Z}/2$ because $\mathbb{R}P^n$ is path-connected. □

10.3 Universal coefficient theorem for cohomology

Define cochains with coefficients in G by

$$C^n(X;G) := \mathrm{Hom}(C_n(X), G), \qquad \delta := \partial^*, \qquad H^n(X;G) := H^n(C^*(X;G)).$$

Theorem 10.6. Universal coefficient theorem for cohomology

There is a natural short exact sequence

$$0 \longrightarrow \mathrm{Ext}(H_{n-1}(X), G) \longrightarrow H^n(X;G) \longrightarrow \mathrm{Hom}(H_n(X), G) \longrightarrow 0,$$

which splits, not canonically in general.

Proof. Consider the short exact sequence $0 \to B_{n-1} \to Z_{n-1} \to H_{n-1}(X) \to 0$. Apply $\mathrm{Hom}(-, G)$ to obtain an exact sequence

$$0 \to \mathrm{Hom}(H_{n-1}(X), G) \to \mathrm{Hom}(Z_{n-1}, G) \to \mathrm{Hom}(B_{n-1}, G) \to \mathrm{Ext}(H_{n-1}(X), G) \to 0,$$

since Z_{n-1} is free abelian and hence $\mathrm{Ext}(Z_{n-1}, G) = 0$. Track the map $\mathrm{Hom}(Z_{n-1}, G) \to \mathrm{Hom}(B_{n-1}, G)$ against the coboundary operator in cochains. Identifying $H^n(X;G)$ as $\ker(\delta\colon C^n \to C^{n+1})/\mathrm{im}(\delta\colon C^{n-1} \to C^n)$ produces the displayed short exact sequence. Naturality follows from functoriality of Hom and of singular chains. A splitting exists after choosing splittings of the short exact sequences of cycles and boundaries inside free groups. □

Example. For $G = \mathbb{Z}$ and a finitely generated abelian group A, one has

$$\mathrm{Hom}(\mathbb{Z}, G) \cong G, \qquad \mathrm{Hom}(\mathbb{Z}/m, G) \cong \{g \in G \mid mg = 0\}, \qquad \mathrm{Ext}(\mathbb{Z}/m, G) \cong G/mG.$$

Thus torsion in $H_{n-1}(X)$ contributes a quotient term to $H^n(X; G)$ through Ext.

10.4 The Künneth theorem

Let R be a principal ideal domain. Assume X and Y are CW complexes. There is a homology cross product

$$\times \colon H_i(X; R) \otimes_R H_j(Y; R) \longrightarrow H_{i+j}(X \times Y; R).$$

Theorem 10.7. Künneth formula over a PID

For each n there is a natural short exact sequence

$$0 \longrightarrow \bigoplus_{i+j=n} H_i(X; R) \otimes_R H_j(Y; R) \longrightarrow H_n(X \times Y; R) \longrightarrow \bigoplus_{i+j=n-1} \mathrm{Tor}_R(H_i(X; R), H_j(Y; R)) \longrightarrow 0,$$

which splits, not canonically in general.

Proof. Apply the algebraic Künneth theorem to the chain complexes $C_*(X; R)$ and $C_*(Y; R)$, which are complexes of free R-modules. The Eilenberg–Zilber theorem identifies $C_*(X \times Y; R)$ with $C_*(X; R) \otimes_R C_*(Y; R)$ up to chain homotopy equivalence, and the statement follows. □

Corollary 10.8. If F is a field, then $\mathrm{Tor}_F(-, -) = 0$ and the cross product induces isomorphisms

$$H_n(X \times Y; F) \cong \bigoplus_{i+j=n} H_i(X; F) \otimes_F H_j(Y; F).$$

Corollary 10.9. If each $H_i(X; R)$ and $H_j(Y; R)$ is torsionfree, then all Tor_R terms vanish and

$$H_n(X \times Y; R) \cong \bigoplus_{i+j=n} H_i(X; R) \otimes_R H_j(Y; R).$$

Proposition 10.10

Let $m, n \geq 1$. With integer coefficients,

$$H_k(S^m \times S^n; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k = 0, m, n, m+n \text{ and } m \neq n, \\ \mathbb{Z} \oplus \mathbb{Z} & k = m = n, \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The homology groups of spheres are free abelian, so Corollary 10.9 applies with $R = \mathbb{Z}$. The only nonzero groups are $H_0(S^m) \cong \mathbb{Z}$ and $H_m(S^m) \cong \mathbb{Z}$, and similarly for S^n. Hence the only degrees that can receive contributions are $k = 0, m, n, m+n$. If $m \neq n$, each of the degrees m and n arises from a single tensor factor, while degree $m+n$ arises from $H_m(S^m) \otimes H_n(S^n) \cong \mathbb{Z}$. If $m = n$, then in degree m there are two contributions, from $(m, 0)$ and $(0, m)$, giving $\mathbb{Z} \oplus \mathbb{Z}$. □

Example. For $T^2 = S^1 \times S^1$, Proposition 10.10 gives

$$H_0(T^2) \cong \mathbb{Z}, \quad H_1(T^2) \cong \mathbb{Z}^2, \quad H_2(T^2) \cong \mathbb{Z},$$

which matches the Mayer–Vietoris computation.

11 Cohomology and the cup product

11.1 Cochains and cohomology

Let R be a commutative ring with 1. For a space X, write $C_n(X) = C_n(X; \mathbb{Z})$ for the singular chains and ∂ for the boundary operator. Define the group of n-cochains with coefficients in R by

$$C^n(X; R) := \mathrm{Hom}(C_n(X), R).$$

The coboundary operator $\delta\colon C^n(X;R) \to C^{n+1}(X;R)$ is defined by

$$(\delta\varphi)(c) := \varphi(\partial c), \qquad \varphi \in C^n(X;R),\ c \in C_{n+1}(X).$$

Then $\delta^2 = 0$ since $\partial^2 = 0$. We define

$$Z^n(X;R) := \ker(\delta\colon C^n \to C^{n+1}), \qquad B^n(X;R) := \operatorname{im}(\delta\colon C^{n-1} \to C^n),$$

and the nth cohomology group

$$H^n(X;R) := Z^n(X;R)/B^n(X;R).$$

A map $f\colon X \to Y$ induces a chain map $f_\#\colon C_*(X) \to C_*(Y)$ and hence a pullback on cochains

$$f^*\colon C^n(Y;R) \to C^n(X;R), \qquad f^*(\varphi) := \varphi \circ f_\#.$$

Since $\partial f_\# = f_\#\partial$, we have $\delta f^* = f^*\delta$, so f^* sends cocycles to cocycles and coboundaries to coboundaries and induces

$$f^*\colon H^n(Y;R) \to H^n(X;R).$$

Thus $H^n(-;R)$ is a contravariant functor.

There is also a pairing with homology. For $\varphi \in C^n(X;R)$ and $c \in C_n(X)$ define $\langle \varphi, c\rangle := \varphi(c) \in R$. If φ is a cocycle and c is a cycle, this depends only on the cohomology class $[\varphi]$ and the homology class $[c]$, yielding the Kronecker pairing

$$\langle\ ,\ \rangle\colon H^n(X;R) \times H_n(X;R) \to R.$$

11.2 The cup product on cochains

Let $\varphi \in C^k(X;R)$ and $\psi \in C^\ell(X;R)$. For a singular simplex $\sigma\colon \Delta^{k+\ell} \to X$ with vertices $v_0, \dots, v_{k+\ell}$, let

$$\sigma|[v_0, \dots, v_k]\colon \Delta^k \to X, \qquad \sigma|[v_k, \dots, v_{k+\ell}]\colon \Delta^\ell \to X$$

be the restrictions to the front and back faces. Define a $(k+\ell)$-cochain $\varphi \smile \psi$ by

$$(\varphi \smile \psi)(\sigma) := \varphi(\sigma|[v_0, \dots, v_k]) \cdot \psi(\sigma|[v_k, \dots, v_{k+\ell}]),$$

extended R-linearly to $C_{k+\ell}(X)$.

Lemma 11.1. Leibniz rule for the coboundary For $\varphi \in C^k(X;R)$ and $\psi \in C^\ell(X;R)$,

$$\delta(\varphi \smile \psi) = (\delta\varphi) \smile \psi + (-1)^k\, \varphi \smile (\delta\psi).$$

Proof. Evaluate both sides on a singular simplex $\sigma\colon \Delta^{k+\ell+1} \to X$ and expand $\delta(\varphi \smile \psi)(\sigma) = (\varphi \smile \psi)(\partial\sigma)$ as an alternating sum over faces. Split the sum into the faces $[v_0, \dots, \widehat{v}_i, \dots, v_{k+1}]$ for $0 \le i \le k+1$ and the faces $[v_k, \dots, \widehat{v}_i, \dots, v_{k+\ell+1}]$ for $k \le i \le k+\ell+1$. The middle terms cancel with the sign $(-1)^k$, and the remaining terms match the two terms on the right-hand side. □

Corollary 11.2. If φ and ψ are cocycles, then $\varphi \smile \psi$ is a cocycle. If either φ or ψ is a coboundary and the other is a cocycle, then $\varphi \smile \psi$ is a coboundary. Hence there is a well-defined product

$$\smile\colon H^k(X;R) \times H^\ell(X;R) \to H^{k+\ell}(X;R), \qquad [\varphi] \smile [\psi] := [\varphi \smile \psi].$$

On cochains the product is bilinear and associative, and these properties pass to cohomology. If R has unit 1, then the cochain in $C^0(X;R)$ taking value 1 on each singular 0-simplex is a cocycle whose class is the unit in $H^0(X;R)$ for $\smile$.

Proposition 11.3. Naturality of the cup product

For $f\colon X \to Y$ and $\alpha \in H^k(Y;R)$, $\beta \in H^\ell(Y;R)$,

$$f^*(\alpha \smile \beta) = f^*\alpha \smile f^*\beta.$$

Proof. On cochains, $f^*(\varphi \smile \psi)$ and $(f^*\varphi) \smile (f^*\psi)$ agree on each simplex since restricting $f \circ \sigma$ to faces is the same as composing f with the face restrictions of σ. □

11.3 Graded commutativity

The cup product is not commutative on the cochain level. The commutativity statement holds in cohomology, with the usual sign.

Theorem 11.4. Graded commutativity

Let R be commutative. For $\alpha \in H^k(X;R)$ and $\beta \in H^\ell(X;R)$,

$$\alpha \smile \beta = (-1)^{k\ell}\, \beta \smile \alpha.$$

Proof. There exists a natural chain map $\rho\colon C_*(X) \to C_*(X)$ which reverses the order of vertices of each simplex with an explicit sign, and which is chain homotopic to the identity. Let $\rho^\#$ be the induced cochain map. Then $\rho^\#$ induces the identity on cohomology, while on cochains one has

$$\rho^\#(\varphi \smile \psi) = (-1)^{k\ell}\, \rho^\#(\psi) \smile \rho^\#(\varphi)$$

for $\varphi \in C^k(X;R)$ and $\psi \in C^\ell(X;R)$. Passing to cohomology gives the stated identity. □

11.4 Cohomology rings as invariants

The graded ring

$$H^*(X;R) := \bigoplus_{n\geq 0} H^n(X;R)$$

with multiplication $\smile$ is often sharper than the graded abelian group alone.

A quick example comes from a surface. For the torus T^2, we have $H^0(T^2;\mathbb{Z}) \cong \mathbb{Z}$, $H^1(T^2;\mathbb{Z}) \cong \mathbb{Z}^2$, and $H^2(T^2;\mathbb{Z}) \cong \mathbb{Z}$. Choosing $\alpha, \beta \in H^1(T^2;\mathbb{Z})$ dual to a basis of $H_1(T^2;\mathbb{Z})$, one can arrange that $\alpha \smile \beta$ is a generator of $H^2(T^2;\mathbb{Z})$, and then graded commutativity forces $\beta \smile \alpha = -(\alpha \smile \beta)$ and $\alpha \smile \alpha = \beta \smile \beta = 0$.

11.5 Projective spaces

$\mathbb{R}P^n$ with $\mathbb{Z}_2$ coefficients

Let $a \in H^1(\mathbb{R}P^n;\mathbb{Z}_2)$ be a generator. Then the graded ring is a truncated polynomial algebra.

Theorem 11.5. The $\mathbb{Z}_2$ cohomology ring of $\mathbb{R}P^n$

For $n \geq 1$,

$$H^*(\mathbb{R}P^n;\mathbb{Z}_2) \cong \mathbb{Z}_2[a]/(a^{n+1}), \qquad |a| = 1.$$

In particular $H^k(\mathbb{R}P^n;\mathbb{Z}_2) \cong \mathbb{Z}_2$ for $0 \leq k \leq n$, with generator a^k, and $H^k = 0$ for $k > n$.

When $n = 2$, this says

$$H^*(\mathbb{R}P^2;\mathbb{Z}_2) \cong \mathbb{Z}_2[a]/(a^3),$$

so $a^2 \neq 0$ even though $H_2(\mathbb{R}P^2;\mathbb{Z}) = 0$.

$\mathbb{C}P^n$ with $\mathbb{Z}$ coefficients

Let $x \in H^2(\mathbb{C}P^n;\mathbb{Z})$ be a generator. Then the graded ring is again truncated polynomial, now with $|x| = 2$.

Theorem 11.6. The integral cohomology ring of $\mathbb{C}P^n$

For $n \geq 1$,

$$H^*(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}), \qquad |x| = 2.$$

Thus $H^{2k}(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$ for $0 \leq k \leq n$, generated by x^k, and all odd-degree cohomology vanishes.

For $n = 2$, one gets

$$H^*(\mathbb{C}P^2;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^3),$$

so x^2 generates $H^4(\mathbb{C}P^2;\mathbb{Z}) \cong \mathbb{Z}$.

12 Degree and a first application

12.1 Degree for self-maps of spheres

Fix $n > 0$. Since $H_n(S^n) \cong \mathbb{Z}$, any homomorphism $H_n(S^n) \to H_n(S^n)$ is multiplication by a unique integer.

Definition 12.1. Degree of a map $S^n \to S^n$

Let $f\colon S^n \to S^n$. Choose a generator $[S^n] \in H_n(S^n)$. There is a unique integer $\deg(f) \in \mathbb{Z}$ such that

$$f_*([S^n]) = \deg(f)\,[S^n].$$

The same integer is obtained from cohomology. Let $u \in H^n(S^n)$ be the generator with $\langle u, [S^n]\rangle = 1$.

Lemma 12.2 For $f\colon S^n \to S^n$ one has

$$f^*(u) = \deg(f)\,u \quad \text{in } H^n(S^n).$$

Proof. For any $x \in H_n(S^n)$, naturality of the Kronecker pairing gives

$$\langle f^*u, x\rangle = \langle u, f_*x\rangle.$$

Apply this with $x = [S^n]$ and use $f_*[S^n] = \deg(f)[S^n]$. □

Proposition 12.3. Basic properties

Let $f, g\colon S^n \to S^n$.

1. $\deg(\mathrm{id}) = 1$.
2. If $f \simeq g$, then $\deg(f) = \deg(g)$.
3. $\deg(g \circ f) = \deg(g)\deg(f)$.
4. If f is not surjective, then $\deg(f) = 0$.

Proof. (1)–(3) follow from functoriality of H_n and the definition of degree.
For (4), choose $x_0 \in S^n \setminus f(S^n)$. Then f factors as $S^n \to S^n \setminus \{x_0\} \hookrightarrow S^n$. Since $S^n \setminus \{x_0\}$ is contractible, $H_n(S^n \setminus \{x_0\}) = 0$, hence $f_* = 0$ on $H_n(S^n)$. □

12.2 Computing degrees

Proposition 12.4

A reflection $r\colon S^n \to S^n$ has $\deg(r) = -1$.

Proof. Realize S^n as two n-simplices glued along their boundary, corresponding to the two hemispheres. A reflection swaps these two top simplices and fixes their common boundary. On the top cellular chain group this sends a fundamental cycle to its negative, hence induces multiplication by -1 on $H_n(S^n)$. □

Corollary 12.5. The antipodal map $a\colon S^n \to S^n$, $a(x) = -x$, satisfies

$$\deg(a) = (-1)^{n+1}.$$

Proof. Write a as the composition of $n+1$ reflections, each changing the sign of one coordinate in $\mathbb{R}^{n+1}$. Apply Proposition 12.3 (3) and Proposition 12.4. □

Example. Let $f_m\colon S^1 \to S^1$ be $f_m(e^{2\pi i t}) = e^{2\pi i m t}$. Then $\deg(f_m) = m$. Indeed, $H_1(S^1) \cong \mathbb{Z}$ is generated by the fundamental class of the standard loop, and f_m wraps that loop around m times.

12.3 Degree and fixed points on spheres

Degree forces fixed points once one rules out the antipodal behavior.

Proposition 12.6

If $f\colon S^n \to S^n$ has no fixed points, then

$$\deg(f) = (-1)^{n+1}.$$

Proof. Assume $f(x) \neq x$ for all $x \in S^n$. Define a homotopy $H\colon S^n \times I \to S^n$ by

$$H(x,t) = \frac{(1-t)f(x) + t(-x)}{\|(1-t)f(x) + t(-x)\|}.$$

The denominator never vanishes: if $(1-t)f(x) + t(-x) = 0$ then $f(x) = \frac{t}{1-t}x$, and taking norms gives $\frac{t}{1-t} = 1$, hence $t = \frac{1}{2}$ and $f(x) = x$, contradicting the hypothesis. Thus H is well-defined and continuous, with $H(-,0) = f$ and $H(-,1) = a$ the antipodal map. Homotopy invariance of degree and Corollary 12.5 give $\deg(f) = \deg(a) = (-1)^{n+1}$. □

Corollary 12.7. If $f\colon S^n \to S^n$ satisfies $\deg(f) \neq (-1)^{n+1}$, then f has a fixed point.

Proof. If f had no fixed points, Proposition 12.6 would force $\deg(f) = (-1)^{n+1}$. □

12.4 Brouwer fixed point theorem

Write $D^n = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ and $\partial D^n = S^{n-1}$.

Lemma 12.8. No retraction There is no retraction $r\colon D^n \to S^{n-1}$.

Proof. Suppose r is a retraction and let $i\colon S^{n-1} \hookrightarrow D^n$ be the inclusion. Then $r \circ i = \mathrm{id}_{S^{n-1}}$. Applying $H_{n-1}(-)$ gives

$$(r \circ i)_* = r_* \circ i_* = \mathrm{id}_{H_{n-1}(S^{n-1})}.$$

However $H_{n-1}(D^n) = 0$ since D^n is contractible, so $i_* = 0$ and hence $r_* \circ i_* = 0$, a contradiction. □

The retraction obstruction is the input for Brouwers theorem.

Theorem 12.9. Brouwer fixed point theorem
Every continuous map $h\colon D^n \to D^n$ has a fixed point.

Proof. Assume $h(x) \neq x$ for all $x \in D^n$. For $x \in D^n$, consider the ray starting at $h(x)$ and passing through x. Since $h(x) \neq x$, this ray meets S^{n-1} in a unique point. Define $r(x)$ to be that intersection point. Then $r\colon D^n \to S^{n-1}$ is continuous, and if $x \in S^{n-1}$ the ray from $h(x)$ through x exits D^n at x, so $r(x) = x$. Thus r is a retraction, contradicting Lemma 12.8. □

Example. For $n = 2$, the same construction gives a retraction $D^2 \to S^1$ from a fixed-point-free map. In dimension 2, one can also rule out retractions by π_1, since a retraction would force a nontrivial loop in S^1 to become nullhomotopic through D^2 while staying inside S^1 after applying the retraction. The argument above replaces π_1 by H_{n-1} and works for all n.

13 Cap products, orientations, and Poincaré duality

13.1 Cap product on chains and cochains

Let R be a commutative ring with 1. Write $C_n(X) = C_n(X;\mathbb{Z})$ for the singular chain groups and $C^k(X;R) = \mathrm{Hom}(C_k(X), R)$ for cochains. The coboundary is $(\delta\varphi)(c) = \varphi(\partial c)$.

Fix $n \geq 0$ and $0 \leq k \leq n$. For a singular n-simplex $\sigma\colon \Delta^n \to X$ with ordered vertices $v_0, \ldots, v_n$, let

$$\sigma|[v_0, \ldots, v_k]\colon \Delta^k \to X, \qquad \sigma|[v_k, \ldots, v_n]\colon \Delta^{n-k} \to X$$

denote the restrictions to the front and back faces.

Definition 13.1. Cap product on chains

For $\varphi \in C^k(X;R)$ and a singular n-simplex σ, define an $(n-k)$-chain with coefficients in R by

$$\sigma \frown \varphi \;:=\; \varphi(\sigma|[v_0, \ldots, v_k]) \cdot \sigma|[v_k, \ldots, v_n] \;\in\; C_{n-k}(X;R).$$

Extend R-linearly in σ to obtain maps

$$\frown : C_n(X;R) \times C^k(X;R) \longrightarrow C_{n-k}(X;R).$$

Lemma 13.2. Boundary formula For $c \in C_n(X;R)$ and $\varphi \in C^k(X;R)$,

$$\partial(c \frown \varphi) = (-1)^k\Big((\partial c) \frown \varphi \;-\; c \frown (\delta\varphi)\Big).$$

Proof. It suffices to check the identity on a generator $c = \sigma$. Expand $\partial(\sigma \frown \varphi)$ as the alternating sum of the faces of $\sigma|[v_k, \dots, v_n]$. On the other hand, the sum $(\partial\sigma) \frown \varphi$ is the alternating sum of $(n-1-k)$-simplices obtained by first taking a face of σ and then restricting to its back face. The difference between these two alternating sums is accounted for by the terms in $\sigma \frown (\delta\varphi)$, coming from the faces of the front k-simplex $\sigma|[v_0, \dots, v_k]$, with an overall factor $(-1)^k$. □

Corollary 13.3. If $z \in Z_n(X;R)$ is a cycle and $\varphi \in B^k(X;R)$ is a coboundary, then $z \frown \varphi$ is a boundary. If z is a boundary and φ is a cocycle, then $z \frown \varphi$ is a boundary. Hence there is an induced cap product

$$\frown : H_n(X;R) \times H^k(X;R) \longrightarrow H_{n-k}(X;R), \qquad [x] \frown [\alpha] := [x \frown \alpha].$$

Proof. Apply Lemma 13.2. □

Proposition 13.4. Naturality

Let $f\colon X \to Y$. For $x \in H_n(X;R)$ and $\alpha \in H^k(Y;R)$,

$$f_*(x) \frown \alpha \;=\; f_*(x \frown f^*\alpha) \;\in H_{n-k}(Y;R).$$

Proof. On the chain level, for a simplex σ in X one has

$$(f_\#\sigma) \frown \varphi \;=\; \sigma \frown (f^*\varphi)$$

for every cochain φ on Y, since restricting $f \circ \sigma$ to a face is the same as composing f with the corresponding restriction of σ. Passing to homology and cohomology gives the stated identity. □

13.2 Relation with the cup product

Write $\langle\ ,\ \rangle$ for the Kronecker pairing $H^m(X;R) \times H_m(X;R) \to R$ induced by evaluation of cochains on cycles.

Proposition 13.5. Cup–cap relation

Let $x \in H_n(X;R)$, $\alpha \in H^k(X;R)$, and $\beta \in H^{n-k}(X;R)$. Then

$$\langle \alpha \smile \beta,\ x\rangle \;=\; \langle \beta,\ x \frown \alpha\rangle.$$

Proof. Choose a cycle $z \in C_n(X;R)$ representing x and cocycles φ, ψ representing α, β. For a singular simplex σ appearing in z, the definitions give

$$(\varphi \smile \psi)(\sigma) \;=\; \varphi(\sigma|[v_0, \dots, v_k]) \cdot \psi(\sigma|[v_k, \dots, v_n]),$$

while

$$\psi(\sigma \frown \varphi) \;=\; \psi\Big(\varphi(\sigma|[v_0, \dots, v_k]) \cdot \sigma|[v_k, \dots, v_n]\Big) \;=\; \varphi(\sigma|[v_0, \dots, v_k]) \cdot \psi(\sigma|[v_k, \dots, v_n]).$$

Summing over the simplices in z gives $\langle \varphi \smile \psi,\ z\rangle = \langle \psi,\ z \frown \varphi\rangle$. □

Proposition 13.6. Associativity with cup

For $x \in H_n(X;R)$, $\alpha \in H^p(X;R)$, and $\beta \in H^q(X;R)$,

$$(x \frown \alpha) \frown \beta \;=\; x \frown (\alpha \smile \beta) \;\in H_{n-p-q}(X;R).$$

Proof. Choose cocycles φ, ψ representing α, β and a cycle z representing x. On a simplex $\sigma\colon [v_0, \dots, v_n] \to X$, the left-hand side extracts the back face twice: first at v_p, then at v_{p+q} inside that back face. The resulting coefficient is

$$\varphi(\sigma|[v_0, \dots, v_p]) \cdot \psi(\sigma|[v_p, \dots, v_{p+q}]),$$

and the remaining simplex is $\sigma|[v_{p+q}, \dots, v_n]$. This matches the definition of $\sigma \frown (\varphi \smile \psi)$. $\square$

13.3 Local homology and orientations

Let M be a topological n-manifold.

Lemma 13.7. Local homology of a manifold For each $x \in M$,

$$H_i(M,\ M \setminus \{x\}; \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & i = n, \\ 0 & i \neq n. \end{cases}$$

Proof. Choose a coordinate chart $U \subset M$ containing x and a smaller open ball $B \subset U$ around x. By excision,

$$H_i(M,\ M \setminus \{x\}) \cong H_i(B,\ B \setminus \{x\}).$$

Pick a homeomorphism $B \cong \mathbb{R}^n$ carrying x to 0. Then $(\mathbb{R}^n, \mathbb{R}^n \setminus \{0\})$ deformation retracts to (D^n, S^{n-1}), and

$$H_i(D^n, S^{n-1}) \cong \begin{cases} \mathbb{Z} & i = n, \\ 0 & i \neq n \end{cases}$$

from the computation of the disk pair. $\square$

Definition 13.8. $\mathbb{Z}$-orientation

A **$\mathbb{Z}$-orientation** of M is a choice, for each $x \in M$, of a generator

$$\mu_x \in H_n(M,\ M \setminus \{x\}; \mathbb{Z}) \cong \mathbb{Z}$$

such that for every coordinate ball $B \subset M$ and every $x, y \in B$, the images of μ_x and μ_y agree under the canonical isomorphisms

$$H_n(M, M \setminus \{x\}) \xleftarrow{\cong} H_n(M, M \setminus B) \xrightarrow{\cong} H_n(M, M \setminus \{y\}).$$

A manifold is **orientable** if it admits such a choice.

When M is connected and orientable, the local generators determine a global class in top homology.

Definition 13.9. Fundamental class

Let M be a connected orientable closed n-manifold. A **fundamental class** is an element $[M] \in H_n(M; \mathbb{Z})$ whose image in $H_n(M, M \setminus \{x\}; \mathbb{Z})$ is μ_x for every $x \in M$.

Lemma 13.10 If M is connected and closed, then a fundamental class $[M] \in H_n(M; \mathbb{Z})$ is determined by its image in $H_n(M, M \setminus \{x\}; \mathbb{Z})$ at one point x.

Proof. Let $x, y \in M$ and choose a path joining them. Cover the path by finitely many coordinate balls $B_1, \dots, B_m$ with consecutive overlaps nonempty, with $x \in B_1$ and $y \in B_m$. For each overlap, the local compatibility in Definition 13.8 identifies the images of $[M]$ in the local homology groups of points in the overlap. Chaining these identifications shows that fixing the image at x forces the image at y. $\square$

13.4 Poincaré duality for closed manifolds

Let R be a commutative ring. If M is orientable, the image of $[M] \in H_n(M; \mathbb{Z})$ in $H_n(M; R)$ under $\mathbb{Z} \to R$ is denoted $[M]_R$.

Theorem 13.11. Poincaré duality

Let M be a connected closed orientable n-manifold, and let R be a commutative ring. Cap product with $[M]_R$ defines homomorphisms

$$D : H^k(M; R) \longrightarrow H_{n-k}(M; R), \qquad D(\alpha) := [M]_R \frown \alpha.$$

Then D is an isomorphism for all k.

Proof. Fix a triangulation of M and take a barycentric subdivision. For each simplex τ of dimension k, let τ^* denote the dual cell of dimension $n-k$ in the dual cell decomposition. The cellular chain group $C^{\mathrm{cell}}_{n-k}(M^*;R)$ is a free R-module on the $(n-k)$-cells τ^*, and the simplicial cochain group $C^k(M;R)$ is the R-module of functions on k-simplices.
Define an R-linear map

$$\Theta_k\colon C^k(M;R) \longrightarrow C^{\mathrm{cell}}_{n-k}(M^*;R)$$

by

$$\Theta_k(\varphi) := \sum_{\tau\in\Sigma_k} \varphi(\tau)\,\tau^*,$$

where Σ_k is the set of oriented k-simplices. The dual cell boundary in M^* and the simplicial coboundary in M are related by the incidence relations of the dual decomposition, giving

$$\partial\,\Theta_k(\varphi) = (-1)^k\,\Theta_{k+1}(\delta\varphi).$$

Hence Θ_* is a chain map from the cochain complex $C^*(M;R)$ (viewed in nonnegative degrees) to the cellular chain complex $C^{\mathrm{cell}}_{n-*}(M^*;R)$ up to the indicated sign, and it induces an isomorphism on homology.
Identify $H_{n-k}(M;R)$ with the cellular homology of M^*. With orientations chosen compatibly, the fundamental class $[M]_R$ is represented by the sum of the oriented n-simplices. On a cochain $\varphi \in C^k(M;R)$, the chain $[M]_R \frown \varphi$ is obtained by evaluating φ on the front k-faces of the n-simplices and retaining the complementary back faces; this produces the same coefficients as $\Theta_k(\varphi)$ on the dual cells τ^*. Thus the map D agrees with the map induced by Θ under the identification of M with its dual cell decomposition. Therefore D is an isomorphism. □

Proposition 13.12. Duality and the cup product pairing

Let M be a connected closed orientable n-manifold and let R be a field. The pairing

$$H^k(M;R)\times H^{n-k}(M;R) \longrightarrow R, \qquad (\alpha,\beta)\longmapsto \langle \alpha\smile\beta,\ [M]_R\rangle$$

is nondegenerate.

Proof. By Proposition 13.5,

$$\langle \alpha\smile\beta,\ [M]_R\rangle = \langle \beta,\ [M]_R\frown\alpha\rangle = \langle \beta,\ D(\alpha)\rangle.$$

Over a field, the Kronecker pairing identifies $H^{n-k}(M;R)$ with $\mathrm{Hom}_R(H_{n-k}(M;R),R)$. Since D is an isomorphism by Theorem 13.11, the bilinear pairing is nondegenerate. □

Example. Let Σ_g be a closed orientable surface of genus g. Then $H_2(\Sigma_g;\mathbb{Z})\cong\mathbb{Z}$ with generator $[\Sigma_g]$, and $D\colon H^1(\Sigma_g;\mathbb{Z})\to H_1(\Sigma_g;\mathbb{Z})$ is an isomorphism. For a standard symplectic basis $a_1,b_1,\dots,a_g,b_g$ of $H_1(\Sigma_g;\mathbb{Z})$, the dual basis in $H^1(\Sigma_g;\mathbb{Z})$ is characterized by the evaluations on these cycles, and the cup product pairing is represented by the usual skew form with $a_i\smile b_i$ pairing to 1 on $[\Sigma_g]$ and $b_i\smile a_i$ pairing to -1.

Example. For $\mathbb{C}P^n$, the integral cohomology ring is $H^*(\mathbb{C}P^n;\mathbb{Z})\cong\mathbb{Z}[x]/(x^{n+1})$ with $|x|=2$. Let $[\mathbb{C}P^n]\in H_{2n}(\mathbb{C}P^n;\mathbb{Z})$ be the fundamental class. Poincaré duality gives isomorphisms

$$D\ :\ H^{2k}(\mathbb{C}P^n;\mathbb{Z}) \xrightarrow{\cong} H_{2n-2k}(\mathbb{C}P^n;\mathbb{Z}), \qquad D(x^k)=[\mathbb{C}P^n]\frown x^k.$$

The class $x^n\in H^{2n}(\mathbb{C}P^n;\mathbb{Z})$ is characterized by $\langle x^n,\ [\mathbb{C}P^n]\rangle = 1$, and then $\langle x^k\smile x^{n-k},\ [\mathbb{C}P^n]\rangle = 1$ for all k.

14 Higher Homotopy Groups and Relative Homotopy

14.1 Based maps from cubes

Fix a pointed space (X,x_0). For $n\geq 1$, write ∂I^n for the boundary of the cube I^n.

Definition 14.1. Based homotopy rel boundary

Let $f, g\colon I^n \to X$ be maps such that $f(\partial I^n) \subseteq \{x_0\}$ and $g(\partial I^n) \subseteq \{x_0\}$. A **based homotopy rel** ∂I^n from f to g is a map $H\colon I^n \times I \to X$ such that

$$H(-,0) = f, \qquad H(-,1) = g, \qquad H(\partial I^n \times I) \subseteq \{x_0\}.$$

We write $f \simeq g \operatorname{rel} \partial I^n$.

Definition 14.2. Higher homotopy groups

For $n \geq 1$, define $\pi_n(X, x_0)$ to be the set of based homotopy classes rel ∂I^n of maps $f\colon I^n \to X$ with $f(\partial I^n) \subseteq \{x_0\}$.

The cube model is convenient for defining operations. It is equivalent to the more geometric sphere model.

Proposition 14.3

For $n \geq 1$, there is a natural bijection

$$\pi_n(X, x_0) \;\cong\; [\,(S^n, *), (X, x_0)\,],$$

where the right-hand side denotes based homotopy classes of maps $S^n \to X$.

Proof. The quotient map $q\colon I^n \to I^n/\partial I^n$ identifies $I^n/\partial I^n$ with S^n. A map $f\colon I^n \to X$ with $f(\partial I^n) \subseteq \{x_0\}$ factors uniquely through q to a based map $\overline{f}\colon S^n \to X$. Conversely, any based map $S^n \to X$ pulls back along q to such an f. A homotopy rel ∂I^n descends to a based homotopy on S^n, and a based homotopy on S^n pulls back to a homotopy rel ∂I^n. □

Example. If X is contractible, then $\pi_n(X, x_0) = 0$ for all $n \geq 1$. Indeed any $f\colon I^n \to X$ with $f(\partial I^n) \subseteq \{x_0\}$ is homotopic rel ∂I^n to the constant map at x_0 by restricting a contraction of X to the image of f.

14.2 The group structure and commutativity for $n \geq 2$

Definition 14.4. Concatenation in the last coordinate

Let $n \geq 1$ and let $f, g\colon I^n \to X$ satisfy $f(\partial I^n) \subseteq \{x_0\}$ and $g(\partial I^n) \subseteq \{x_0\}$. Define $f \cdot g\colon I^n \to X$ by

$$(f \cdot g)(t_1, \ldots, t_{n-1}, t_n) = \begin{cases} f(t_1, \ldots, t_{n-1}, 2t_n) & 0 \leq t_n \leq \frac{1}{2}, \\ g(t_1, \ldots, t_{n-1}, 2t_n - 1) & \frac{1}{2} \leq t_n \leq 1. \end{cases}$$

Proposition 14.5

For $n \geq 1$, the operation induced by $\cdot$ makes $\pi_n(X, x_0)$ a group. The identity is the class of the constant map c_{x_0}. An inverse of $[f]$ is represented by the reflection

$$f^{-1}(t_1, \ldots, t_{n-1}, t_n) = f(t_1, \ldots, t_{n-1}, 1 - t_n).$$

Proof. First, $f \cdot g$ sends ∂I^n to x_0 since each restriction to a boundary face factors through a boundary face of either f or g.

Well-definedness on homotopy classes is proved by pasting homotopies. If $f_0 \simeq f_1 \operatorname{rel} \partial I^n$ via H and $g_0 \simeq g_1 \operatorname{rel} \partial I^n$ via K, define $L\colon I^n \times I \to X$ by using H on the lower half $t_n \in [0, \frac{1}{2}]$ and K on the upper half $t_n \in [\frac{1}{2}, 1]$ with the same reparametrization as in Definition 14.4. Then L is continuous and gives $f_0 \cdot g_0 \simeq f_1 \cdot g_1 \operatorname{rel} \partial I^n$.

For the identity, the maps $f \cdot c_{x_0}$ and $c_{x_0} \cdot f$ are reparametrizations of f in the last coordinate, with a constant segment inserted. A straight-line homotopy between the two parametrizations gives homotopies rel ∂I^n to f.

Associativity is similar. The maps $(f \cdot g) \cdot h$ and $f \cdot (g \cdot h)$ traverse f, then g, then h, and differ only by a piecewise-linear reparametrization of the last coordinate. A homotopy between these reparametrizations yields a homotopy rel ∂I^n between the two composites.

For inverses, the concatenations $f \cdot f^{-1}$ and $f^{-1} \cdot f$ collapse to c_{x_0} by shrinking the portion of the last coordinate in which f is traversed. This uses only a homotopy of the parameter interval and respects the boundary condition. □

For $n = 1$, this agrees with the fundamental group operation. For $n \geq 2$, the group is abelian.

Proposition 14.6

For $n \geq 2$, the group $\pi_n(X, x_0)$ is abelian.

Proof. Define a second operation $\star$ by concatenation in the $(n-1)$st coordinate:

$$(f \star g)(t_1, \ldots, t_{n-2}, t_{n-1}, t_n) = \begin{cases} f(t_1, \ldots, t_{n-2}, 2t_{n-1}, t_n) & 0 \leq t_{n-1} \leq \frac{1}{2}, \\ g(t_1, \ldots, t_{n-2}, 2t_{n-1} - 1, t_n) & \frac{1}{2} \leq t_{n-1} \leq 1. \end{cases}$$

This induces another group law on $\pi_n(X, x_0)$, with the same identity class $[c_{x_0}]$.
Now check the interchange relation at the level of representatives:

$$(f \cdot g) \star (f' \cdot g') \;\simeq\; (f \star f') \cdot (g \star g') \qquad \operatorname{rel} \partial I^n.$$

Both sides cut the square in the (t_{n-1}, t_n)-plane into four rectangles and place the four maps f, f', g, g' on these rectangles in the same pattern, differing only by how the cuts are parametrized. A homotopy of the partition lines gives a based homotopy rel boundary between the two maps.
Apply the interchange relation with $f' = c_{x_0}$ and $g = c_{x_0}$. In $\pi_n(X, x_0)$ we get

$$[f] \star [g] \;=\; ([f] \cdot [c_{x_0}]) \star ([c_{x_0}] \cdot [g]) \;=\; ([f] \star [c_{x_0}]) \cdot ([c_{x_0}] \star [g]) \;=\; [f] \cdot [g].$$

So the two operations coincide on $\pi_n(X, x_0)$.
Finally, using the interchange relation again,

$$[f] \cdot [g] \;=\; [f] \star [g] \;=\; ([c_{x_0}] \cdot [f]) \star ([g] \cdot [c_{x_0}]) \;=\; ([c_{x_0}] \star [g]) \cdot ([f] \star [c_{x_0}]) \;=\; [g] \cdot [f].$$

Hence $\pi_n(X, x_0)$ is abelian. □

Example. For $n \geq 2$, one has $\pi_n(S^1, 1) = 0$. Let $f\colon (S^n, *) \to (S^1, 1)$ be a based map. Since S^n is simply connected for $n \geq 2$, the map f lifts to a map $\widetilde{f}\colon S^n \to \mathbb{R}$ along the universal covering $\mathbb{R} \to S^1$. The space $\mathbb{R}$ is contractible, so $\widetilde{f}$ is homotopic to a constant map, and composing with the covering map shows that f is nullhomotopic.

14.3 Functoriality and products

Proposition 14.7

Let $f\colon (X, x_0) \to (Y, y_0)$ be a based map. Then there is an induced homomorphism

$$f_*\colon \pi_n(X, x_0) \to \pi_n(Y, y_0), \qquad f_*([g]) = [f \circ g].$$

Moreover $(\mathrm{id}_X)_* = \mathrm{id}$ and $(f \circ g)_* = f_* \circ g_*$. If $f \simeq g$ through a based homotopy, then $f_* = g_*$.

Proof. If $g_0 \simeq g_1 \operatorname{rel} \partial I^n$ via H, then $f \circ H$ is a homotopy rel ∂I^n from $f \circ g_0$ to $f \circ g_1$. Thus f_* is well defined. It preserves products because $f \circ (g_1 \cdot g_2) = (f \circ g_1) \cdot (f \circ g_2)$ as maps of cubes. The identity and composition statements are immediate. If $f \simeq g$ via a based homotopy $K\colon X \times I \to Y$, then $K \circ (h \times \mathrm{id}_I)$ gives a based homotopy between $f \circ h$ and $g \circ h$ for any $h\colon I^n \to X$, so $f_* = g_*$. □

Proposition 14.8

For based spaces (X, x_0) and (Y, y_0), the map

$$\pi_n(X \times Y, (x_0, y_0)) \to \pi_n(X, x_0) \times \pi_n(Y, y_0), \qquad [f] \mapsto ([\mathrm{pr}_X \circ f], [\mathrm{pr}_Y \circ f]),$$

is an isomorphism of groups.

Proof. Define Φ by projections as above. Given classes $[a] \in \pi_n(X, x_0)$ and $[b] \in \pi_n(Y, y_0)$, represented by maps $a\colon I^n \to X$ and $b\colon I^n \to Y$ sending ∂I^n to the basepoints, define $f = (a, b)\colon I^n \to X \times Y$. Then $f(\partial I^n) \subseteq \{(x_0, y_0)\}$ and $\Phi([f]) = ([a], [b])$, so Φ is surjective.
If $\Phi([f]) = \Phi([g])$, then there are homotopies rel ∂I^n from $\mathrm{pr}_X \circ f$ to $\mathrm{pr}_X \circ g$ and from $\mathrm{pr}_Y \circ f$ to $\mathrm{pr}_Y \circ g$. Taking their product gives a homotopy rel ∂I^n from f to g in $X \times Y$. Thus Φ is injective. Compatibility with the group law follows from compatibility of projections with concatenation. □

14.4 Relative homotopy groups and the long exact sequence

Let (X, A, x_0) be a pair with $x_0 \in A \subseteq X$. For $n \geq 1$, define the subspace

$$J^{n-1} = (\partial I^{n-1} \times I) \ \cup \ (I^{n-1} \times \{0\}) \ \subseteq \ \partial I^n.$$

Thus ∂I^n is the union of J^{n-1} and the top face $I^{n-1} \times \{1\}$, and their intersection is $\partial I^{n-1} \times \{1\}$.

Definition 14.9. Relative homotopy groups

For $n \geq 1$, define $\pi_n(X, A, x_0)$ to be the set of homotopy classes of maps $f\colon I^n \to X$ such that

$$f(J^{n-1}) \subseteq \{x_0\}, \qquad f(I^{n-1} \times \{1\}) \subseteq A,$$

where homotopies $H\colon I^n \times I \to X$ are required to satisfy

$$H(J^{n-1} \times I) \subseteq \{x_0\}, \qquad H(I^{n-1} \times \{1\} \times I) \subseteq A.$$

Remark. When $A = \{x_0\}$, this reduces to $\pi_n(X, x_0)$. For $n \geq 2$, $\pi_n(X, A, x_0)$ is a group under concatenation in the last coordinate. For $n = 1$, it is a pointed set.

There are natural maps

$$\pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0),$$

where i_* is induced by inclusion $A \hookrightarrow X$, and j_* regards a map with all of ∂I^n sent to x_0 as a relative map.

Definition 14.10. Boundary map

For $n \geq 2$, define

$$\partial\colon \pi_n(X, A, x_0) \to \pi_{n-1}(A, x_0)$$

by restriction to the top face. If $[f] \in \pi_n(X, A, x_0)$, set

$$\partial([f]) = \left[\, f|_{I^{n-1} \times \{1\}} \,\right].$$

This is well defined since $f(\partial I^{n-1} \times \{1\}) \subseteq f(J^{n-1}) = \{x_0\}$.

To prove exactness, we will use the standard extension property of CW pairs.

Definition 14.11. Homotopy extension property

A pair (X, A) has the **homotopy extension property** if for every space Y and every diagram

$$\begin{tikzcd} A \arrow[r, "F_0"] \arrow[d, hook] & Y^I \\ X \arrow[ur, dashed, "\widetilde{F}_0"'] & \end{tikzcd}$$

in which F_0 is a homotopy on A, there exists a homotopy $\widetilde{F}\colon X \times I \to Y$ extending it. Equivalently, given $f\colon X \to Y$ and a homotopy $H\colon A \times I \to Y$ with $H(-, 0) = f|_A$, there exists $\widetilde{H}\colon X \times I \to Y$ with $\widetilde{H}(-, 0) = f$ and $\widetilde{H}|_{A \times I} = H$.

Proposition 14.12

If (X, A) is a CW pair, then (X, A) has the homotopy extension property.

Proof. It suffices to show that $X \times \{0\} \cup A \times I$ is a deformation retract of $X \times I$. Write X as the union of its skeleta $X^{(n)}$. For each n-cell, the product $D^n \times I$ deformation retracts onto $D^n \times \{0\} \cup \partial D^n \times I$ by a radial deformation in $D^n \times \mathbb{R}$. Attach these deformations along cells, inductively over n, to obtain deformation retractions of $X^{(n)} \times I$ onto $X^{(n)} \times \{0\} \cup (X^{(n-1)} \cup A) \times I$. Concatenating these homotopies over n yields a deformation retraction of $X \times I$ onto $X \times \{0\} \cup A \times I$. The weak topology of CW complexes ensures continuity because the construction is continuous on each skeleton. $\square$

Theorem 14.13. Long exact sequence in homotopy for a pair

et (X, A, x_0) be a CW pair with $x_0 \in A$. Then there is a long exact sequence

$$\cdots \to \pi_n(A, x_0) \xrightarrow{i_*} \pi_n(X, x_0) \xrightarrow{j_*} \pi_n(X, A, x_0) \xrightarrow{\partial} \pi_{n-1}(A, x_0) \to \cdots$$

continuing down to

$$\pi_1(X, x_0) \xrightarrow{j_*} \pi_1(X, A, x_0) \xrightarrow{\partial} \pi_0(A) \xrightarrow{i_*} \pi_0(X).$$

For $n \geq 2$, all groups in sight are groups and all maps are homomorphisms.

Proof. We first check that consecutive maps compose to the trivial map.
(i) $j_* \circ i_* = 0$. Let $[f] \in \pi_n(A, x_0)$ be represented by $f\colon I^n \to A$ with $f(\partial I^n) \subseteq \{x_0\}$. Viewed as a map into X, it represents $i_*[f] \in \pi_n(X, x_0)$. Viewed further as a relative map, it represents $j_* i_*[f] \in \pi_n(X, A, x_0)$. Since I^n is contractible, there is a homotopy $H\colon I^n \times I \to A$ from f to the constant map c_{x_0}. This homotopy satisfies $H(I^{n-1} \times \{1\} \times I) \subseteq A$ and $H(J^{n-1} \times I) \subseteq \{x_0\}$ because $f(\partial I^n) = x_0$ and we may take H to contract the domain to a boundary point. Hence $j_* i_*[f] = 0$ in $\pi_n(X, A, x_0)$.
(ii) $\partial \circ j_* = 0$. If $[f] \in \pi_n(X, x_0)$, then $j_*[f]$ is represented by the same f. Its restriction to $I^{n-1} \times \{1\}$ is constant at x_0, so $\partial(j_*[f]) = 0$.
(iii) $i_* \circ \partial = 0$. Let $[f] \in \pi_n(X, A, x_0)$. Then $\partial([f])$ is represented by $f|_{I^{n-1} \times \{1\}}\colon I^{n-1} \to A$. Composing with inclusion gives a map into X which extends to I^n by f itself, hence represents the trivial element in $\pi_{n-1}(X, x_0)$. Therefore $i_* \partial([f]) = 0$.
Now we prove exactness.
Exactness at $\pi_n(X, x_0)$. We show $\ker(j_*) = \operatorname{im}(i_*)$. If $[f] \in \operatorname{im}(i_*)$, then $j_*[f] = 0$ by (i), so $\operatorname{im}(i_*) \subseteq \ker(j_*)$. Conversely, suppose $[f] \in \pi_n(X, x_0)$ and $j_*[f] = 0$. By definition of the relative group, there is a homotopy $H\colon I^n \times I \to X$ from f to the constant map c_{x_0} such that

$$H(J^{n-1} \times I) \subseteq \{x_0\}, \qquad H(I^{n-1} \times \{1\} \times I) \subseteq A.$$

Restricting H to $\partial I^n \times I$ gives a homotopy in A from the constant map on ∂I^n to itself. Using the homotopy extension property for the CW pair $(I^n, \partial I^n)$, we may adjust H through a homotopy of homotopies so that throughout the deformation we have

$$H(\partial I^n \times I) \subseteq A.$$

Now define $g\colon I^n \to A$ by $g(u) = H(u, 1)$. Then $g(\partial I^n) \subseteq A$ and in fact $g(\partial I^n) = x_0$ because $H(J^{n-1} \times I) \subseteq \{x_0\}$ and ∂I^n deformation retracts onto $J^{n-1} \cup (\partial I^{n-1} \times \{1\})$. The original homotopy shows that f is based homotopic to g rel ∂I^n as maps into X. Thus $[f] = i_*[g]$, so $[f] \in \operatorname{im}(i_*)$.
Exactness at $\pi_n(X, A, x_0)$. We show $\ker(\partial) = \operatorname{im}(j_*)$. If $[f] \in \operatorname{im}(j_*)$, then $\partial([f]) = 0$ by (ii), so $\operatorname{im}(j_*) \subseteq \ker(\partial)$. Conversely, let $[f] \in \pi_n(X, A, x_0)$ with $\partial([f]) = 0$. Then the restriction $f|_{I^{n-1} \times \{1\}}$ represents 0 in $\pi_{n-1}(A, x_0)$, so there is a homotopy

$$K\colon (I^{n-1} \times \{1\}) \times I \to A$$

from $f|_{I^{n-1} \times \{1\}}$ to the constant map at x_0, rel $\partial I^{n-1} \times \{1\}$. By the homotopy extension property for the CW pair $(I^n, I^{n-1} \times \{1\})$, we can extend this to a homotopy $\widetilde{K}\colon I^n \times I \to X$ such that $\widetilde{K}(-, 0) = f$ and $\widetilde{K}|_{(I^{n-1} \times \{1\}) \times I} = K$, and such that $\widetilde{K}(J^{n-1} \times I) \subseteq \{x_0\}$ since $f(J^{n-1}) = \{x_0\}$ and K fixes the boundary. Let $f' = \widetilde{K}(-, 1)$. Then $f'(I^{n-1} \times \{1\}) = \{x_0\} \subseteq A$, and still $f'(J^{n-1}) = \{x_0\}$. So f' sends all of ∂I^n to x_0, hence represents an element of $\pi_n(X, x_0)$. Moreover f is homotopic to f' through a homotopy defining equality in $\pi_n(X, A, x_0)$. Therefore $[f] = j_*([f'])$, proving $\ker(\partial) \subseteq \operatorname{im}(j_*)$.
Exactness at $\pi_{n-1}(A, x_0)$. We show $\ker(i_*) = \operatorname{im}(\partial)$. If $[f] \in \pi_n(X, A, x_0)$, then $i_* \partial([f]) = 0$ by (iii), so $\operatorname{im}(\partial) \subseteq \ker(i_*)$. Conversely, let $[\alpha] \in \pi_{n-1}(A, x_0)$ with $i_*[\alpha] = 0$ in $\pi_{n-1}(X, x_0)$. Choose a representative $\alpha\colon I^{n-1} \to A$ with $\alpha(\partial I^{n-1}) \subseteq \{x_0\}$. The condition $i_*[\alpha] = 0$ means there is a based homotopy rel ∂I^{n-1} in X from α to c_{x_0}. Regard this homotopy as a map

$$H\colon I^{n-1} \times I \to X$$

with $H(I^{n-1} \times \{0\}) \subseteq A$, $H(\partial I^{n-1} \times I) \subseteq \{x_0\}$, and $H(I^{n-1} \times \{1\}) = \{x_0\}$. Extend H to a map $f\colon I^n \to X$ by letting f be constant x_0 on the remaining boundary faces, that is, on J^{n-1}. Then $f(J^{n-1}) \subseteq \{x_0\}$ and $f(I^{n-1} \times \{1\}) \subseteq A$. So f represents a class in $\pi_n(X, A, x_0)$, and by construction $\partial([f]) = [\alpha]$. Thus $[\alpha] \in \operatorname{im}(\partial)$. This proves exactness for $n \geq 2$. The low-dimensional tail with π_1 and π_0 is obtained by the same restriction and extension arguments, with sets instead of groups. $\square$

15 Serre Fibrations and Fiber Sequences

15.1 Serre fibrations and lifting

Definition 15.1. Serre fibration

A map $p\colon E \to B$ is a **Serre fibration** if it has the homotopy lifting property for cubes. This means that for every $n \geq 0$ and every commutative square

$$\begin{tikzcd} I^n \times \{0\} \arrow[r, "\widetilde{g}_0"] \arrow[d, hook] & E \arrow[d, "p"] \\ I^n \times I \arrow[r, "g"] & B \end{tikzcd}$$

there exists a lift $\widetilde{g}\colon I^n \times I \to E$ with $\widetilde{g}|_{I^n \times \{0\}} = \widetilde{g}_0$ and $p \circ \widetilde{g} = g$.

Remark. If p is a fiber bundle, then p is a Serre fibration. Local trivializations reduce the lifting problem to the product projection $U \times F \to U$.

Proposition 15.2. Pullbacks preserve Serre fibrations

Let $p\colon E \to B$ be a Serre fibration and let $f\colon A \to B$ be any map. Form the pullback

$$f^*E = \{(a, e) \in A \times E \mid f(a) = p(e)\}$$

with projection $\pi\colon f^*E \to A$, $\pi(a, e) = a$. Then π is a Serre fibration.

Proof. Consider a lifting problem for π with data

$$\begin{tikzcd} I^n \times \{0\} \arrow[r, "\widetilde{g}_0"] \arrow[d, hook] & f^*E \arrow[d, "\pi"] \\ I^n \times I \arrow[r, "g"] & A. \end{tikzcd}$$

Write $\widetilde{g}_0 = (g_0, \widetilde{h}_0)$ under the inclusion $f^*E \subset A \times E$. Then $\pi \circ \widetilde{g}_0 = g_0 = g|_{I^n \times \{0\}}$ and the defining condition for the pullback says

$$p \circ \widetilde{h}_0 = f \circ g_0.$$

Compose g with f to obtain a map $f \circ g\colon I^n \times I \to B$. The square

$$\begin{tikzcd} I^n \times \{0\} \arrow[r, "\widetilde{h}_0"] \arrow[d, hook] & E \arrow[d, "p"] \\ I^n \times I \arrow[r, "f \circ g"] & B \end{tikzcd}$$

commutes, so the Serre fibration property produces $\widetilde{h}\colon I^n \times I \to E$ lifting $f \circ g$ and extending $\widetilde{h}_0$. Then $\widetilde{g} = (g, \widetilde{h})$ lands in f^*E and solves the original problem. $\square$

15.2 Fibers and fiber sequences

Fix a basepoint $b_0 \in B$ and a point $e_0 \in E$ with $p(e_0) = b_0$. Write $F = p^{-1}(b_0)$ for the fiber, and let $\iota\colon F \hookrightarrow E$ be the inclusion.

Definition 15.3. Fiber sequence

A **fiber sequence** is a sequence of based maps

$$(F, e_0) \xrightarrow{\iota} (E, e_0) \xrightarrow{p} (B, b_0)$$

where p is a Serre fibration and $F = p^{-1}(b_0)$.

The map p determines an identification of certain relative homotopy groups. This is the point where the lifting property turns into algebra.

Proposition 15.4

Let $p\colon (E, e_0) \to (B, b_0)$ be a Serre fibration and let $F = p^{-1}(b_0)$. For each $n \geq 1$, the induced map

$$p_*\colon \pi_n(E, F, e_0) \longrightarrow \pi_n(B, b_0)$$

is an isomorphism.

Proof. Surjectivity. Let $[\alpha] \in \pi_n(B, b_0)$. Choose a representative $\alpha\colon I^n \to B$ with $\alpha(\partial I^n) = b_0$. View α as a homotopy $\alpha_t\colon I^{n-1} \to B$ by slicing in the last coordinate, $\alpha_t(x) = \alpha(x, t)$. Then α_0 and α_1 are constant at b_0, and each α_t is constant on ∂I^{n-1}. Apply the lifting property to the square

$$\begin{tikzcd} I^{n-1} \times \{0\} \arrow[r, "c_{e_0}"] \arrow[d, hook] & E \arrow[d, "p"] \\ I^{n-1} \times I \arrow[r, "\alpha"] & B \end{tikzcd}$$

where c_{e_0} is the constant map at e_0. This yields a lift $\widetilde{\alpha}\colon I^{n-1} \times I \to E$ with $p \circ \widetilde{\alpha} = \alpha$ and $\widetilde{\alpha}(-, 0) = c_{e_0}$. Define $\beta\colon I^n \to E$ by $\beta(x, t) = \widetilde{\alpha}(x, t)$. Then $\beta(\partial I^n) \subseteq F$ and $\beta(I^{n-1} \times \{0\}) = \{e_0\}$, so β represents a class in $\pi_n(E, F, e_0)$. By construction $p_*[\beta] = [\alpha]$.
Injectivity. Suppose $[\beta_0], [\beta_1] \in \pi_n(E, F, e_0)$ satisfy $p_*[\beta_0] = p_*[\beta_1]$ in $\pi_n(B, b_0)$. Then there is a homotopy $H\colon I^n \times I \to B$ rel ∂I^n from $p \circ \beta_0$ to $p \circ \beta_1$. Restrict H to $I^{n-1} \times I \times I$ by slicing the last coordinate of I^n as above. The boundary conditions ensure that H takes $\partial I^{n-1} \times I \times I$ and $I^{n-1} \times \{0, 1\} \times I$ to b_0. Apply the lifting property to lift this homotopy starting at β_0. The resulting lifted homotopy shows $\beta_0 \simeq \beta_1$ rel the appropriate boundary conditions in (E, F). Hence $[\beta_0] = [\beta_1]$ in $\pi_n(E, F, e_0)$. □

Theorem 15.5. Long exact sequence of a Serre fibration

Let $F \to E \xrightarrow{p} B$ be a fiber sequence. Then there is a long exact sequence

$$\cdots \to \pi_n(F, e_0) \xrightarrow{\iota_*} \pi_n(E, e_0) \xrightarrow{p_*} \pi_n(B, b_0) \xrightarrow{\partial} \pi_{n-1}(F, e_0) \to \cdots$$

continuing down to

$$\pi_1(E, e_0) \xrightarrow{p_*} \pi_1(B, b_0) \xrightarrow{\partial} \pi_0(F) \xrightarrow{\iota_*} \pi_0(E) \xrightarrow{p_*} \pi_0(B).$$

For $n \geq 2$ all groups are groups and all maps are homomorphisms.

Proof. Apply the long exact sequence of the pair (E, F). It gives exactness of

$$\cdots \to \pi_n(F, e_0) \xrightarrow{\iota_*} \pi_n(E, e_0) \xrightarrow{j_*} \pi_n(E, F, e_0) \xrightarrow{\delta} \pi_{n-1}(F, e_0) \to \cdots$$

where j_* is the map from the absolute to the relative group. Compose j_* with the isomorphism $p_*\colon \pi_n(E, F, e_0) \to \pi_n(B, b_0)$ from Proposition 15.4. This produces the map $\pi_n(E, e_0) \to \pi_n(B, b_0)$ in the displayed sequence. Define ∂ to be the composite

$$\pi_n(B, b_0) \xrightarrow[\cong]{(p_*)^{-1}} \pi_n(E, F, e_0) \xrightarrow{\delta} \pi_{n-1}(F, e_0).$$

Exactness at each term follows from exactness of the pair sequence and the fact that $(p_*)^{-1}$ is an isomorphism. The low-dimensional tail is treated in the same way, with pointed sets at the π_1 and π_0 end. □

15.3 The path fibration

Let (X, x_0) be a based space. Define

$$PX = \{\lambda\colon I \to X \mid \lambda(0) = x_0\}, \qquad \mathrm{ev}_1\colon PX \to X,\ \mathrm{ev}_1(\lambda) = \lambda(1).$$

Proposition 15.6

The evaluation map $\mathrm{ev}_1\colon PX \to X$ is a Serre fibration. Its fiber over x_0 is ΩX.

Proof. Let $g\colon I^n \times I \to X$ and $\widetilde{g}_0\colon I^n \times \{0\} \to PX$ form a commutative square for ev_1. Write $\widetilde{g}_0(u) = \lambda_u$ so that $\lambda_u(0) = x_0$ and $\lambda_u(1) = g(u, 0)$. Define $\widetilde{g}\colon I^n \times I \to PX$ by letting $\widetilde{g}(u, t)$ be the path obtained by first following λ_u and then following the path $s \mapsto g(u, ts)$. A linear reparametrization produces a continuous path on $[0, 1]$ with initial point x_0 and endpoint $g(u, t)$. Then $\mathrm{ev}_1(\widetilde{g}(u, t)) = g(u, t)$ and $\widetilde{g}(u, 0) = \widetilde{g}_0(u)$, so $\widetilde{g}$ is a lift.

The fiber over x_0 consists of those λ with $\lambda(1) = x_0$, which is exactly ΩX. $\square$

Proposition 15.7

The space PX is contractible.

Proof. Define $H\colon PX \times I \to PX$ by

$$H(\lambda, t)(s) = \lambda(ts).$$

Then $H(\lambda, 0)$ is the constant path at x_0 and $H(\lambda, 1) = \lambda$. $\square$

Corollary 15.8. For $n \geq 1$ there are isomorphisms

$$\pi_n(X, x_0) \cong \pi_{n-1}(\Omega X, c_{x_0}).$$

Proof. Apply Theorem 15.5 to the fiber sequence $\Omega X \to PX \to X$. Since PX is contractible, $\pi_k(PX) = 0$ for all $k \geq 1$, so the connecting maps give the displayed isomorphisms. $\square$

15.4 The Hopf fibration

Write

$$S^3 = \{(z_0, z_1) \in \mathbb{C}^2 \mid |z_0|^2 + |z_1|^2 = 1\}.$$

Define $p\colon S^3 \to \mathbb{C}P^1$ by $p(z_0, z_1) = [z_0 : z_1]$. The circle S^1 acts on S^3 by scalar multiplication $\lambda \cdot (z_0, z_1) = (\lambda z_0, \lambda z_1)$. Two points map to the same point of $\mathbb{C}P^1$ exactly when they lie in the same S^1-orbit, so p has fiber S^1.

Proposition 15.9

The map $p\colon S^3 \to \mathbb{C}P^1$ is a fiber bundle with fiber S^1. Using $\mathbb{C}P^1 \cong S^2$, this gives a fiber sequence

$$S^1 \longrightarrow S^3 \xrightarrow{p} S^2.$$

Proof. Consider the standard open sets $U_0 = \{[z_0 : z_1] \mid z_0 \neq 0\}$ and $U_1 = \{[z_0 : z_1] \mid z_1 \neq 0\}$ covering $\mathbb{C}P^1$. On U_0, every line has a representative of the form $(1, w)$ with $w = z_1/z_0$. Define a local section $\sigma_0\colon U_0 \to S^3$ by

$$\sigma_0([1 : w]) = \frac{(1, w)}{\sqrt{1 + |w|^2}}.$$

Then every point in $p^{-1}(U_0)$ can be uniquely written as $\lambda \cdot \sigma_0([1 : w])$ with $\lambda \in S^1$. This gives a homeomorphism $U_0 \times S^1 \to p^{-1}(U_0)$ commuting with projections. The same construction on U_1 provides a trivialization over U_1. Hence p is a fiber bundle with fiber S^1.

The identification $\mathbb{C}P^1 \cong S^2$ is standard, for instance by stereographic projection after viewing $\mathbb{C}P^1$ as the Riemann sphere. $\square$

Proposition 15.10

The long exact sequence for $S^1 \to S^3 \to S^2$ yields

$$\pi_2(S^2) \cong \pi_1(S^1) \cong \mathbb{Z}, \qquad \pi_n(S^3) \cong \pi_n(S^2) \ \text{ for all } n \geq 3.$$

In particular, $\pi_3(S^2) \cong \mathbb{Z}$.

Proof. Since $\pi_k(S^1) = 0$ for $k \geq 2$, the segment of the long exact sequence for $n \geq 3$ reads

$$0 = \pi_{n-1}(S^1) \to \pi_n(S^3) \to \pi_n(S^2) \to \pi_{n-2}(S^1) = 0,$$

so $\pi_n(S^3) \to \pi_n(S^2)$ is an isomorphism.

For $n = 2$ the exact sequence gives

$$\pi_2(S^3) \to \pi_2(S^2) \xrightarrow{\partial} \pi_1(S^1) \to \pi_1(S^3).$$

Here $\pi_2(S^3) = 0$ and $\pi_1(S^3) = 0$, so ∂ is an isomorphism. Since $\pi_1(S^1) \cong \mathbb{Z}$, it follows that $\pi_2(S^2) \cong \mathbb{Z}$.

For $n = 3$, exactness gives

$$\pi_3(S^3) \to \pi_3(S^2) \to \pi_2(S^1) = 0,$$

so $\pi_3(S^3) \to \pi_3(S^2)$ is surjective. The previous term is $\pi_3(S^1) = 0$, hence the map is injective as well. Since $\pi_3(S^3) \cong \mathbb{Z}$, this gives $\pi_3(S^2) \cong \mathbb{Z}$. $\square$

16 Hurewicz and Whitehead

16.1 Connectivity and the Hurewicz map

Definition 16.1. Connectivity

Let X be a path-connected space and let $n \geq 0$. The space X is **n-connected** if $\pi_i(X, x_0) = 0$ for all $0 \leq i \leq n$ (and hence for every choice of basepoint x_0). Equivalently, X is 0-connected if it is path-connected, and 1-connected if it is simply connected.

Definition 16.2. The Hurewicz homomorphism

Let $n \geq 1$ and let (X, x_0) be a based space. The **Hurewicz homomorphism**

$$h\colon \pi_n(X, x_0) \longrightarrow H_n(X; \mathbb{Z})$$

is defined as follows. A class $[f] \in \pi_n(X, x_0)$ is represented by a based map $f\colon S^n \to X$. Let $[S^n] \in H_n(S^n; \mathbb{Z})$ be a chosen generator. Set $h([f]) = f_*([S^n])$.

Remark. Changing the choice of generator $[S^n]$ changes h by an overall sign. In statements below, the sign is irrelevant.

16.2 The Hurewicz theorem

Theorem 16.3. Hurewicz

Let $n \geq 2$ and let X be $(n-1)$-connected. Then $H_i(X; \mathbb{Z}) = 0$ for $i < n$, and the Hurewicz homomorphism

$$h\colon \pi_n(X, x_0) \longrightarrow H_n(X; \mathbb{Z})$$

is an isomorphism.

For $n = 1$ and path-connected X, the map $\pi_1(X, x_0) \to H_1(X; \mathbb{Z})$ induces an isomorphism

$$\pi_1(X, x_0)^{\mathrm{ab}} \;\cong\; H_1(X; \mathbb{Z}).$$

Proof. The case $n = 1$ is standard.

Assume $n \geq 2$. Replace X by a homotopy equivalent CW complex. Since X is $(n-1)$-connected, there is a CW model with one 0-cell and no cells in dimensions $1, \ldots, n-1$. Collapsing $X^{(n-1)}$ to a point does not change π_n or H_n, so we may assume $X^{(n-1)}$ is a point.

Cells of dimension $\geq n+2$ do not affect π_n or H_n, so we further replace X by its $(n+1)$-skeleton. Thus X is obtained from its n-skeleton by attaching $(n+1)$-cells. Because there are no cells below dimension n, the n-skeleton is a wedge of n-spheres,

$$X^{(n)} \;\cong\; \bigvee_{\alpha} S^n_\alpha.$$

Let the $(n+1)$-cells be e^{n+1}_β, with attaching maps

$$\varphi_\beta\colon S^n \longrightarrow X^{(n)}.$$

Consider the long exact sequence of the pair $(X, X^{(n)})$. The relative group $\pi_{n+1}(X, X^{(n)}, x_0)$ is freely generated by the $(n+1)$-cells, hence is isomorphic to $\bigoplus_\beta \mathbb{Z}$. Likewise $\pi_n(X^{(n)}, x_0) \cong \bigoplus_\alpha \mathbb{Z}$, with the αth copy of $\mathbb{Z}$ identified with $\pi_n(S^n_\alpha) \cong \mathbb{Z}$ via degree. Exactness gives

$$\pi_{n+1}(X, X^{(n)}, x_0) \xrightarrow{\partial} \pi_n(X^{(n)}, x_0) \longrightarrow \pi_n(X, x_0) \longrightarrow 0,$$

so

$$\pi_n(X, x_0) \;\cong\; \operatorname{coker}(\partial).$$

Under the identifications above, the homomorphism $\partial\colon \bigoplus_\beta \mathbb{Z} \to \bigoplus_\alpha \mathbb{Z}$ is determined by the degrees of the composites

$$S^n \xrightarrow{\varphi_\beta} X^{(n)} \longrightarrow S^n_\alpha,$$

where the last map collapses all wedge summands except S^n_α.

Now compare with homology. The cellular chain groups satisfy $C_{n+1}^{\mathrm{cell}}(X) \cong \bigoplus_\beta \mathbb{Z}$ and $C_n^{\mathrm{cell}}(X) \cong \bigoplus_\alpha \mathbb{Z}$, and the cellular differential

$$d_{n+1}\colon C_{n+1}^{\mathrm{cell}}(X) \longrightarrow C_n^{\mathrm{cell}}(X)$$

is given by the same degree matrix coming from the attaching maps φ_β. Because there are no cells in dimensions $< n$, one has $H_i(X;\mathbb{Z}) = 0$ for $i < n$, and

$$H_n(X;\mathbb{Z}) \;\cong\; \mathrm{coker}(d_{n+1}).$$

It remains to compare ∂ with d_{n+1}. Under the identification $\pi_n(S^n) \cong \mathbb{Z}$ by degree, the Hurewicz map $\pi_n(S^n) \to H_n(S^n)$ sends 1 to a generator. Naturality then shows that, for each attaching map, the induced map on π_n and the induced map on H_n record the same degrees on wedge summands. Hence ∂ and d_{n+1} agree under the identifications, so their cokernels are canonically isomorphic. This identifies $\pi_n(X, x_0)$ with $H_n(X;\mathbb{Z})$ and the identification is induced by the Hurewicz map. □

The theorem turns the first nontrivial homotopy group of a highly connected space into a homology group. That often reduces questions about maps to questions about degrees.

Example. For $n \geq 2$, the sphere S^n is $(n-1)$-connected, and $H_n(S^n;\mathbb{Z}) \cong \mathbb{Z}$. Theorem 16.3 gives $\pi_n(S^n) \cong \mathbb{Z}$. A based map $f\colon S^n \to S^n$ represents $k \in \pi_n(S^n)$ exactly when $\deg(f) = k$.

16.3 A relative form

Lemma 16.4 Let (X, A) be a CW pair and let $n \geq 2$. Then the quotient map $X \to X/A$ induces isomorphisms

$$\pi_i(X, A, x_0) \;\cong\; \pi_i(X/A, *) \quad \text{for } i \leq n, \qquad H_i(X, A;\mathbb{Z}) \;\cong\; \widetilde{H}_i(X/A;\mathbb{Z}) \quad \text{for all } i.$$

Proof. The homology statement is standard. For homotopy, a map $(D^i, S^{i-1}) \to (X, A)$ with $i \leq n$ can be pushed off the lower skeleta and arranged to meet A only on S^{i-1}. The quotient then identifies based homotopy classes with based maps into X/A. □

Theorem 16.5. Relative Hurewicz

Let $n \geq 2$ and let (X, A) be a CW pair that is $(n-1)$-connected, meaning that $\pi_i(X, A, x_0) = 0$ for $i \leq n-1$. Then $H_i(X, A;\mathbb{Z}) = 0$ for $i < n$, and the relative Hurewicz homomorphism

$$h\colon \pi_n(X, A, x_0) \longrightarrow H_n(X, A;\mathbb{Z})$$

is an isomorphism.

Proof. By Lemma 16.4, the hypotheses imply that X/A is $(n-1)$-connected. Apply Theorem 16.3 to X/A. Translate back using Lemma 16.4. □

Example. Let X be simply connected and built from a point by attaching n-cells and $(n+1)$-cells, with $n \geq 2$. Then X is $(n-1)$-connected. Theorem 16.3 says $\pi_n(X) \cong H_n(X)$. In this range, the attaching maps of the $(n+1)$-cells determine $\pi_n(X)$ and $H_n(X)$ through the same degree matrix, exactly as in the proof of Theorem 16.3.

16.4 The Whitehead theorem

Definition 16.6. Weak homotopy equivalence

A map $f\colon X \to Y$ is a **weak homotopy equivalence** if it induces a bijection on path components and, for every basepoint $x_0 \in X$, it induces isomorphisms

$$f_*\colon \pi_n(X, x_0) \longrightarrow \pi_n(Y, f(x_0)) \qquad \text{for all } n \geq 1.$$

Theorem 16.7. Whitehead

Let $f\colon X \to Y$ be a weak homotopy equivalence between CW complexes. Then f is a homotopy equivalence.

Proof. Replace f by a cellular map using the cellular approximation theorem.
A map $g\colon Y \to X$ is constructed inductively on skeleta so that $f \circ g$ and $g \circ f$ become homotopic to the identities. Assume g is defined on $Y^{(k-1)}$ and that $(f \circ g)|_{Y^{(k-1)}} \simeq \mathrm{id}$. Consider a k-cell e^k of Y with attaching map $\psi\colon S^{k-1} \to Y^{(k-1)}$. The map $g \circ \psi\colon S^{k-1} \to X$ represents a class in $\pi_{k-1}(X)$. Since $f_*\colon \pi_{k-1}(X) \to \pi_{k-1}(Y)$ is surjective, there exists a class $[\eta] \in \pi_{k-1}(X)$ with

$$f_*[\eta] = [(f \circ g \circ \psi)] \in \pi_{k-1}(Y).$$

Because $(f \circ g)|_{Y^{(k-1)}} \simeq \mathrm{id}$, the class of $f \circ g \circ \psi$ equals the class of ψ in $\pi_{k-1}(Y)$. Thus $[\psi] = f_*[\eta]$. Choose a representative $\eta\colon S^{k-1} \to X$. Then $f \circ \eta \simeq \psi$ in $Y^{(k-1)}$.
Attach the k-cell by extending g over e^k so that on ∂e^k it agrees with η. Such an extension exists because e^k is a disk. Doing this for every k-cell defines g on $Y^{(k)}$.
Now compare $f \circ g$ with id on $Y^{(k)}$. On $Y^{(k-1)}$ they are homotopic by the inductive hypothesis. On each k-cell, the boundary maps $S^{k-1} \to Y^{(k-1)}$ are homotopic. By the homotopy extension property for CW pairs, these homotopies extend across $Y^{(k)}$. This yields $(f \circ g)|_{Y^{(k)}} \simeq \mathrm{id}_{Y^{(k)}}$.
Passing to the colimit over k gives $f \circ g \simeq \mathrm{id}_Y$. A symmetric induction, using injectivity and surjectivity of f_* on the relevant homotopy groups, gives $g \circ f \simeq \mathrm{id}_X$. Hence f is a homotopy equivalence with homotopy inverse g. □

Corollary 16.8. Let $f\colon X \to Y$ be a map between simply connected CW complexes. If

$$f_*\colon H_n(X;\mathbb{Z}) \longrightarrow H_n(Y;\mathbb{Z})$$

is an isomorphism for all n, then f is a homotopy equivalence.

Proof. Replace f by the inclusion $X \hookrightarrow M_f$ into its mapping cylinder and then by the inclusion into Y. Thus assume f is an inclusion $X \subseteq Y$ of CW complexes. The hypothesis implies $H_n(Y, X; \mathbb{Z}) = 0$ for all n by the long exact sequence of the pair.
Since X and Y are simply connected, the pair (Y, X) is 1-connected. If $\pi_n(Y, X) \neq 0$ for some $n \geq 2$, choose the smallest such n. Then (Y, X) is $(n-1)$-connected, so Theorem 16.5 gives an isomorphism

$$\pi_n(Y, X) \;\cong\; H_n(Y, X; \mathbb{Z}).$$

The right-hand side is zero, a contradiction. Therefore $\pi_n(Y, X) = 0$ for all $n \geq 2$, and also $\pi_1(Y, X) = 0$ since X and Y are simply connected. Thus $X \hookrightarrow Y$ is a weak homotopy equivalence. Apply Theorem 16.7. □

17 The Leray–Serre Spectral Sequence I

Let $F \hookrightarrow E \xrightarrow{\pi} B$ be a Serre fibration with B path-connected and F connected. Fix a basepoint $b_0 \in B$ and a point $e_0 \in E$ with $\pi(e_0) = b_0$. Write $F = \pi^{-1}(b_0)$ and $i\colon F \hookrightarrow E$ for the inclusion.

17.1 Local coefficients from monodromy

For each $q \geq 0$, the fibration determines a local system on B with fiber $H^q(F; R)$, where R is a commutative ring with unit. Concretely, a loop $\gamma\colon I \to B$ based at b_0 admits lifts starting at points of F. Up to homotopy, this produces a self-map of the fiber F, hence an automorphism of $H^q(F; R)$. In this way, $\pi_1(B, b_0)$ acts on $H^q(F; R)$. We write $H^p(B; H^q(F; R))$ for cohomology of B with local coefficients in this $\pi_1(B, b_0)$-module.

Remark. If B is simply connected, then the action of $\pi_1(B, b_0)$ is trivial. In this case the local system is constant and $H^p(B; H^q(F; R))$ is ordinary cohomology with coefficients in the R-module $H^q(F; R)$. For many computations in this course, that is the relevant case.

17.2 Statement of the spectral sequence

Theorem 17.1. The cohomology Leray–Serre spectral sequence

Let R be a commutative ring with unit. For a Serre fibration $F \hookrightarrow E \xrightarrow{\pi} B$ with B path-connected and F connected, there exists a first quadrant cohomological spectral sequence

$$\{E_r^{p,q}, d_r\}_{r \geq 2}$$

with the following properties.

(1) The E_2-page is
$$E_2^{p,q} \cong H^p(B; H^q(F; R)).$$
(2) The differential has bidegree $(r, 1-r)$
$$d_r \colon E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}.$$
(3) The spectral sequence converges to $H^*(E; R)$ in the following sense. For each $n \geq 0$ there is a decreasing filtration
$$H^n(E; R) = F^0 H^n(E; R) \supseteq F^1 H^n(E; R) \supseteq \cdots \supseteq F^{n+1} H^n(E; R) = 0$$
such that
$$E_\infty^{p,q} \cong F^p H^{p+q}(E; R)/F^{p+1} H^{p+q}(E; R).$$
(4) The construction is natural with respect to maps of fibrations.

If the cohomology groups are taken with their cup product, then the spectral sequence is multiplicative and the multiplication on E_r is compatible with the differential via the Leibniz rule.

Remark. The filtration in Item (3) can be obtained from the skeletal filtration of the base. Choose a CW structure on B, and set $B^{(p)}$ to be its p-skeleton. Then one may take
$$F^p H^n(E; R) = \ker\Big(H^n(E; R) \longrightarrow H^n(\pi^{-1}(B^{(p-1)}); R)\Big).$$
This makes the edge maps in the next subsection transparent.

17.3 Filtration and edge homomorphisms

Along the bottom row, $E_2^{p,0} \cong H^p(B; H^0(F; R))$. Since F is connected, $H^0(F; R) \cong R$ and so $E_2^{p,0} \cong H^p(B; R)$. Along the left column, $E_2^{0,q} \cong H^0(B; H^q(F; R))$. If the local system is constant, then $E_2^{0,q} \cong H^q(F; R)$.

Proposition 17.2. The edge maps

Assume the local system $H^q(F; R)$ on B is constant. Then the composites along the bottom row
$$H^p(B; R) = E_2^{p,0} \longrightarrow E_\infty^{p,0} \hookrightarrow H^p(E; R)$$
agree with $\pi^* \colon H^p(B; R) \to H^p(E; R)$.

The composites along the left column
$$H^q(E; R) \twoheadrightarrow E_\infty^{0,q} \hookrightarrow E_2^{0,q} = H^q(F; R)$$
agree with $i^* \colon H^q(E; R) \to H^q(F; R)$.

Proof. Use the filtration described above. If $\alpha \in H^p(B; R)$, then $\pi^*\alpha$ vanishes on $\pi^{-1}(B^{(p-1)})$, hence lies in $F^p H^p(E; R)$. Its class in the quotient $F^p H^p(E; R)/F^{p+1} H^p(E; R) \cong E_\infty^{p,0}$ is the image of α along the bottom row.

For the second statement, the restriction map i^* factors through the quotient by $F^1 H^q(E; R)$ because classes in F^1 restrict trivially to the fiber. The quotient $H^q(E; R)/F^1 H^q(E; R)$ identifies with $E_\infty^{0,q}$, and the inclusion $E_\infty^{0,q} \hookrightarrow E_2^{0,q}$ matches this quotient with $H^q(F; R)$ when the local system is constant. □

17.4 A first computation, the Hopf fibration

Consider the Hopf fibration
$$S^1 \longrightarrow S^3 \xrightarrow{\pi} S^2,$$
where $\pi(z_0, z_1) = [z_0 : z_1]$ under the identification $S^2 \cong \mathbb{C}P^1$.

We compute $H^*(S^3; \mathbb{Z})$ from the Leray–Serre spectral sequence. Since S^2 is simply connected, the local system is constant. We have
$$E_2^{p,q} \cong H^p(S^2; H^q(S^1; \mathbb{Z})) \cong H^p(S^2; \mathbb{Z}) \otimes H^q(S^1; \mathbb{Z}),$$
because $H^q(S^1; \mathbb{Z})$ is free abelian. Let $a \in H^2(S^2; \mathbb{Z})$ be a generator and let $b \in H^1(S^1; \mathbb{Z})$ be a generator. Then
$$H^*(S^2; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}a, \qquad a^2 = 0, \qquad H^*(S^1; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}b, \qquad b^2 = 0.$$

Therefore the only nonzero groups on the E_2-page are

$$E_2^{0,0} \cong \mathbb{Z}, \quad E_2^{2,0} \cong \mathbb{Z}\langle a\rangle, \quad E_2^{0,1} \cong \mathbb{Z}\langle b\rangle, \quad E_2^{2,1} \cong \mathbb{Z}\langle ab\rangle,$$

and all other $E_2^{p,q}$ vanish.

Now check which differentials can possibly be nonzero. By bidegree considerations,

$$d_2 \colon E_2^{0,1} \to E_2^{2,0}$$

is the only differential with a nontrivial target or source. Every d_r with $r \geq 3$ either leaves the first quadrant or lands in a bidegree where $E_r^{p,q} = 0$ for dimensional reasons.

Thus the entire calculation reduces to understanding $d_2(b)$. There are only three possibilities

$$d_2 = 0, \qquad d_2 \text{ is multiplication by } \pm 1, \qquad d_2 \text{ is multiplication by } \pm m \text{ with } |m| \geq 2.$$

We determine which one occurs by comparing with known cohomology of the total space.

Since S^3 has a CW structure with one 0-cell and one 3-cell, we know

$$H^0(S^3;\mathbb{Z}) \cong \mathbb{Z}, \quad H^3(S^3;\mathbb{Z}) \cong \mathbb{Z}, \quad H^1(S^3;\mathbb{Z}) = H^2(S^3;\mathbb{Z}) = 0.$$

Convergence implies $E_\infty^{p,q}$ is the associated graded for $H^{p+q}(S^3;\mathbb{Z})$.

If $d_2 = 0$, then $E_\infty^{0,1} = E_2^{0,1} \cong \mathbb{Z}$, which would force $H^1(S^3;\mathbb{Z}) \neq 0$, a contradiction. If d_2 is multiplication by $\pm m$ with $|m| \geq 2$, then

$$E_3^{2,0} \cong \operatorname{coker}(d_2) \cong \mathbb{Z}/m\mathbb{Z},$$

which would force torsion in $H^2(S^3;\mathbb{Z})$, again a contradiction. Hence d_2 must be an isomorphism. After choosing generators, we may arrange

$$d_2(b) = a.$$

Taking homology with respect to d_2 gives

$$E_3^{0,0} \cong \mathbb{Z}, \qquad E_3^{2,1} \cong \mathbb{Z}\langle ab\rangle,$$

and all other $E_3^{p,q}$ vanish. Since there are no further differentials, $E_3 = E_\infty$. The filtration statement Item (3) now forces

$$H^0(S^3;\mathbb{Z}) \cong \mathbb{Z}, \qquad H^3(S^3;\mathbb{Z}) \cong \mathbb{Z},$$

with the generator in degree 3 represented on the E_∞-page by the class $ab \in E_\infty^{2,1}$.

Multiplicativity also matches the ring structure. Since there is only one nonzero class in positive degree and it sits in degree 3, the cup product forces its square to vanish for degree reasons. Thus

$$H^*(S^3;\mathbb{Z}) \cong \Lambda(u), \qquad |u| = 3,$$

and the class u corresponds to the permanent cycle represented by ab.

Finally, the edge maps behave as expected. The map $\pi^* \colon H^2(S^2;\mathbb{Z}) \to H^2(S^3;\mathbb{Z})$ is zero because $H^2(S^3;\mathbb{Z}) = 0$, and on the spectral sequence this is visible because $a \in E_2^{2,0}$ is hit by d_2. The restriction $i^* \colon H^1(S^3;\mathbb{Z}) \to H^1(S^1;\mathbb{Z})$ is zero because $H^1(S^3;\mathbb{Z}) = 0$, and on the spectral sequence this is visible because $b \in E_2^{0,1}$ does not survive to E_∞.

18 The Leray–Serre Spectral Sequence II

Let $F \hookrightarrow E \xrightarrow{\pi} B$ be a Serre fibration, with B path-connected and F connected. We continue with cohomology with coefficients in a commutative ring R.

18.1 Sphere fibers and the first possible differential

Assume for a moment that $F \simeq S^m$ with $m \geq 1$ and that the local system on B is constant. Then

$$H^q(F;R) = \begin{cases} R & q = 0, \\ R\langle u\rangle & q = m, \\ 0 & \text{otherwise}, \end{cases}$$

so the E_2-page has only two nonzero rows, $q = 0$ and $q = m$. Bidegree considerations force a very rigid shape.

Lemma 18.1 In the situation above, the only differential that can be nonzero is

$$d_{m+1} \colon E_{m+1}^{p,m} \longrightarrow E_{m+1}^{p+m+1,0}.$$

All differentials d_r with $2 \le r \le m$ have zero target, and all d_r with $r \ge m+2$ leave the first quadrant.

Proof. For $2 \le r \le m$, the target of $d_r\colon E_r^{p,m} \to E_r^{p+r,m-r+1}$ has second index $m-r+1$ with $1 \le m-r+1 \le m-1$, hence it is zero already on the E_2-page. For $r = m+1$ the target is the bottom row. For $r \ge m+2$ the target has second index $m-r+1 \le -1$, so it vanishes. □

Write

$$\kappa \;=\; d_{m+1}(u) \in E_{m+1}^{m+1,0} \cong H^{m+1}(B;R).$$

This class controls the entire differential on the top row.

Proposition 18.2

Let $x \in H^p(B;R) = E_{m+1}^{p,0}$ and view $x \cdot u \in E_{m+1}^{p,m}$. Then

$$d_{m+1}(x \cdot u) = (-1)^p\, x \cdot \kappa.$$

In particular, the fate of every class on the $q = m$ row is determined by κ.

Proof. The spectral sequence is multiplicative, and d_{m+1} satisfies the Leibniz rule with respect to total degree. Since d_{m+1} raises total degree by 1 and d_{m+1} vanishes on the $q = 0$ row for degree reasons, we have

$$d_{m+1}(x \cdot u) = d_{m+1}(x) \cdot u + (-1)^{|x|}\, x \cdot d_{m+1}(u) = (-1)^p\, x \cdot \kappa.$$

□

When $m = 1$ this is the familiar differential d_2 for circle bundles. In Lecture 17 we saw this in the Hopf fibration $S^1 \to S^3 \to S^2$, where $d_2(b)$ was a generator of $H^2(S^2;\mathbb{Z})$.

18.2 From E_∞ to cohomology and extension issues

Convergence gives, for each n, a decreasing filtration

$$H^n(E;R) = F^0H^n(E;R) \supseteq F^1H^n(E;R) \supseteq \cdots \supseteq F^{n+1}H^n(E;R) = 0$$

with

$$E_\infty^{p,q} \cong F^pH^{p+q}(E;R)/F^{p+1}H^{p+q}(E;R).$$

So E_∞ determines the associated graded module of $H^*(E;R)$. Reconstructing $H^*(E;R)$ from these graded pieces can require choices.

Proposition 18.3

Fix n. Assume that each $E_\infty^{p,n-p}$ is a free abelian group. Then $H^n(E;\mathbb{Z})$ is free abelian, with rank

$$\operatorname{rank} H^n(E;\mathbb{Z}) = \sum_{p=0}^{n} \operatorname{rank} E_\infty^{p,n-p}.$$

Moreover the filtration splits as a filtration of abelian groups.

Proof. Each step in the filtration gives a short exact sequence

$$0 \to F^{p+1}H^n(E;\mathbb{Z}) \to F^pH^n(E;\mathbb{Z}) \to E_\infty^{p,n-p} \to 0.$$

By downward induction on p, all $F^pH^n(E;\mathbb{Z})$ are free abelian. An extension of a free abelian group by a free abelian group splits since $\operatorname{Ext}^1_{\mathbb{Z}}(\mathbb{Z}^a,\mathbb{Z}^b) = 0$. This gives the splitting and the rank formula. □

When torsion appears on the E_∞-page, there can be nontrivial extensions. In practice, multiplicativity often pins down what happens, because the filtration is compatible with products. Reducing coefficients mod p can also separate possibilities.

18.3 Computing $H^*(\mathbb{C}P^n;\mathbb{Z})$ from $S^1 \to S^{2n+1} \to \mathbb{C}P^n$

Consider the Hopf fibration

$$S^1 \longrightarrow S^{2n+1} \xrightarrow{\pi} \mathbb{C}P^n,$$

where $\pi(z_0,\dots,z_n) = [z_0 : \cdots : z_n]$. Since $\mathbb{C}P^n$ is simply connected for $n \ge 1$, the local system is constant. Let $b \in H^1(S^1;\mathbb{Z})$ be a generator. Write $E_r^{p,q}$ for the cohomology Leray–Serre spectral sequence of this fibration.

The fiber has $H^0(S^1;\mathbb{Z}) \cong \mathbb{Z}$ and $H^1(S^1;\mathbb{Z}) \cong \mathbb{Z}\langle b\rangle$, so

$$E_2^{p,0} \cong H^p(\mathbb{C}P^n;\mathbb{Z}), \qquad E_2^{p,1} \cong H^p(\mathbb{C}P^n;\mathbb{Z}) \cdot b,$$

and all other rows vanish. By Creflem:two-row with $m = 1$, the only potentially nonzero differential is

$$d_2 \colon E_2^{p,1} \to E_2^{p+2,0}.$$

Set

$$x = d_2(b) \in E_2^{2,0} \cong H^2(\mathbb{C}P^n;\mathbb{Z}).$$

By Proposition 18.2, for $u \in H^p(\mathbb{C}P^n;\mathbb{Z})$ we have

$$d_2(u \cdot b) = (-1)^p\, u \cdot x.$$

After replacing b by $-b$ if necessary, we may take $d_2(u \cdot b) = u \cdot x$.

Now use convergence to the cohomology of the total space. Since S^{2n+1} has

$$H^0(S^{2n+1};\mathbb{Z}) \cong \mathbb{Z}, \qquad H^{2n+1}(S^{2n+1};\mathbb{Z}) \cong \mathbb{Z}, \qquad H^k(S^{2n+1};\mathbb{Z}) = 0 \text{ for } 0 < k < 2n+1,$$

the E_∞-page must vanish in every total degree k with $0 < k < 2n+1$.

Lemma 18.4 For $1 \le j \le n$, one has $H^{2j-1}(\mathbb{C}P^n;\mathbb{Z}) = 0$.

Proof. Consider total degree $2j-1$. The only possible bidegree is $(p,q) = (2j-1,0)$ since q is 0 or 1. There is no differential hitting $E_2^{2j-1,0}$, because that would have to come from $E_2^{2j-3,1}$ and $2j-3$ is odd, so such a source would itself be forced to vanish by descending induction starting at $H^1(\mathbb{C}P^n)$. Hence $E_\infty^{2j-1,0} = E_2^{2j-1,0} = H^{2j-1}(\mathbb{C}P^n;\mathbb{Z})$. But E_∞ in this total degree must be zero since $H^{2j-1}(S^{2n+1};\mathbb{Z}) = 0$. □

Lemma 18.5 For $0 \le j \le n$, the group $H^{2j}(\mathbb{C}P^n;\mathbb{Z})$ is cyclic. Moreover multiplication by x gives isomorphisms

$$\cdot x \colon H^{2j}(\mathbb{C}P^n;\mathbb{Z}) \xrightarrow{\cong} H^{2j+2}(\mathbb{C}P^n;\mathbb{Z}) \qquad \text{for } 0 \le j < n.$$

Proof. Fix $0 \le j < n$. Look at total degree $2j+2$ in the E_2-page. The only bidegree contributing is $(2j+2,0)$. To make $E_\infty^{2j+2,0}$ vanish, the map

$$d_2 \colon E_2^{2j,1} \to E_2^{2j+2,0}$$

must be surjective. Thus $\cdot x \colon H^{2j}(\mathbb{C}P^n) \to H^{2j+2}(\mathbb{C}P^n)$ is surjective.

Now look at total degree $2j+1$. The only bidegree contributing is $(2j,1)$. To make $E_\infty^{2j,1}$ vanish, we need $E_3^{2j,1} = 0$. But

$$E_3^{2j,1} \cong \ker(d_2 \colon E_2^{2j,1} \to E_2^{2j+2,0}),$$

so d_2 is injective on $E_2^{2j,1}$. Hence $\cdot x$ is injective from $H^{2j}(\mathbb{C}P^n)$ to $H^{2j+2}(\mathbb{C}P^n)$. Therefore it is an isomorphism. Starting from $H^0(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$, this forces each $H^{2j}(\mathbb{C}P^n;\mathbb{Z})$ to be cyclic, generated by x^j. □

In top degree, total degree $2n+1$ comes only from bidegree $(2n,1)$. Since there is no target for $d_2 \colon E_2^{2n,1} \to E_2^{2n+2,0}$, we have

$$E_\infty^{2n,1} = E_2^{2n,1} \cong H^{2n}(\mathbb{C}P^n;\mathbb{Z}) \cdot b.$$

Convergence forces $E_\infty^{2n,1} \cong H^{2n+1}(S^{2n+1};\mathbb{Z}) \cong \mathbb{Z}$, so $H^{2n}(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}$, and x^n is a generator.

Putting this together, we obtain the full ring.

Theorem 18.6

Let $x \in H^2(\mathbb{C}P^n;\mathbb{Z})$ be the element $x = d_2(b)$ in the Serre spectral sequence of $S^1 \to S^{2n+1} \to \mathbb{C}P^n$. Then

$$H^*(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}), \qquad |x| = 2.$$

Proof. Lemma 18.4 gives $H^{\mathrm{odd}}(\mathbb{C}P^n;\mathbb{Z}) = 0$. Lemma 18.5 gives $H^{2j}(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}\langle x^j\rangle$ for $0 \le j \le n$. Since $\dim \mathbb{C}P^n = 2n$, one has $H^{2n+2}(\mathbb{C}P^n;\mathbb{Z}) = 0$, hence $x^{n+1} = 0$. These statements determine the graded ring as $\mathbb{Z}[x]/(x^{n+1})$. □

Remark. The same calculation works with coefficients in any ring R. If one replaces the total space S^{2n+1} by a lens space, the same two-row shape persists, but the differential becomes $d_2(b) = mx$ for an integer m, which is visible as torsion in the cohomology of the total space.

19 Multiplicative Structure and Transgression

19.1 A product on the Leray–Serre spectral sequence

Let $F \hookrightarrow E \xrightarrow{\pi} B$ be a Serre fibration with B path-connected and F connected. Assume the local systems $H^q(F;R)$ are constant, so that

$$E_2^{p,q} \cong H^p(B;R) \otimes H^q(F;R).$$

The cup product on $H^*(E;R)$ is reflected on each page of the Leray–Serre spectral sequence.

Theorem 19.1. Multiplicativity and the Leibniz rule

For the cohomology Leray–Serre spectral sequence of $\pi\colon E \to B$ with constant local coefficients, there are bilinear pairings

$$E_r^{p,q} \otimes E_r^{p',q'} \longrightarrow E_r^{p+p',\,q+q'}$$

making each E_r into a graded-commutative algebra with respect to total degree. Moreover, for homogeneous classes x, y one has

$$d_r(xy) = d_r(x)\,y + (-1)^{|x|}\,x\,d_r(y).$$

Under the identifications $E_2^{p,0} \cong H^p(B;R)$ and $E_2^{0,q} \cong H^q(F;R)$, the product on E_2 agrees with the tensor product of cup products.

Proof. Choose a CW structure on B and filter E by $E_p = \pi^{-1}(B^{(p)})$. The induced filtration on cochains is

$$F^pC^n(E;R) = \{\, \alpha \in C^n(E;R) \mid \alpha|_{E_{p-1}} = 0 \,\}.$$

If $\alpha \in F^pC^*(E;R)$ and $\beta \in F^{p'}C^*(E;R)$, then $\alpha \smile \beta$ vanishes on $E_{p+p'-1}$, hence $\alpha \smile \beta \in F^{p+p'}C^*(E;R)$. So $F^\bullet C^*(E;R)$ is a multiplicative filtration.

The Leray–Serre spectral sequence is the spectral sequence of the filtered cochain complex $F^\bullet C^*(E;R)$. A multiplicative filtration induces products on the associated graded groups and hence on every E_r. The cup product is graded-commutative, so the induced products on E_r are graded-commutative with respect to total degree. The Leibniz rule follows from the fact that the cochain differential is a graded derivation for the cup product. On the E_2-page, the identification with $H^p(B;R) \otimes H^q(F;R)$ is compatible with products when the local system is constant. □

A first use is to control differentials on generators. If the E_2-page is generated as an algebra by a small list of classes, it suffices to understand the differentials on those generators.

Example. In Lecture 17, for the Hopf fibration $S^1 \to S^3 \to S^2$, the E_2-page was generated by $a \in E_2^{2,0}$ and $b \in E_2^{0,1}$. Once $d_2(b) = a$ was known, Theorem 19.1 forced $d_2(ab) = a^2 = 0$ for degree reasons, and there were no other possible differentials.

19.2 Transgressive classes

Transgression is the differential that starts in the left column and lands on the bottom row. It is most useful when it lands in $H^*(B;R)$ rather than in a higher row.

Definition 19.2. Transgressive classes

Let $r \geq 2$. A class $u \in E_r^{0,r-1}$ is **transgressive** if it survives to the r-page, and the differential

$$d_r(u) \in E_r^{r,0}$$

is defined. In this case the **transgression** of u is

$$\tau(u) = d_r(u) \in E_r^{r,0} \subseteq E_2^{r,0} \cong H^r(B;R).$$

Proposition 19.3. Naturality

Given a map of Serre fibrations with constant local coefficients

$$\begin{tikzcd} F \arrow[r, hook] \arrow[d, "f_F"'] & E \arrow[r, "\pi"] \arrow[d, "f_E"] & B \arrow[d, "f_B"] \\ F' \arrow[r, hook] & E' \arrow[r, "\pi'"] & B', \end{tikzcd}$$

the induced map of Leray–Serre spectral sequences satisfies

$$f_B^*(\tau(u')) = \tau(f_F^*(u'))$$

whenever $u' \in (E'_r)^{0,r-1}$ is transgressive.

Proof. Naturality of the spectral sequence gives $f^*(d_r(u')) = d_r(f^*(u'))$ on the r-page. If u' lies in bidegree $(0, r-1)$, then $f^*(u')$ lies in the same bidegree, and d_r lands in bidegree $(r, 0)$. Under the identification of $(r, 0)$ with a subgroup of $H^r(B; R)$, this is exactly the stated formula. □

The edge map $i^*\colon H^{r-1}(E; R) \to H^{r-1}(F; R)$ controls when transgression can be nonzero.

Proposition 19.4

Let $u \in E_2^{0,r-1} \cong H^{r-1}(F; R)$. If u lies in the image of $i^*\colon H^{r-1}(E; R) \to H^{r-1}(F; R)$, then u is not transgressive. Equivalently, for every representative of u on the E_r-page one has $d_r(u) = 0$.

Proof. If $u = i^*(\alpha)$ for some $\alpha \in H^{r-1}(E; R)$, then α represents a class in the spectral sequence whose restriction to the fiber gives u along the left edge. Since α is an actual cohomology class on E, it survives to E_∞. Hence its restriction u survives as well and supports no nonzero differential. □

19.3 Sphere bundles and a first characteristic class

Assume now that $F \simeq S^m$ with $m \geq 1$ and that the local system is constant. Let $u \in H^m(S^m; R)$ be a chosen generator, viewed as a class in $E_2^{0,m}$. By the two-row shape, the first possible differential on u is d_{m+1}.

Definition 19.5. Euler class of an oriented sphere bundle

Let $S^m \hookrightarrow E \xrightarrow{\pi} B$ be a Serre fibration with constant local coefficients. Assume the top cohomology of the fiber is identified with R so that a preferred generator $u \in E_2^{0,m}$ is specified. Define the **Euler class** of the fibration to be

$$e(\pi) = d_{m+1}(u) \in E_{m+1}^{m+1,0} \subseteq H^{m+1}(B; R).$$

Proposition 19.6

In the situation of Definition 19.5, the only potentially nonzero differential is d_{m+1}, and for $x \in H^p(B; R)$ one has

$$d_{m+1}(x \cdot u) = (-1)^p\, x \cdot e(\pi).$$

Proof. The bidegree forces $d_r = 0$ for $2 \leq r \leq m$ on the top row and $d_r = 0$ for $r \geq m+2$ for degree reasons, so only d_{m+1} can be nonzero. The formula follows from Theorem 19.1 applied to $x \in E_{m+1}^{p,0}$ and $u \in E_{m+1}^{0,m}$. □

A section forces this class to vanish.

Proposition 19.7

If $\pi\colon E \to B$ admits a section $s\colon B \to E$, then $e(\pi) = 0$.

Proof. The restriction $i^*\colon H^m(E; R) \to H^m(S^m; R)$ factors as $i^* = (s \circ i)^*$ on cohomology of the fiber, and s^* is a left inverse to π^*. In particular, i^* is surjective in degree m. So the generator $u \in H^m(S^m; R)$ lies in the image of i^*. By Proposition 19.4, u supports no nonzero differential. Hence $e(\pi) = d_{m+1}(u) = 0$. □

When $m = 1$, this recovers the class $d_2(b)$ from Lecture 18.

Example. For $S^1 \to S^{2n+1} \to \mathbb{C}P^n$, the generator $b \in H^1(S^1;\mathbb{Z})$ satisfies $d_2(b) = x \in H^2(\mathbb{C}P^n;\mathbb{Z})$. With Definition 19.5, the class x is the Euler class of this circle bundle. The calculation in Lecture 18 then shows

$$H^*(\mathbb{C}P^n;\mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}).$$

Example. For the quaternionic Hopf fibration $S^3 \to S^7 \to S^4$, the E_2-page has two nonzero rows $q = 0$ and $q = 3$. Let $u \in H^3(S^3;\mathbb{Z})$ be a generator. The only possible differential is $d_4\colon E_4^{0,3} \to E_4^{4,0}$. Since $H^*(S^7;\mathbb{Z})$ vanishes in degrees $1, 2, 3, 4, 5, 6$, the class u must be killed and $d_4(u)$ must generate $H^4(S^4;\mathbb{Z})$. Thus the Euler class of $S^3 \to S^7 \to S^4$ is a generator of $H^4(S^4;\mathbb{Z})$.

19.4 Transgression in a universal fibration

Let G be a topological group and consider a universal principal G-bundle

$$G \longrightarrow EG \longrightarrow BG$$

with EG contractible. Then $H^{>0}(EG;R) = 0$, so every positive-degree class on the E_2-page must be hit by a differential. In this situation transgression is the mechanism that creates the basic cohomology classes on BG.

Proposition 19.8

Assume EG is contractible and the local system is constant. If $x \in H^q(G;R)$ represents a class in $E_2^{0,q}$ that survives to the first page where it can support a nonzero differential, then its differential is a class in $H^{q+1}(BG;R)$. In particular, transgressive generators in $H^*(G;R)$ give canonical classes in $H^*(BG;R)$.

Proof. Since $H^q(EG;R) = 0$ for $q > 0$, the associated graded groups $E_\infty^{p,q}$ vanish for $q > 0$. Hence each nonzero class in the left column must be killed by some differential. If a class in bidegree $(0, q)$ is first killed by d_{q+1}, then its image lies in bidegree $(q + 1, 0)$, which is identified with a subgroup of $H^{q+1}(BG;R)$. □

Example. For $G = S^1$, one may take $EG = S^\infty$ and $BG = \mathbb{C}P^\infty$. Let $b \in H^1(S^1;\mathbb{Z})$ be a generator. Then $d_2(b)$ is a generator $c \in H^2(\mathbb{C}P^\infty;\mathbb{Z})$. For any principal S^1-bundle over B classified by a map $f\colon B \to \mathbb{C}P^\infty$, the class $f^*(c) \in H^2(B;\mathbb{Z})$ is the Euler class of the bundle in the sense of Definition 19.5.

20 Eilenberg–Mac Lane Spaces and Classifying Spaces

20.1 Eilenberg–Mac Lane spaces

Definition 20.1. Eilenberg–Mac Lane space

Let π be a group and let $n \geq 1$. An **Eilenberg–Mac Lane space of type** (π, n) (or simply a $K(\pi, n)$) if it is path-connected and

$$\pi_q(X, x_0) = \begin{cases} \pi & q = n, \\ 0 & q \neq n, \end{cases}$$

for some (hence any) basepoint $x_0 \in X$.

Example. S^1 is a $K(\mathbb{Z}, 1)$. More generally, a connected graph is a $K(\pi, 1)$, where π is its fundamental group.

Example. $\mathbb{R}P^\infty$ is a $K(\mathbb{Z}_2, 1)$. For each $m \geq 2$, the infinite lens space $L^\infty(m)$ is a $K(\mathbb{Z}_m, 1)$.

Definition 20.2. Aspherical space

A path-connected space X is **aspherical** if $\pi_q(X, x_0) = 0$ for all $q \geq 2$. Equivalently, X is a $K(\pi, 1)$ with $\pi = \pi_1(X, x_0)$.

A $K(\pi, 1)$ is the same thing as a space whose universal cover is contractible.

Proposition 20.3

Let X be a path-connected CW complex with fundamental group π. Then X is a $K(\pi, 1)$ if and only if its universal covering space $\widetilde{X}$ is contractible.

Proof. If $\widetilde{X}$ is contractible, then $\pi_q(\widetilde{X}) = 0$ for all $q \geq 1$. For $q \geq 2$, covering maps induce isomorphisms on π_q, so $\pi_q(X) = 0$ for $q \geq 2$. Also $\pi_1(X) \cong \pi$ by definition.
Conversely, if X is a $K(\pi, 1)$, then $\pi_q(\widetilde{X}) = 0$ for $q \geq 1$ since $\widetilde{X}$ is simply connected and $\pi_q(\widetilde{X}) \cong \pi_q(X)$ for $q \geq 2$. By Whitehead's theorem in the CW setting, a simply connected CW complex with trivial homotopy groups is contractible. □

When a group π is fixed, any $K(\pi, 1)$ is often written $B\pi$.

20.2 Principal bundles and the discrete case

Definition 20.4. Principal G-bundle

Let G be a topological group. A **principal G-bundle** over B is a map $p\colon P \to B$ equipped with a free right action of G on P such that p is G-invariant and for each $b \in B$ there is an open neighborhood U of b and a G-equivariant homeomorphism

$$p^{-1}(U) \;\cong\; U \times G$$

covering the identity on U, where G acts on $U \times G$ by right multiplication on the G-factor.

Example. If G is discrete, then a principal G-bundle $p\colon P \to B$ is the same as a covering map whose fibers are G-torsors. In this situation the G-action identifies G with the deck transformation group acting freely and transitively on each fiber.

The first appearance of $B\pi$ comes from this discrete case.

Proposition 20.5

Let B be path-connected, locally path-connected, and semi-locally simply connected, and let $\pi = \pi_1(B, b_0)$. The universal cover $\widetilde{B} \to B$ is a principal π-bundle, with π acting by deck transformations.

Proof. Deck transformations act freely on $\widetilde{B}$, and the action is transitive on each fiber since any two lifts of b_0 are related by a unique deck transformation. Local path-connectedness and the standard lifting properties give evenly covered neighborhoods, hence local trivializations. □

Example. For $B = S^1$ with $\pi \cong \mathbb{Z}$, the universal cover $\mathbb{R} \to S^1$ is a principal $\mathbb{Z}$-bundle. The connected n-sheeted covers correspond to the subgroups $n\mathbb{Z} \subset \mathbb{Z}$.

20.3 A concrete model for $EG \to BG$

A useful construction of a classifying space starts from a simplicial model.

For a topological group G, define spaces

$$E_n(G) = G^{n+1}, \qquad B_n(G) = G^n \qquad (n \geq 0),$$

and a map $p_n\colon E_n(G) \to B_n(G)$ by forgetting the last coordinate,

$$p_n(g_0, \ldots, g_n) = (g_0, \ldots, g_{n-1}).$$

The face maps d_i and degeneracy maps s_i are given by

$$d_i(g_0, \ldots, g_n) = \begin{cases} (g_1, \ldots, g_n) & i = 0, \\ (g_0, \ldots, g_{i-1}, g_i g_{i+1}, g_{i+2}, \ldots, g_n) & 1 \leq i \leq n, \end{cases} \qquad s_i(g_0, \ldots, g_n) = (g_0, \ldots, g_i, e, g_{i+1}, \ldots, g_n),$$

and on $B_n(G)$ the same formulas apply after omitting the last coordinate (so d_n just deletes the last coordinate). These maps make $E_\bullet(G)$ and $B_\bullet(G)$ into simplicial spaces, and $p_\bullet$ into a simplicial map.

Let

$$EG = |E_\bullet(G)|, \qquad BG = |B_\bullet(G)|, \qquad p = |p_\bullet|\colon EG \to BG$$

be their geometric realizations. Right multiplication on the last coordinate gives a free right G-action on $E_n(G)$

commuting with all faces and degeneracies, hence a free right G-action on EG. Moreover $BG \cong EG/G$.

The point that matters most for homotopy theory is the contractibility of EG.

Lemma 20.6 The space EG is contractible.

Proof. Define an "extra degeneracy" $\sigma\colon E_n(G) \to E_{n+1}(G)$ by

$$\sigma(g_0, \dots, g_n) = (e, g_0, \dots, g_n).$$

One checks directly from the simplicial identities that σ provides a simplicial contraction of $E_\bullet(G)$. After geometric realization, this gives a homotopy from id_{EG} to a constant map. □

Proposition 20.7

The map $p\colon EG \to BG$ is a principal G-bundle.

Proof. The G-action on EG is free and $BG \cong EG/G$ by construction. Local triviality can be verified along the standard filtration of BG coming from the skeleta of the simplices in the realization: over each simplex, the bundle is visibly a product, and these product charts are compatible on overlaps. □

Example. Take $G = S^1$. A standard model has $ES^1 = S^\infty$ and $BS^1 = \mathbb{C}P^\infty$, with S^1 acting freely on S^∞ by scalar multiplication. Since S^∞ is contractible, the long exact sequence of the fibration $S^1 \to S^\infty \to \mathbb{C}P^\infty$ gives

$$\pi_2(\mathbb{C}P^\infty) \cong \pi_1(S^1) \cong \mathbb{Z}, \qquad \pi_q(\mathbb{C}P^\infty) = 0 \text{ for } q \neq 2,$$

so $\mathbb{C}P^\infty$ is a $K(\mathbb{Z}, 2)$.

20.4 Classification of principal bundles

The bundle $EG \to BG$ is universal in the sense that every principal G-bundle over a CW complex is obtained as a pullback.

Theorem 20.8. Classification theorem

Let G be a topological group and let B be a CW complex. For every principal G-bundle $p\colon P \to B$ there exists a map $f\colon B \to BG$ such that P is isomorphic to the pullback bundle $f^*(EG)$. If f and f' are homotopic, then $f^*(EG)$ and $(f')^*(EG)$ are isomorphic as principal G-bundles. Thus there is a natural bijection

$$[B, BG] \;\cong\; \{\text{isomorphism classes of principal } G\text{-bundles over } B\}.$$

Proof. Choose a numerable trivializing open cover $\{U_i\}$ of B for P, with transition functions $g_{ij}\colon U_i \cap U_j \to G$. Choose a partition of unity $\{\varphi_i\}$ subordinate to $\{U_i\}$. For each $b \in B$, only finitely many $\varphi_i(b)$ are nonzero, so the indices with $\varphi_i(b) \neq 0$ determine a simplex in the nerve of the cover. The functions $g_{ij}(b)$ provide the G-coordinates needed to place b into the corresponding piece of the simplicial model of BG. This constructs a continuous map $f\colon B \to BG$, well defined up to homotopy by standard refinements of covers and partitions of unity.

Pulling back $EG \to BG$ along f yields a principal G-bundle over B with transition functions g_{ij} on the chosen cover, hence it is isomorphic to P. Homotopies of f induce isomorphisms of pullbacks since pullback is functorial and the universal bundle admits local trivializations compatible with homotopies. □

Remark. When G is discrete, Theorem 20.8 says that principal G-bundles over B are classified by homotopy classes of maps $B \to BG$, and Lemma 20.6 implies BG is a $K(G, 1)$. Thus the classifying space of a discrete group is an Eilenberg–Mac Lane space.

20.5 Covering spaces as the first case of $B\pi$

Let B be path-connected, locally path-connected, and semi-locally simply connected, and write $\pi = \pi_1(B, b_0)$. The universal cover $\widetilde{B} \to B$ is a principal π-bundle by Proposition 20.5. By Theorem 20.8, it is classified by a map

$$c_B\colon B \to B\pi,$$

unique up to homotopy. This map is the **classifying map** of the universal cover.

For a subgroup $H \leq \pi$, the quotient $\widetilde{B}/H \to B$ is a connected covering space. Two subgroups give isomorphic

connected coverings if and only if they are conjugate. When H is normal, the covering is regular and the deck transformation group is π/H. This recovers the classical classification of connected covering spaces while also explaining why $B\pi$ is the natural recipient of the classifying map: pullbacks along maps into $B\pi$ generate principal π-bundles, hence regular coverings with group π.

21 Vector Bundles and Classifying Maps

21.1 Vector bundles and basic operations

Definition 21.1. Real and complex vector bundles

A **real vector bundle of rank** n over a space B is a map $\pi\colon E \to B$ together with a choice of n-dimensional real vector space structure on each fiber $E_b = \pi^{-1}(b)$ such that every $b \in B$ has an open neighborhood U and a homeomorphism

$$\varphi\colon \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^n$$

with $\pi = \mathrm{pr}_U \circ \varphi$ and such that for each $u \in U$ the restriction $\varphi\colon E_u \to \{u\} \times \mathbb{R}^n$ is linear. A **complex vector bundle of rank** n is defined similarly with $\mathbb{C}^n$. Two bundles over B are **isomorphic** if there is a homeomorphism between total spaces over B that is fiberwise linear.

Definition 21.2. Pullback bundle

Given a vector bundle $\pi\colon E \to B$ of rank n and a map $f\colon A \to B$, the **pullback bundle** $f^*E \to A$ is the subspace

$$f^*E = \{(a, e) \in A \times E \mid f(a) = \pi(e)\},$$

with projection $(a, e) \mapsto a$ and fiberwise linear structure induced from E.

Proposition 21.3

Pullback is functorial up to canonical isomorphism. That is, $(\mathrm{id}_B)^*E \cong E$ and $(g \circ f)^*E \cong f^*(g^*E)$. If $f_0 \simeq f_1$, then f_0^*E and f_1^*E are isomorphic.

Proof. The first two statements are checked directly from the defining pullback set. For the homotopy invariance, a homotopy $H\colon A \times I \to B$ gives a bundle $H^*E \to A \times I$. Restricting to $A \times \{0\}$ and $A \times \{1\}$ yields f_0^*E and f_1^*E. Since $A \times I$ deformation retracts onto either end, the two restrictions are isomorphic. □

Definition 21.4. Restriction, sum, dual

Let $\xi \to B$ and $\eta \to B$ be vector bundles.

(1) For $A \subseteq B$, the **restriction** $\xi|_A$ is the pullback along $A \hookrightarrow B$.
(2) The **Whitney sum** $\xi \oplus \eta$ is the bundle over B with fiber $(\xi \oplus \eta)_b = \xi_b \oplus \eta_b$.
(3) The **dual bundle** $\xi^\vee$ has fiber $(\xi^\vee)_b = \mathrm{Hom}_{\mathbb{F}}(\xi_b, \mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

Proposition 21.5

For a map $f\colon A \to B$ one has natural isomorphisms

$$f^*(\xi \oplus \eta) \cong f^*\xi \oplus f^*\eta, \qquad f^*(\xi^\vee) \cong (f^*\xi)^\vee, \qquad f^*(\xi|_A) \cong (f^*\xi)|_A.$$

Proof. These identifications are fiberwise linear and are defined by the universal property of pullbacks. For example, at a point $a \in A$ with $b = f(a)$,

$$(f^*(\xi \oplus \eta))_a \cong (\xi \oplus \eta)_b \cong \xi_b \oplus \eta_b \cong (f^*\xi)_a \oplus (f^*\eta)_a.$$

Continuity follows from local trivializations. □

Example. If M is a smooth manifold and $U \subseteq M$ is open, then $TM|_U$ is a rank $\dim(M)$ bundle over U. If $f\colon N \to M$ is a smooth map, then $f^*(TM)$ is the bundle over N whose fiber at y is $T_{f(y)}M$.

21.2 Clutching on spheres

Write $S^k = D^k_+ \cup D^k_-$ as the union of the upper and lower hemispheres, each homeomorphic to D^k, with intersection $D^k_+ \cap D^k_- = S^{k-1}$.

Definition 21.6. Clutching function

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and let $n \geq 1$. A **clutching function** is a map

$$g\colon S^{k-1} \longrightarrow \mathrm{GL}_n(\mathbb{F}).$$

It determines a rank-n $\mathbb{F}$-vector bundle over S^k by gluing the trivial bundles $D^k_+ \times \mathbb{F}^n$ and $D^k_- \times \mathbb{F}^n$ along the overlap via

$$(x, v) \sim (x, g(x)v) \qquad (x \in S^{k-1}).$$

We denote the resulting bundle by $\xi_g \to S^k$.

Proposition 21.7

If $g_0 \simeq g_1$ as maps $S^{k-1} \to \mathrm{GL}_n(\mathbb{F})$, then $\xi_{g_0} \cong \xi_{g_1}$ as bundles over S^k.

Proof. A homotopy $G\colon S^{k-1} \times I \to \mathrm{GL}_n(\mathbb{F})$ gives a bundle over $S^k \times I$ obtained by gluing $D^k_\pm \times I \times \mathbb{F}^n$ along $S^{k-1} \times I$ using G. Restricting to $S^k \times \{0\}$ and $S^k \times \{1\}$ gives ξ_{g_0} and ξ_{g_1}. Since $S^k \times I$ deformation retracts onto either end, the two restrictions are isomorphic. □

Theorem 21.8. Clutching classification over spheres

For $k \geq 1$ and $n \geq 1$, the assignment $[g] \mapsto [\xi_g]$ induces a bijection

$$[S^{k-1}, \mathrm{GL}_n(\mathbb{F})] \;\cong\; \{\text{isomorphism classes of rank-}n\ \mathbb{F}\text{-bundles over } S^k\}.$$

If $\mathbb{F} = \mathbb{R}$, one may replace $\mathrm{GL}_n(\mathbb{R})$ by $O(n)$, and if $\mathbb{F} = \mathbb{C}$, one may replace $\mathrm{GL}_n(\mathbb{C})$ by $U(n)$.

Proof. Existence. Let $\xi \to S^k$ be a rank-n bundle. Each hemisphere $D^k_\pm$ is contractible, hence $\xi|_{D^k_\pm}$ is trivial. Choose trivializations

$$\varphi_\pm\colon \xi|_{D^k_\pm} \xrightarrow{\cong} D^k_\pm \times \mathbb{F}^n.$$

On the overlap S^{k-1}, the transition map $\varphi_+ \circ \varphi_-^{-1}$ has the form

$$(x, v) \longmapsto (x, g(x)v)$$

for a continuous $g\colon S^{k-1} \to \mathrm{GL}_n(\mathbb{F})$. By construction, $\xi \cong \xi_g$.

Uniqueness up to homotopy. If we change trivializations by maps $a_\pm\colon D^k_\pm \to \mathrm{GL}_n(\mathbb{F})$, then the clutching function changes by

$$g \longmapsto (a_+|_{S^{k-1}})\, g\, (a_-|_{S^{k-1}})^{-1}.$$

Since each $a_\pm$ is defined on a disk, the restrictions $a_\pm|_{S^{k-1}}$ are nullhomotopic. Thus the new clutching function is homotopic to g. Conversely, Proposition 21.7 shows that homotopic clutching functions give isomorphic bundles.

Finally, replacing $\mathrm{GL}_n(\mathbb{F})$ by $O(n)$ or $U(n)$ follows by choosing fiberwise inner products and applying Gram–Schmidt fiberwise inside a trivialization. □

Example. Real line bundles over S^1 correspond to $[S^0, O(1)]$. Since $O(1) = \{\pm 1\}$, there are exactly two isomorphism classes. One is the trivial bundle ε^1, and the other is the Möbius bundle.

Example. Complex line bundles over S^2 correspond to $[S^1, U(1)]$. Since $U(1) \cong S^1$, this set is in bijection with $\pi_1(S^1) \cong \mathbb{Z}$. The Hopf line bundle over $S^2 \cong \mathbb{C}P^1$ corresponds to a generator.

21.3 Grassmannians and the tautological bundle

For $k \geq 0$, let $\mathrm{Gr}_n(\mathbb{R}^{n+k})$ be the set of n-dimensional linear subspaces of $\mathbb{R}^{n+k}$, with its standard topology. Define the **tautological bundle** $\gamma^{n+k}_n \to \mathrm{Gr}_n(\mathbb{R}^{n+k})$ by

$$E(\gamma^{n+k}_n) = \{(V, v) \in \mathrm{Gr}_n(\mathbb{R}^{n+k}) \times \mathbb{R}^{n+k} \mid v \in V\}, \qquad (V, v) \longmapsto V.$$

Proposition 21.9

The projection $\gamma_n^{n+k} \to \mathrm{Gr}_n(\mathbb{R}^{n+k})$ is a rank-n real vector bundle.

Proof. Fix $V_0 \in \mathrm{Gr}_n(\mathbb{R}^{n+k})$. Let U be the set of all V such that orthogonal projection $\mathbb{R}^{n+k} \to V_0$ restricts to an isomorphism $V \to V_0$. Then each $V \in U$ is the graph of a linear map $V_0 \to V_0^\perp$, and this gives a homeomorphism $U \cong \mathrm{Hom}(V_0, V_0^\perp) \cong \mathbb{R}^{nk}$. Over U, the assignment $(V, x) \mapsto (V, \text{the unique } v \in V \text{ projecting to } x)$ gives a trivialization $U \times V_0 \cong (\gamma_n^{n+k})|_U$. □

Proposition 21.10

Let $\xi \to B$ be a rank-n real vector bundle over a compact space B. Then for some $k \geq 0$ there exists a fiberwise linear injective map

$$\Phi\colon \xi \longrightarrow \varepsilon^{n+k}.$$

Such a choice determines a map $c_\xi\colon B \to \mathrm{Gr}_n(\mathbb{R}^{n+k})$ by

$$c_\xi(b) = \Phi(\xi_b) \subseteq \mathbb{R}^{n+k},$$

and there is a canonical isomorphism of bundles

$$\xi \;\cong\; c_\xi^*(\gamma_n^{n+k}).$$

Proof. Choose finitely many local trivializations $\xi|_{U_i} \cong U_i \times \mathbb{R}^n$ covering B. Using a partition of unity subordinate to $\{U_i\}$, one constructs maps $\Phi_i\colon \xi \to \varepsilon^n$ that are fiberwise linear and supported over U_i. Taking the direct sum $\Phi = (\Phi_1, \ldots, \Phi_r)\colon \xi \to \varepsilon^{nr}$ gives a fiberwise linear injective map. Setting $n + k = nr$ gives the desired Φ.

For the second statement, the map $b \mapsto \Phi(\xi_b)$ is continuous because it can be checked in local trivializations, where it becomes the graph of a continuous family of injective linear maps. The pullback $c_\xi^*(\gamma_n^{n+k})$ has fiber over b equal to $c_\xi(b)$, which is exactly the image of $\Phi\colon \xi_b \to \mathbb{R}^{n+k}$. This identifies ξ with the pullback tautological bundle. □

Example. If a smooth n-manifold M is embedded in $\mathbb{R}^{n+k}$, then each tangent space T_xM determines an n-plane in $\mathbb{R}^{n+k}$. This gives a map $M \to \mathrm{Gr}_n(\mathbb{R}^{n+k})$, and Proposition 21.10 recovers the tangent bundle as a pullback of γ_n^{n+k}.

21.4 The classifying space $BO(n)$ and the standard maps

Let $\mathrm{Gr}_n(\mathbb{R}^n) \subset \mathrm{Gr}_n(\mathbb{R}^{n+1}) \subset \cdots$ be the inclusions induced by $\mathbb{R}^{n+k} \subset \mathbb{R}^{n+k+1}$. Set

$$BO(n) = \mathrm{Gr}_n(\mathbb{R}^\infty) = \mathrm{colim}_k \mathrm{Gr}_n(\mathbb{R}^{n+k}),$$

and let $\gamma^n \to BO(n)$ be the colimit of the tautological bundles γ_n^{n+k}.

Theorem 21.11. Classification of real vector bundles

Let B be a CW complex. Sending a homotopy class $[f] \in [B, BO(n)]$ to the pullback bundle $f^*(\gamma^n)$ induces a bijection

$$[B, BO(n)] \;\cong\; \{\text{isomorphism classes of rank-}n\text{ real vector bundles over } B\}.$$

Moreover, if $\xi \cong f^*(\gamma^n)$, then for any map $g\colon A \to B$ one has $g^*\xi \cong (f \circ g)^*(\gamma^n)$.

Proof. Given f, the pullback $f^*(\gamma^n)$ is a rank-n bundle. Homotopic maps give isomorphic pullbacks by Proposition 21.3, so the assignment is well defined on $[B, BO(n)]$.

Conversely, given a rank-n bundle $\xi \to B$, apply Proposition 21.10 on each finite subcomplex of B and pass to the colimit over skeleta. This gives a map $c_\xi\colon B \to BO(n)$ with $\xi \cong c_\xi^*(\gamma^n)$. Different choices of embeddings into trivial bundles produce homotopic maps, again using Proposition 21.3. This gives the inverse correspondence. □

There are two maps that will be used repeatedly.

Definition 21.12. Stabilization and sum maps

The inclusion $\mathbb{R}^{n+k} \hookrightarrow \mathbb{R}^{n+k+1}$ induces maps

$$i_n \colon BO(n) \longrightarrow BO(n+1),$$

corresponding to $\xi \mapsto \xi \oplus \varepsilon^1$.
Given $m, n \geq 0$, the direct sum of subspaces induces maps

$$p_{m,n} \colon BO(m) \times BO(n) \longrightarrow BO(m+n),$$

corresponding to $(\xi, \eta) \mapsto \xi \oplus \eta$.

Proposition 21.13

Let ξ be a rank-n bundle over B with classifying map $c_\xi \colon B \to BO(n)$. Then $\xi \oplus \varepsilon^1$ is classified by $i_n \circ c_\xi$.
If ξ and η have classifying maps $c_\xi \colon B \to BO(m)$ and $c_\eta \colon B \to BO(n)$, then $\xi \oplus \eta$ is classified by

$$p_{m,n} \circ (c_\xi, c_\eta) \colon B \to BO(m+n).$$

Proof. By construction, $i_n^*(\gamma^{n+1}) \cong \gamma^n \oplus \varepsilon^1$ and $p_{m,n}^*(\gamma^{m+n}) \cong \mathrm{pr}_1^*(\gamma^m) \oplus \mathrm{pr}_2^*(\gamma^n)$. Pulling back along c_ξ and (c_ξ, c_η) gives the stated bundle isomorphisms. □

Remark. The complex story is parallel. Replacing $\mathbb{R}$ by $\mathbb{C}$ gives spaces $BU(n)$ and a universal bundle over $BU(n)$ that classifies rank-n complex vector bundles over CW complexes.

22 Thom Spaces, Thom Isomorphism, Euler Classes, and the Gysin Sequence

22.1 Thom spaces and relative cohomology

Let $\xi \to B$ be a real vector bundle of rank n, with total space $E = E(\xi)$ and projection $\pi \colon E \to B$. Write $B \subset E$ for the zero section, and let

$$E_0 = E \setminus B$$

be the complement of the zero section.

Definition 22.1. Thom space

The **Thom space** of ξ is the quotient

$$\mathrm{Th}(\xi) = E/E_0,$$

where all of E_0 is collapsed to a single point.

Since E_0 is open in E, the quotient map identifies reduced cohomology of $\mathrm{Th}(\xi)$ with relative cohomology of (E, E_0).

Proposition 22.2

For any commutative ring R there are natural isomorphisms

$$\widetilde{H}^k(\mathrm{Th}(\xi); R) \;\cong\; H^k(E, E_0; R) \qquad \text{for all } k.$$

Proof. Let $q \colon E \to E/E_0 = \mathrm{Th}(\xi)$ be the quotient map. The cofiber sequence $E_0 \hookrightarrow E \to \mathrm{Th}(\xi)$ yields the long exact sequence of the pair (E, E_0) and, by the definition of reduced cohomology, the connecting identifications

$$\widetilde{H}^k(\mathrm{Th}(\xi); R) \cong H^k(E, E_0; R).$$

Naturality follows from functoriality of quotient maps and of relative cohomology. □

Example. If $\xi = \varepsilon^n$ is the trivial rank-n bundle over B, then $E = B \times \mathbb{R}^n$ and $E_0 = B \times (\mathbb{R}^n \setminus \{0\})$. Collapsing E_0 identifies $\mathrm{Th}(\varepsilon^n)$ with the quotient of $B \times S^n$ obtained by collapsing $\{*\} \times S^n$ and $B \times \{*\}$ to a point, hence

$$\mathrm{Th}(\varepsilon^n) \;\cong\; B_+ \wedge S^n,$$

where $B_+ = B \sqcup \{*\}$ is B with a disjoint basepoint.

22.2 Thom classes and the Thom isomorphism

For each $b \in B$, the fiber ξ_b is isomorphic to $\mathbb{R}^n$. Let $D(\xi) \subset E$ be the unit disk bundle and $S(\xi) = \partial D(\xi)$ the sphere bundle, after choosing a fiberwise inner product. Then $(D(\xi), S(\xi))$ is a fiber bundle of pairs with typical fiber (D^n, S^{n-1}). Using excision, $H^*(E, E_0; R) \cong H^*(D(\xi), S(\xi); R)$, but it is convenient to keep (E, E_0) in the notation.

Definition 22.3. Thom class

A class $u \in H^n(E, E_0; R)$ is an R-**Thom class** for ξ if for every $b \in B$ its restriction to the fiber pair

$$(E_b, (E_0)_b) \;\cong\; (D^n, S^{n-1})$$

is a generator of $H^n(D^n, S^{n-1}; R) \cong R$.

Definition 22.4. Thom isomorphism

Given a Thom class u for ξ, define

$$\Phi_u \colon H^k(B; R) \longrightarrow H^{k+n}(E, E_0; R), \qquad \Phi_u(x) = \pi^*(x) \smile u.$$

Theorem 22.5. Thom isomorphism

If u is a Thom class for ξ, then Φ_u is an isomorphism for all k.

Proof. Over a trivializing open set $U \subset B$, the pair $(E|_U, (E_0)|_U)$ is isomorphic to $(U \times D^n, U \times S^{n-1})$. The restriction of u to U agrees with $1 \otimes u_0$ for a generator $u_0 \in H^n(D^n, S^{n-1}; R)$, so on U the map Φ_u becomes the Künneth-type identification

$$H^k(U; R) \xrightarrow{\cong} H^{k+n}(U \times D^n, U \times S^{n-1}; R).$$

Choose a numerable trivializing cover and use Mayer–Vietoris on B together with excision on the corresponding cover of E to patch these local isomorphisms to a global one. □

Proposition 22.6. Naturality

Let $f \colon B' \to B$ be a map and let $f^*\xi$ be the pullback bundle with induced map of pairs

$$\widetilde{f} \colon (E(f^*\xi), E(f^*\xi)_0) \longrightarrow (E(\xi), E(\xi)_0).$$

If u is a Thom class for ξ, then $\widetilde{f}^*(u)$ is a Thom class for $f^*\xi$, and the Thom isomorphisms satisfy

$$\widetilde{f}^*(\Phi_u(x)) = \Phi_{\widetilde{f}^*u}(f^*x).$$

Proof. Restriction of $\widetilde{f}^*(u)$ to a fiber over $b' \in B'$ identifies with restriction of u to the fiber over $f(b')$, so it is a generator. For the formula,

$$\widetilde{f}^*(\pi^*(x) \smile u) = (\pi \circ \widetilde{f})^*(x) \smile \widetilde{f}^*(u) = \pi'^*(f^*x) \smile \widetilde{f}^*(u),$$

where π' is the projection of $f^*\xi$. □

22.3 Euler classes

The inclusion of pairs $(E, \varnothing) \hookrightarrow (E, E_0)$ induces a restriction map

$$j^* \colon H^*(E, E_0; R) \longrightarrow H^*(E; R), \qquad y \longmapsto y|_E.$$

Also, the bundle projection $\pi \colon E \to B$ has a section (the zero section), hence π is a homotopy equivalence. In particular, $\pi^* \colon H^*(B; R) \to H^*(E; R)$ is an isomorphism.

Definition 22.7. Euler class

Assume ξ admits an R-Thom class $u \in H^n(E, E_0; R)$. The **Euler class** of ξ is the unique element $e(\xi) \in H^n(B; R)$ such that

$$\pi^*(e(\xi)) = u|_E \in H^n(E; R).$$

Proposition 22.8

Let ξ be an R-oriented rank-n bundle with Thom class u.

(1) (**Naturality**) If $f\colon B' \to B$, then $e(f^*\xi) = f^*e(\xi)$.

(2) If ξ is trivial and $n > 0$, then $e(\xi) = 0$.

(3) Reversing the orientation multiplies $e(\xi)$ by -1.

Proof. For Item (1), apply Proposition 22.6 and then restrict to $E(f^*\xi)$:

$$\pi'^*e(f^*\xi) = (\widetilde{f}^*u)|_{E(f^*\xi)} = \widetilde{f}^*(u|_E) = \widetilde{f}^*(\pi^*e(\xi)) = \pi'^*(f^*e(\xi)).$$

Since π'^* is an isomorphism, $e(f^*\xi) = f^*e(\xi)$.

For Item (2), in the trivial case $\mathrm{Th}(\varepsilon^n) \cong B_+ \wedge S^n$, and we may take $u = 1 \otimes u_0$. The restriction $u|_E$ vanishes because $H^n(B \times \mathbb{R}^n; R) \cong H^n(B; R)$ and the class restricts to zero along the deformation to the zero section when $n > 0$. Equivalently, the zero section provides a nowhere-zero section, so the obstruction measured by $e(\xi)$ vanishes.

For Item (3), reversing orientation replaces the chosen generator on each fiber by its negative, hence replaces u by $-u$, so $u|_E$ changes sign and so does $e(\xi)$. □

Proposition 22.9

If n is odd, then $e(\xi) = -e(\xi)$. In particular, when $R = \mathbb{Z}$ one has $2e(\xi) = 0$.

Proof. On each fiber, the map $v \mapsto -v$ reverses orientation when n is odd. This gives an orientation-reversing bundle automorphism $\xi \to \xi$ covering id_B. By Item (3) and Item (1), we get

$$e(\xi) = \mathrm{id}_B^* e(\xi) = -e(\xi).$$

□

22.4 The Gysin sequence for an oriented sphere bundle

Let $\xi \to B$ be an R-oriented rank-n bundle with Thom class u. Let $D(\xi) \subset E(\xi)$ be the unit disk bundle and $S(\xi) = \partial D(\xi)$ the sphere bundle. Write $p\colon S(\xi) \to B$ for the projection.

By excision and Theorem 22.5, we have an isomorphism

$$\Phi_u\colon H^{k-n}(B; R) \xrightarrow{\cong} H^k(D(\xi), S(\xi); R), \qquad \Phi_u(x) = \pi^*(x) \smile u,$$

where $\pi\colon D(\xi) \to B$ is the disk bundle projection.

Consider the long exact cohomology sequence of the pair $(D(\xi), S(\xi))$:

$$\cdots \to H^k(D(\xi), S(\xi); R) \xrightarrow{j^*} H^k(D(\xi); R) \xrightarrow{i^*} H^k(S(\xi); R) \xrightarrow{\delta} H^{k+1}(D(\xi), S(\xi); R) \to \cdots$$

Here $i\colon S(\xi) \hookrightarrow D(\xi)$ is inclusion. Since $D(\xi)$ deformation retracts onto the zero section, $\pi^*\colon H^*(B; R) \to H^*(D(\xi); R)$ is an isomorphism.

Proposition 22.10

Under the identifications $H^k(D(\xi); R) \cong H^k(B; R)$ and $H^k(D(\xi), S(\xi); R) \cong H^{k-n}(B; R)$ via π^* and Φ_u, the map

$$j^*\colon H^k(D(\xi), S(\xi); R) \longrightarrow H^k(D(\xi); R)$$

corresponds to cup product with the Euler class:

$$H^{k-n}(B; R) \xrightarrow{\smile e(\xi)} H^k(B; R).$$

Proof. By definition, $\Phi_u(x) = \pi^*(x) \smile u$. Applying j^* gives

$$j^*\Phi_u(x) = \pi^*(x) \smile (u|_{D(\xi)}).$$

By Definition 22.7, the class $u|_{D(\xi)}$ equals $\pi^*e(\xi)$ under the identification $H^n(D(\xi); R) \cong H^n(B; R)$. Hence

$$j^*\Phi_u(x) = \pi^*(x) \smile \pi^*(e(\xi)) = \pi^*(x \smile e(\xi)).$$

Since π^* is an isomorphism, j^* corresponds to $x \mapsto x \smile e(\xi)$. □

Theorem 22.11. Gysin sequence

Let $\xi \to B$ be an R-oriented rank-n bundle and let $S(\xi) \xrightarrow{p} B$ be its sphere bundle. There is a long exact sequence

$$\cdots \to H^{k-n}(B;R) \xrightarrow{\smile e(\xi)} H^k(B;R) \xrightarrow{p^*} H^k(S(\xi);R) \xrightarrow{p_!} H^{k-n+1}(B;R) \to \cdots$$

where $p_!$ is the composite

$$H^k(S(\xi);R) \xrightarrow{\delta} H^{k+1}(D(\xi),S(\xi);R) \xrightarrow[\cong]{\Phi_u^{-1}} H^{k-n+1}(B;R).$$

Proof. Start from the long exact sequence of the pair $(D(\xi),S(\xi))$ and use the identifications $H^*(D(\xi);R) \cong H^*(B;R)$ and $H^*(D(\xi),S(\xi);R) \cong H^{*-n}(B;R)$ given by π^* and Φ_u. By Proposition 22.10, the map j^* becomes cup product with $e(\xi)$. The map i^* becomes p^* after identifying $D(\xi) \simeq B$ through the zero section. Defining $p_!$ as stated yields the displayed sequence. □

Example. If $\xi = \varepsilon^n$ is trivial with $n > 0$, then $e(\xi) = 0$ by Proposition 22.8. The Gysin sequence breaks into short exact sequences

$$0 \to H^k(B;R) \xrightarrow{p^*} H^k(B \times S^{n-1};R) \xrightarrow{p_!} H^{k-n+1}(B;R) \to 0,$$

and this matches the Künneth description when $H^*(S^{n-1};R)$ is free.

Example. For an oriented circle bundle $S^1 \to E \xrightarrow{p} B$, the Gysin sequence reads

$$\cdots \to H^{k-2}(B;R) \xrightarrow{\smile e} H^k(B;R) \xrightarrow{p^*} H^k(E;R) \xrightarrow{p_!} H^{k-1}(B;R) \to \cdots$$

with $e = e(\xi) \in H^2(B;R)$. In Lecture 18, for $S^1 \to S^{2n+1} \to \mathbb{C}P^n$, the class e is the element $x \in H^2(\mathbb{C}P^n;\mathbb{Z})$ detected by the differential d_2 in the Leray–Serre spectral sequence, and the cohomology of the base is generated by powers of x.

23 Stiefel–Whitney Classes

Throughout this section cohomology is taken with coefficients in $\mathbb{Z}_2$. For a real rank-n vector bundle $\xi \to B$ we will write

$$w(\xi) = 1 + w_1(\xi) + w_2(\xi) + \cdots + w_n(\xi) \in H^*(B;\mathbb{Z}_2),$$

and call this the **total Stiefel–Whitney class**.

23.1 Axioms and first consequences

Definition 23.1. Stiefel–Whitney classes (axioms)

A **Stiefel–Whitney theory** assigns to each real vector bundle $\xi \to B$ a class

$$w(\xi) = 1 + w_1(\xi) + \cdots + w_n(\xi) \in H^*(B;\mathbb{Z}_2), \qquad w_i(\xi) \in H^i(B;\mathbb{Z}_2),$$

with the following properties.

(1) $w_0(\xi) = 1$ and $w_i(\xi) = 0$ for $i > \mathrm{rk}(\xi)$.

(2) (**Naturality**) For any map $f\colon B' \to B$,

$$w(f^*\xi) = f^*w(\xi).$$

(3) (**Whitney product**) For bundles ξ, η over the same base,

$$w(\xi \oplus \eta) = w(\xi) \smile w(\eta).$$

(4) (**Normalization**) For the canonical real line bundle $\gamma^1 \to \mathbb{R}P^1$, the class $w_1(\gamma^1) \in H^1(\mathbb{R}P^1;\mathbb{Z}_2) \cong \mathbb{Z}_2$ is nonzero.

Theorem 23.2. Existence

There exists a Stiefel–Whitney theory.

Remark. A standard construction uses the Thom class of a bundle and cohomology operations on $\mathbb{Z}_2$-cohomology. We will use the properties listed in Definition 23.1 as the starting point.

The axioms immediately force the usual formal identities.

Proposition 23.3

Let $\xi, \eta \to B$ be real bundles.

(1) If ε^n is trivial, then $w(\varepsilon^n) = 1$.

(2) $w(\xi \oplus \varepsilon^k) = w(\xi)$ for all $k \geq 0$.

(3) The Whitney product formula expands as

$$w_r(\xi \oplus \eta) = \sum_{i+j=r} w_i(\xi) \smile w_j(\eta).$$

Proof. For Item (1), apply naturality Item (2) to the unique map $B \to *$ and use Item (1). Then Item (2) follows from Item (3). Finally, Item (3) is the degree-r component of Item (3). □

The first class detects orientability.

Proposition 23.4

Let $\xi \to B$ be a real rank-n bundle. Then ξ is orientable if and only if $w_1(\xi) = 0$.

Proof. Choose a bundle atlas for ξ with transition functions $g_{ij} \colon U_i \cap U_j \to \mathrm{GL}_n(\mathbb{R})$. Reducing mod 2, the determinant sign $\det(g_{ij})/|\det(g_{ij})| \in \{\pm 1\}$ defines a Čech 1-cocycle with values in $\mathbb{Z}_2$. Its cohomology class is $w_1(\xi)$. This cocycle is a coboundary exactly when the transition maps can be chosen in $\mathrm{GL}_n^+(\mathbb{R})$, which is exactly orientability. □

The top class controls existence of linearly independent sections.

Proposition 23.5

Let $\xi \to B$ be a real rank-n bundle. If ξ admits a nowhere zero section, then $w_n(\xi) = 0$. More generally, if ξ admits k everywhere linearly independent sections, then

$$w_{n-k+1}(\xi) = w_{n-k+2}(\xi) = \cdots = w_n(\xi) = 0.$$

Proof. If ξ has k everywhere linearly independent sections, then ξ splits as $\varepsilon^k \oplus \xi'$ with $\mathrm{rk}(\xi') = n - k$. By Item (3) and Item (1),

$$w(\xi) = w(\varepsilon^k)\, w(\xi') = w(\xi'),$$

so $w_i(\xi) = 0$ for $i > n - k$ by Item (1). □

23.2 Uniqueness

Theorem 23.6. Uniqueness

There exists at most one Stiefel–Whitney theory satisfying Definition 23.1.

Proof. Let w and $\widetilde{w}$ be two assignments satisfying Definition 23.1. By Item (4), the canonical line bundle over $\mathbb{R}P^1$ satisfies

$$w(\gamma^1) = \widetilde{w}(\gamma^1) = 1 + a$$

where a is the nonzero class in $H^1(\mathbb{R}P^1; \mathbb{Z}_2)$.

Embed $\mathbb{R}P^1$ into $\mathbb{R}P^\infty$ in the standard way and use naturality to conclude that

$$w(\gamma^1) = \widetilde{w}(\gamma^1) = 1 + a \quad \text{in } H^*(\mathbb{R}P^\infty; \mathbb{Z}_2) \cong \mathbb{Z}_2[a].$$

Let $p_i \colon (\mathbb{R}P^\infty)^n \to \mathbb{R}P^\infty$ be projection onto the ith factor, and set

$$\xi = \bigoplus_{i=1}^n p_i^*(\gamma^1).$$

Then Items (2) and (3) imply

$$w(\xi) = \widetilde{w}(\xi) = \prod_{i=1}^n (1 + a_i) \qquad \text{in } H^*\big((\mathbb{R}P^\infty)^n; \mathbb{Z}_2\big),$$

where $a_i = p_i^*(a)$.
Now consider the universal rank-n bundle $\gamma^n \to BO(n)$. There exists a bundle map $\xi \to \gamma^n$, equivalently a classifying map $f\colon (\mathbb{R}P^\infty)^n \to BO(n)$ with $f^*(\gamma^n) \cong \xi$. Moreover, the induced map

$$f^*\colon H^*(BO(n);\mathbb{Z}_2) \longrightarrow H^*((\mathbb{R}P^\infty)^n;\mathbb{Z}_2)$$

is injective. Since $w(\gamma^n)$ and $\widetilde{w}(\gamma^n)$ have the same pullback along f, they must agree in $H^*(BO(n);\mathbb{Z}_2)$. Finally, for any rank-n bundle $\eta \to B$ with classifying map $c_\eta\colon B \to BO(n)$, naturality gives

$$w(\eta) = c_\eta^* w(\gamma^n) = c_\eta^* \widetilde{w}(\gamma^n) = \widetilde{w}(\eta).$$

Thus $w = \widetilde{w}$ on all bundles. □

23.3 Projective space calculations

Let $\mathbb{R}P^n$ be real projective space, and let $a \in H^1(\mathbb{R}P^n;\mathbb{Z}_2)$ denote its standard generator. Then

$$H^*(\mathbb{R}P^n;\mathbb{Z}_2) \cong \mathbb{Z}_2[a]/(a^{n+1}).$$

Proposition 23.7

Let $\gamma_n^1 \to \mathbb{R}P^n$ be the canonical real line bundle. Then

$$w(\gamma_n^1) = 1 + a.$$

Proof. The bundle $\gamma_1^1 \to \mathbb{R}P^1$ is the restriction of γ_n^1 along the standard inclusion $j\colon \mathbb{R}P^1 \hookrightarrow \mathbb{R}P^n$. Naturality gives

$$j^*(w_1(\gamma_n^1)) = w_1(\gamma_1^1) \neq 0,$$

so $w_1(\gamma_n^1)$ is the unique nonzero element of $H^1(\mathbb{R}P^n;\mathbb{Z}_2)$, namely a. All higher classes vanish by Item (1). □

Let $\gamma^\perp \to \mathbb{R}P^n$ be the orthogonal complement of γ_n^1 inside the trivial bundle ε^{n+1}. Thus

$$\gamma_n^1 \oplus \gamma^\perp \cong \varepsilon^{n+1}.$$

Proposition 23.8

The total class of $\gamma^\perp$ is

$$w(\gamma^\perp) = (1+a)^{-1} = 1 + a + a^2 + \cdots + a^n \in H^*(\mathbb{R}P^n;\mathbb{Z}_2).$$

Proof. By Item (3) and Item (1),

$$w(\gamma_n^1)\, w(\gamma^\perp) = w(\varepsilon^{n+1}) = 1,$$

hence $w(\gamma^\perp) = (w(\gamma_n^1))^{-1} = (1+a)^{-1}$. In the ring $\mathbb{Z}_2[a]/(a^{n+1})$ one checks directly that

$$(1+a)(1 + a + a^2 + \cdots + a^n) = 1 + a^{n+1} = 1.$$

□

Now consider the tangent bundle τ of $\mathbb{R}P^n$.

Proposition 23.9

There is a bundle isomorphism

$$\tau \;\cong\; \mathrm{Hom}(\gamma_n^1, \gamma^\perp).$$

Proof. A point of $\mathbb{R}P^n$ is a line $L \subset \mathbb{R}^{n+1}$. A tangent vector at L describes an infinitesimal change of the line, which is determined by a linear map $L \to L^\perp$. This identifies the fiber τ_L with $\mathrm{Hom}(L, L^\perp)$, and these identifications vary continuously with L. □

Since $\varepsilon^{n+1} \cong \gamma_n^1 \oplus \gamma^\perp$, applying $\mathrm{Hom}(\gamma_n^1, -)$ gives

$$\mathrm{Hom}(\gamma_n^1, \varepsilon^{n+1}) \;\cong\; \mathrm{Hom}(\gamma_n^1, \gamma_n^1) \;\oplus\; \mathrm{Hom}(\gamma_n^1, \gamma^\perp) \;\cong\; \varepsilon^1 \oplus \tau.$$

On the other hand, since ε^{n+1} is trivial,

$$\mathrm{Hom}(\gamma_n^1, \varepsilon^{n+1}) \;\cong\; (\gamma_n^1)^{\oplus(n+1)}.$$

Therefore

$$\tau \oplus \varepsilon^1 \;\cong\; (\gamma_n^1)^{\oplus(n+1)}.$$

Theorem 23.10

Let τ be the tangent bundle of $\mathbb{R}P^n$. Then

$$w(\tau) = (1+a)^{n+1} \in H^*(\mathbb{R}P^n; \mathbb{Z}_2).$$

Proof. Using Item (2) and Item (3),

$$w(\tau) = w(\tau \oplus \varepsilon^1) = w\big((\gamma_n^1)^{\oplus(n+1)}\big) = w(\gamma_n^1)^{n+1} = (1+a)^{n+1},$$

where the last step uses Proposition 23.7. □

Example. For $\mathbb{R}P^2$, the formula gives $w(\tau) = (1+a)^3 = 1 + a + a^2$. In particular, $w_1(\tau) = a \neq 0$, so $\mathbb{R}P^2$ is not orientable. For $\mathbb{R}P^3$, one gets $w(\tau) = (1+a)^4 = 1 + a^4 = 1$, since $a^4 = 0$ in $H^*(\mathbb{R}P^3; \mathbb{Z}_2)$. Thus all Stiefel–Whitney classes of $\mathbb{R}P^3$ vanish, a first hint that $\mathbb{R}P^3$ behaves unusually well among projective spaces.

24 Chern Classes and the Projective Bundle Theorem

24.1 Projectivization and the tautological line bundle

Let $\xi \to B$ be a complex vector bundle of rank n. Define its projectivization $q\colon P(\xi) \to B$ by

$$P(\xi) = \bigsqcup_{b \in B} \mathbb{C}P(\xi_b),$$

the space of complex lines in each fiber.

There is a canonical complex line bundle on $P(\xi)$.

Definition 24.1. Tautological line bundle

Let $q\colon P(\xi) \to B$ be as above. The **tautological line bundle** $\lambda \to P(\xi)$ is the subbundle $\lambda \subset q^*\xi$ whose fiber at a point $\ell \in \mathbb{C}P(\xi_b)$ is the line $\ell \subset \xi_b$. We write

$$x = c_1(\lambda^\vee) \in H^2(P(\xi); \mathbb{Z}).$$

Remark. Some authors define the generator on $\mathbb{C}P^n$ using $c_1(\lambda)$ rather than $c_1(\lambda^\vee)$. The two choices differ by a sign.

24.2 The projective bundle theorem

The key structural result is that the cohomology of $P(\xi)$ is built from that of B by adjoining the class x.

Theorem 24.2. Projective bundle theorem

Let $\xi \to B$ be a rank-n complex vector bundle over a CW complex. With $x = c_1(\lambda^\vee) \in H^2(P(\xi); \mathbb{Z})$ as in Definition 24.1, the groups

$$1,\ x,\ x^2,\ \ldots,\ x^{n-1}$$

form a basis of the $H^*(B; \mathbb{Z})$-module $H^*(P(\xi); \mathbb{Z})$, where the module structure is via q^*. In particular, $q^*\colon H^*(B; \mathbb{Z}) \to H^*(P(\xi); \mathbb{Z})$ is injective.

Since x^n lies in the $H^*(B)$-span of $1, x, \ldots, x^{n-1}$, there are uniquely determined classes in $H^{2i}(B; \mathbb{Z})$ that record this dependence. These classes are the Chern classes.

Definition 24.3. Chern classes via the projective bundle relation

Let $\xi \to B$ be a rank-n complex vector bundle and write $x = c_1(\lambda^\vee) \in H^2(P(\xi); \mathbb{Z})$. There exist unique classes $c_i(\xi) \in H^{2i}(B; \mathbb{Z})$ with $c_0(\xi) = 1$ such that

$$x^n - q^*c_1(\xi)\, x^{n-1} + q^*c_2(\xi)\, x^{n-2} + \cdots + (-1)^n q^*c_n(\xi) = 0 \quad \text{in } H^{2n}(P(\xi); \mathbb{Z}).$$

The **total Chern class** is
$$c(\xi) = 1 + c_1(\xi) + \cdots + c_n(\xi) \in H^*(B;\mathbb{Z}).$$

Remark. The sign pattern in Definition 24.3 is arranged so that for a direct sum of line bundles $L_1 \oplus \cdots \oplus L_n$ the relation becomes
$$\prod_{i=1}^{n}(x - c_1(L_i^\vee)) = 0 \quad \text{in } H^*(P(L_1 \oplus \cdots \oplus L_n);\mathbb{Z}),$$
so $c_k(L_1 \oplus \cdots \oplus L_n)$ is the kth elementary symmetric polynomial in the classes $c_1(L_i^\vee)$.

24.3 Basic properties and the Whitney sum formula

The definition already gives naturality.

Proposition 24.4. Naturality

For a map $f\colon B' \to B$ one has
$$c_i(f^*\xi) = f^*c_i(\xi) \quad \text{for all } i.$$

Proof. There is an induced map $P(f^*\xi) \to P(\xi)$ commuting with projections to the base, and it pulls back λ to the tautological line bundle over $P(f^*\xi)$. Therefore it pulls back x to the corresponding class x'. Pulling back the defining relation in Definition 24.3 gives the defining relation for $f^*\xi$, and uniqueness of the coefficients forces $c_i(f^*\xi) = f^*c_i(\xi)$. □

The Whitney sum formula is best understood after splitting.

Theorem 24.5. Splitting principle

Let $\xi \to B$ be a rank-n complex vector bundle. There exists a map $p\colon F(\xi) \to B$ such that

(1) $p^*\colon H^*(B;\mathbb{Z}) \to H^*(F(\xi);\mathbb{Z})$ is injective,

(2) $p^*\xi$ splits as a direct sum of line bundles,
$$p^*\xi \cong L_1 \oplus \cdots \oplus L_n.$$

Write $x_i = c_1(L_i^\vee) \in H^2(F(\xi);\mathbb{Z})$. Then by Definition 24.3,
$$p^*c(\xi) = \prod_{i=1}^{n}(1 + x_i).$$
This way of writing $c(\xi)$ is often called the expression in *Chern roots*.

Theorem 24.6. Whitney sum formula

For complex vector bundles $\xi, \eta \to B$ one has
$$c(\xi \oplus \eta) = c(\xi) \smile c(\eta).$$
Equivalently,
$$c_k(\xi \oplus \eta) = \sum_{i+j=k} c_i(\xi) \smile c_j(\eta).$$

Proof. Apply Theorem 24.5 to $\xi \oplus \eta$ and pull back further along the projection to ensure both ξ and η split into line bundles. Then $p^*\xi \cong \bigoplus_i L_i$ and $p^*\eta \cong \bigoplus_j M_j$, so
$$p^*c(\xi \oplus \eta) = c\left(\bigoplus_i L_i \oplus \bigoplus_j M_j\right) = \prod_i (1 + c_1(L_i^\vee)) \prod_j (1 + c_1(M_j^\vee)) = p^*c(\xi)\, p^*c(\eta).$$
Since p^* is injective on cohomology, the claimed identity holds on B. □

24.4 Computations on complex projective space

Take $B = *$. For the trivial bundle $\xi = \varepsilon^{n+1}$ over a point, we have $P(\xi) = \mathbb{C}P^n$ and λ is the canonical line bundle over $\mathbb{C}P^n$. The relation in Definition 24.3 becomes

$$x^{n+1} = 0.$$

Since Theorem 24.2 says $1, x, \ldots, x^n$ is a $\mathbb{Z}$-basis, this gives

$$H^*(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}[x]/(x^{n+1}), \qquad |x| = 2.$$

Let Q be the quotient bundle over $\mathbb{C}P^n$ defined by the short exact sequence

$$0 \longrightarrow \lambda \longrightarrow \varepsilon^{n+1} \longrightarrow Q \longrightarrow 0.$$

Using Theorem 24.6 and $c(\varepsilon^{n+1}) = 1$, we get

$$c(Q) = c(\lambda)^{-1} = (1 - c_1(\lambda^\vee))^{-1} = (1-x)^{-1} = 1 + x + x^2 + \cdots + x^n,$$

since $x^{n+1} = 0$.

The tangent bundle of $\mathbb{C}P^n$ can be identified as

$$T\mathbb{C}P^n \cong \mathrm{Hom}_{\mathbb{C}}(\lambda, Q) \cong \lambda^\vee \otimes Q.$$

From the decomposition

$$\mathrm{Hom}_{\mathbb{C}}(\lambda, \varepsilon^{n+1}) \cong \lambda^{\vee \oplus (n+1)} \cong \mathrm{Hom}_{\mathbb{C}}(\lambda, \lambda) \ \oplus \ \mathrm{Hom}_{\mathbb{C}}(\lambda, Q),$$

we obtain a stable isomorphism

$$T\mathbb{C}P^n \oplus \varepsilon^1 \cong \lambda^{\vee \oplus (n+1)}.$$

Applying Theorem 24.6 gives

$$c(T\mathbb{C}P^n) = c(\lambda^{\vee \oplus (n+1)}) = (1+x)^{n+1} \quad \text{in } H^*(\mathbb{C}P^n; \mathbb{Z}).$$

25 Pontryagin Classes and Further Uses of the Euler Class

25.1 Complexification and the definition of Pontryagin classes

Let $\xi \to B$ be a real vector bundle of rank n. Its *complexification* is the complex rank-n bundle

$$\xi \otimes \mathbb{C} \ \to \ B$$

whose fiber over b is $\xi_b \otimes_{\mathbb{R}} \mathbb{C}$.

Definition 25.1. Pontryagin classes

For $i \geq 1$, the i**th Pontryagin class** of ξ is

$$p_i(\xi) = (-1)^i \, c_{2i}(\xi \otimes \mathbb{C}) \in H^{4i}(B; \mathbb{Z}).$$

The **total Pontryagin class** is

$$p(\xi) = 1 + p_1(\xi) + p_2(\xi) + \cdots \in H^*(B; \mathbb{Z}),$$

with $p_i(\xi) = 0$ for $2i > n$.

Proposition 25.2. Basic properties

(1) (**Naturality**) For any map $f\colon B' \to B$, one has $p_i(f^*\xi) = f^* p_i(\xi)$.
(2) (**Stability**) $p(\xi \oplus \varepsilon^k) = p(\xi)$ for all $k \geq 0$.
(3) (**Whitney sum**) $p(\xi \oplus \eta) = p(\xi) \smile p(\eta)$.

Proof. For Item (1), complexification commutes with pullback, and Chern classes are natural by Proposition 24.4.
For Item (2), the complexification of ε^k is a trivial complex bundle, so all its positive Chern classes vanish, hence complexification does not change the even Chern classes.

For Item (3), one has $(\xi \oplus \eta) \otimes \mathbb{C} \cong (\xi \otimes \mathbb{C}) \oplus (\eta \otimes \mathbb{C})$. Apply the Whitney sum formula for Chern classes Theorem 24.6 and then take the even components with the sign $(-1)^i$ built into Definition 25.1. □

Example. If n is any positive integer, then the tangent bundle of the sphere satisfies

$$p(TS^n) = 1.$$

Indeed $TS^n \oplus \varepsilon^1$ is trivial, so Item (2) gives $p(TS^n) = 1$.

25.2 Comparing Pontryagin and Chern classes

Let $\omega \to B$ be a complex vector bundle. Write $\omega_{\mathbb{R}}$ for the underlying real bundle, and $\overline{\omega}$ for the conjugate complex bundle.

Lemma 25.3 There is a canonical complex bundle isomorphism

$$\omega_{\mathbb{R}} \otimes \mathbb{C} \;\cong\; \omega \oplus \overline{\omega}.$$

Proof. Fiberwise, if F is a complex vector space, then $F_{\mathbb{R}} \otimes \mathbb{C}$ splits as $F \oplus \overline{F}$ by sending $x+iy$ to $(x-iy,\ x+iy)$. These splittings vary continuously in trivializations, hence assemble to the stated bundle isomorphism. □

Proposition 25.4. Pontryagin classes of an underlying real bundle

Let ω be a complex rank-n bundle. Then

$$1 - p_1(\omega_{\mathbb{R}}) + p_2(\omega_{\mathbb{R}}) - \cdots + (-1)^n p_n(\omega_{\mathbb{R}}) \;=\; c(\omega) \smile c(\overline{\omega}).$$

Proof. By Lemma 25.3,

$$c(\omega_{\mathbb{R}} \otimes \mathbb{C}) = c(\omega \oplus \overline{\omega}) = c(\omega)\, c(\overline{\omega})$$

by Theorem 24.6. Now take the degree $4i$ part and use Definition 25.1, which says $p_i(\omega_{\mathbb{R}}) = (-1)^i c_{2i}(\omega_{\mathbb{R}} \otimes \mathbb{C})$. □

A useful special case is rank 1.

Proposition 25.5

Let $L \to B$ be a complex line bundle and let $L_{\mathbb{R}}$ be its underlying real 2-plane bundle. Then

$$p_1(L_{\mathbb{R}}) = c_1(L)^2.$$

If $L_{\mathbb{R}}$ is oriented using its complex structure, then

$$e(L_{\mathbb{R}}) = c_1(L) \qquad \text{and hence} \qquad p_1(L_{\mathbb{R}}) = e(L_{\mathbb{R}})^2.$$

Proof. For a line bundle, $c(L) = 1 + c_1(L)$ and $c(\overline{L}) = 1 - c_1(L)$, so

$$c(L)c(\overline{L}) = 1 - c_1(L)^2.$$

Apply Proposition 25.4 with $n = 1$, which reads $1 - p_1(L_{\mathbb{R}}) = c(L)c(\overline{L})$. This gives $p_1(L_{\mathbb{R}}) = c_1(L)^2$.
For the Euler class, a complex structure on a real 2-plane bundle is the same as an orientation together with a compatible fiberwise linear operator J with $J^2 = -1$. In that situation the Thom class of the oriented real bundle corresponds to the Thom class of the complex line bundle, and the Euler class agrees with the first Chern class. Combine this with the first part. □

The last identity is a template for higher rank as well.

Proposition 25.6

Let $\xi \to B$ be an oriented real vector bundle of rank $2k$. Then

$$p_k(\xi) = e(\xi)^2 \in H^{4k}(B; \mathbb{Z}).$$

Proof. Complexify ξ and regard $(\xi \otimes \mathbb{C})_{\mathbb{R}}$ as a real bundle of rank $4k$. The complex structure on $\xi \otimes \mathbb{C}$ induces a preferred orientation on $(\xi \otimes \mathbb{C})_{\mathbb{R}}$. By definition,

$$p_k(\xi) = (-1)^k c_{2k}(\xi \otimes \mathbb{C}).$$

On the other hand, the top Chern class $c_{2k}(\xi \otimes \mathbb{C})$ agrees with the Euler class of $(\xi \otimes \mathbb{C})_{\mathbb{R}}$ up to the same sign convention, so $p_k(\xi)$ is the Euler class of $(\xi \otimes \mathbb{C})_{\mathbb{R}}$.

Now use the fiberwise identification $(\xi \otimes \mathbb{C})_{\mathbb{R}} \cong \xi \oplus \xi$. With the induced orientation, the Euler class satisfies

$$e(\xi \oplus \xi) = e(\xi) \smile e(\xi)$$

by the multiplicativity of Euler classes for oriented bundles, proved from Thom classes in Lecture 22. Putting these together gives $p_k(\xi) = e(\xi)^2$. □

25.3 A model computation: $\mathbb{C}P^n$

Let $x \in H^2(\mathbb{C}P^n; \mathbb{Z})$ be the class from Lecture 24, defined by $x = c_1(\lambda^\vee)$ for the tautological line bundle λ on $\mathbb{C}P^n$. From Lecture 24 we have

$$c(T\mathbb{C}P^n) = (1 + x)^{n+1}.$$

Theorem 25.7

For the underlying real tangent bundle $(T\mathbb{C}P^n)_{\mathbb{R}}$ one has

$$p\big((T\mathbb{C}P^n)_{\mathbb{R}}\big) = (1 + x^2)^{n+1}.$$

Equivalently,

$$p_k\big((T\mathbb{C}P^n)_{\mathbb{R}}\big) = \binom{n+1}{k} x^{2k} \qquad \text{for } 1 \le k \le \lfloor n/2 \rfloor.$$

Proof. Apply Proposition 25.4 with $\omega = T\mathbb{C}P^n$. Complex conjugation on $\mathbb{C}P^n$ sends $\lambda^\vee$ to λ, hence sends $x = c_1(\lambda^\vee)$ to $-x$. Therefore

$$c(\overline{T\mathbb{C}P^n}) = (1 - x)^{n+1}.$$

Then Proposition 25.4 gives

$$1 - p_1 + p_2 - \cdots = (1 + x)^{n+1}(1 - x)^{n+1} = (1 - x^2)^{n+1}.$$

Comparing coefficients of x^{2k} yields $(-1)^k p_k = (-1)^k \binom{n+1}{k} x^{2k}$, hence $p_k = \binom{n+1}{k} x^{2k}$. This is the stated total class. □

Example. For $\mathbb{C}P^2$, the formula gives

$$p_1\big((T\mathbb{C}P^2)_{\mathbb{R}}\big) = 3x^2 \in H^4(\mathbb{C}P^2; \mathbb{Z}).$$

For $\mathbb{C}P^3$, it gives $p_1 = 4x^2$ and $p_2 = 6x^4$.

Euler classes can also be read cleanly for complex manifolds.

Proposition 25.8

The Euler class of the underlying real bundle satisfies

$$e\big((T\mathbb{C}P^n)_{\mathbb{R}}\big) = c_n(T\mathbb{C}P^n) = \binom{n+1}{n} x^n = (n+1)x^n.$$

Proof. A complex rank-n bundle has a canonical orientation on its underlying real rank-$2n$ bundle, and the Euler class agrees with the top Chern class. From $c(T\mathbb{C}P^n) = (1 + x)^{n+1}$, the degree $2n$ term is $(n + 1)x^n$. □

Example. Pairing Proposition 25.8 with the fundamental class gives

$$\langle e\big((T\mathbb{C}P^n)_{\mathbb{R}}\big), [\mathbb{C}P^n] \rangle = n + 1,$$

which matches the Euler characteristic computed from the CW structure with one cell in each even dimension.

25.4 Stable calculations in practice

Stable equivalences are often the shortest route to Pontryagin classes because of Item (2). For instance, Lecture 24 gave a stable isomorphism

$$T\mathbb{C}P^n \oplus \varepsilon^1 \cong (\lambda^\vee)^{\oplus(n+1)}$$

as complex bundles. Passing to underlying real bundles and using Item (2) and Item (3), one gets

$$p\big((T\mathbb{C}P^n)_{\mathbb{R}}\big) = p\big((\lambda^\vee_{\mathbb{R}})^{\oplus(n+1)}\big) = p(\lambda^\vee_{\mathbb{R}})^{n+1}.$$

Now Proposition 25.5 says $p(\lambda^\vee_{\mathbb{R}}) = 1 + x^2$, and this recovers Theorem 25.7 immediately.

The same pattern works for many Grassmannians and flag manifolds, where tangent bundles admit stable

descriptions in terms of the tautological bundles and their orthogonal complements.

26 Geometric Consequences of Characteristic Classes

26.1 Nontriviality and the vanishing tests

A first use of characteristic classes is negative: nonzero classes rule out overly simple geometric behavior.

Proposition 26.1

Let $\xi \to B$ be a real vector bundle and $\eta \to B$ a complex vector bundle.

(1) If ξ is trivial, then $w(\xi) = 1$.
(2) If η is trivial, then $c(\eta) = 1$.
(3) If ξ is trivial, then $p(\xi) = 1$.
(4) If ξ is oriented and trivial of positive rank, then $e(\xi) = 0$.

Consequently, if any of these classes is nonzero in positive degree, the bundle is nontrivial.

Proof. For Item (1), trivial bundles admit a global frame, hence split as sums of trivial line bundles; apply the Whitney product formula and $w(\varepsilon^1) = 1$.
For Item (2), the same argument applies with complex line bundles and the Whitney sum formula for Chern classes.
For Item (3), use Definition 25.1 and that the complexification of a trivial real bundle is a trivial complex bundle, so its Chern classes vanish.
For Item (4), a trivial oriented bundle of positive rank has a nowhere zero section, so its Euler class vanishes by Proposition 22.8Item (2). □

Example. Let $x \in H^2(\mathbb{C}P^n; \mathbb{Z})$ be the generator from Definition 24.1. Since $x = c_1(\lambda^\vee)$, the tautological line bundle λ on $\mathbb{C}P^n$ is not trivial.

26.2 Sections, Euler classes, and the Gysin sequence

A section of a bundle is often the geometric object one wants; characteristic classes constrain its existence.

Proposition 26.2

Let $\xi \to B$ be an oriented real vector bundle of positive rank. If ξ admits a nowhere zero section, then $e(\xi) = 0$.

Proof. A nowhere zero section $s\colon B \to E(\xi)$ lands in $E(\xi)_0 = E(\xi)\setminus B$. Pull back ξ along s. The bundle $s^*\xi$ has a tautological nowhere zero section, hence $e(s^*\xi) = 0$. By naturality of Euler classes Proposition 22.8Item (1), $e(s^*\xi) = s^*e(\xi)$.
On the other hand, the composite $B \xrightarrow{s} E(\xi)_0 \hookrightarrow E(\xi) \xrightarrow{\pi} B$ is the identity. Since $\pi^*\colon H^*(B;\mathbb{Z}) \to H^*(E(\xi);\mathbb{Z})$ is an isomorphism, the restriction map $H^*(E(\xi);\mathbb{Z}) \to H^*(E(\xi)_0;\mathbb{Z})$ is injective in degree equal to $\mathrm{rk}(\xi)$ whenever cup product by $e(\xi)$ is injective on $H^0(B;\mathbb{Z})$. Equivalently, the first arrow in the Gysin sequence Theorem 22.11 detects whether $e(\xi)$ is zero. Since $s^*e(\xi) = 0$ and s^* is left inverse to π^*, it follows that $e(\xi) = 0$. □

The Gysin sequence packages many quick deductions. For an oriented sphere bundle $p\colon S(\xi) \to B$ of fiber dimension $n-1$, exactness in Theorem 22.11 gives:

Corollary 26.3. Let $\xi \to B$ be oriented of rank $n \geq 1$ and let $p\colon S(\xi) \to B$ be its sphere bundle.

(1) If $e(\xi) \smile -\colon H^{k-n}(B) \to H^k(B)$ is injective, then $p^*\colon H^k(B) \to H^k(S(\xi))$ is injective.
(2) If $e(\xi) \smile -\colon H^{k-n}(B) \to H^k(B)$ is surjective, then $p^*\colon H^k(B) \to H^k(S(\xi))$ is zero.

Proof. Exactness at $H^k(B) \xrightarrow{p^*} H^k(S(\xi))$ in Theorem 22.11 says

$$\ker(p^*) = \mathrm{im}(H^{k-n}(B) \xrightarrow{\smile e(\xi)} H^k(B)).$$

This gives Items (1) and (2). □

26.3 The hairy ball phenomenon

Let TS^n be the tangent bundle of the sphere. A nowhere zero tangent vector field on S^n is exactly a nowhere zero section of TS^n.

Theorem 26.4

Let $u \in H^n(S^n;\mathbb{Z})$ be a generator. Then

$$e(TS^n) = \begin{cases} 2u & n \text{ even}, \\ 0 & n \text{ odd}. \end{cases}$$

Proof. For a closed oriented smooth manifold M, the pairing $\langle e(TM), [M]\rangle$ equals the Euler characteristic $\chi(M)$. Applying this to $M = S^n$, one has $\chi(S^n) = 2$ for even n and $\chi(S^n) = 0$ for odd n. Since $H^n(S^n;\mathbb{Z}) \cong \mathbb{Z}$ is generated by u, the Euler class must be $\chi(S^n)\, u$. □

Corollary 26.5. If n is even, there is no nowhere zero continuous tangent vector field on S^n.

Proof. If such a vector field existed, Proposition 26.2 would give $e(TS^n) = 0$. This contradicts Theorem 26.4 when n is even. □

Example. For odd n, there is a concrete nowhere zero tangent vector field. Write $S^{2m+1} \subset \mathbb{C}^{m+1}$ and define

$$v(z) = iz.$$

Then $v(z)$ is tangent to the sphere at z because $\operatorname{Re}\langle iz, z\rangle = 0$, and it never vanishes. This matches $e(TS^{2m+1}) = 0$ in Theorem 26.4.

26.4 Real projective space and vector fields

Let τ be the tangent bundle of $\mathbb{R}P^n$ and let $a \in H^1(\mathbb{R}P^n;\mathbb{Z}_2)$ be the standard generator. Lecture 23 computed

$$w(\tau) = (1+a)^{n+1}.$$

Proposition 26.6

$\mathbb{R}P^n$ is orientable if and only if n is odd.

Proof. By Proposition 23.4, orientability is equivalent to $w_1(\tau) = 0$. From $w(\tau) = (1+a)^{n+1}$, one has $w_1(\tau) = (n+1)a$. In $\mathbb{Z}_2$-coefficients this vanishes exactly when $n+1$ is even. □

Proposition 26.7

If n is even, then $\mathbb{R}P^n$ admits no nowhere zero tangent vector field.

Proof. If $\mathbb{R}P^n$ had a nowhere zero tangent vector field, then τ would split as $\varepsilon^1 \oplus \tau'$. By Proposition 23.5, this would force $w_n(\tau) = 0$.
But $w_n(\tau)$ is the degree-n term of $(1+a)^{n+1}$, namely

$$w_n(\tau) = \binom{n+1}{n} a^n = (n+1)a^n = a^n,$$

which is nonzero in $H^n(\mathbb{R}P^n;\mathbb{Z}_2) \cong \mathbb{Z}_2$. This contradiction rules out a nowhere zero section. □

Example. For $n = 2$, the computation gives $w(\tau) = 1 + a + a^2$. The top class $w_2(\tau) = a^2 \neq 0$ already forces the absence of a nowhere zero tangent vector field on $\mathbb{R}P^2$.

26.5 Stable triviality tests via p and c

Stable equivalence is often the right notion for tangent bundles, and Pontryagin classes are stable invariants.

Proposition 26.8

If a real bundle ξ is stably trivial, then $p(\xi) = 1$. If a complex bundle η is stably trivial, then $c(\eta) = 1$.

Proof. If $\xi \oplus \varepsilon^k$ is trivial, then $p(\xi \oplus \varepsilon^k) = 1$ by Proposition 26.1Item (3). By stability Proposition 25.2Item (2), $p(\xi \oplus \varepsilon^k) = p(\xi)$. The complex statement is the same, using the Whitney sum formula for Chern classes and that $c(\varepsilon^k) = 1$. $\square$

Example. For $\mathbb{C}P^2$, Theorem 25.7 gives

$$p_1\big((T\mathbb{C}P^2)_{\mathbb{R}}\big) = 3x^2 \in H^4(\mathbb{C}P^2; \mathbb{Z}),$$

which is nonzero. By Proposition 26.8, the tangent bundle of $\mathbb{C}P^2$ is not stably trivial.